Probing Non-Gaussian Cosmic Fields with Critical-Point Statistics

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To 呆呆金, my dearest friend.

Abstract

We extend the leading-order analytic formalism for the number density of critical points in weakly non-Gaussian smoothed cosmic fields and validate it by direct numerical counting based on Newton's method. The observables are cumulative and differential number densities as functions of threshold for peaks, voids, saddles in two-dimensional (2D) space, peaks, voids, filaments, and walls in three-dimensional (3D) space, and the Euler characteristic. We test the formulas using 2D local-type non-Gaussian maps and 3D matter-density fields from N-body simulations. The comparison shows that the leading-order bispectrum correction gives a substantially better description than the Gaussian prediction in the weakly non-Gaussian regime and at sufficiently large smoothing scales, while higher-order corrections are required for more nonlinear fields, especially in the tails corresponding to high-threshold peaks and low-threshold voids. Besides, We also find that the abundances of saddles are better described by weakly non-Gaussian predictions than peaks and voids, which is caused by the saddles are predominantly reside in the mildly regime. Within the valid regime, the analytic formula can be used to estimate parameter responses of critical-point statistics efficiently, reducing the need for separate large N-body simulation suites for every parameter variation. Then, we use this as probe to test how the constrain power for the standard cosmological paramters. Finally, we provide a first attempt at the kurtosis order correction in the appendix and hope to address the fully next-to-leading order in the future.

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List of Abbreviations

BAO baryon acoustic oscillation
BBKS Bardeen–Bond–Kaiser–Szalay framework
CDM cold dark matter
CPs critical points
LO leading order
LSS large-scale structure
NG non-Gaussian
NLO next-to-leading order
PT perturbation theory
SPT standard perturbation theory

Chapter 1

Introduction

The large-scale structure (LSS) of the Universe provides one of the main observational windows for understanding cosmology. Small primordial fluctuations generated in the early Universe were amplified by gravitational instability, eventually forming the late-time cosmic web of clusters, filaments, sheets, and voids. Conventional cosmology is built around N-point statistics, such as correlation functions and power spectra. These statistics are powerful, but incomplete. Nonlinear gravitational evolution, projection effects, galaxy bias, and possible primordial non-Gaussianity all generate information that is not fully captured by two-point statistics. Higher-order correlations, such as the bispectrum and trispectrum, are useful, but they still do not necessarily capture all of the practically relevant information in a cosmological field.

This motivates the use of geometric and topological statistics of cosmological fields. Peak statistics, Minkowski functionals, Betti numbers, persistent homology, and related descriptors respond directly to the morphology of a field and therefore probe information beyond its variance and correlation length. The main focus of this thesis is peak and critical-point statistics, which have a long history [1–3]. In Gaussian random fields, the theory is well developed and has been widely used to connect the statistics of peaks in the initial density field to the formation of dark-matter halos; see Ref. [4] for a review. Once non-Gaussian initial conditions [5] or nonlinear gravitational evolution [6] are included, higher-order correlations become important. Applications range from primordial black holes [7–9] to weak-lensing maps [10–12].

Peak statistics can also be generalized to all critical points, abbreviated here as CPs, of a scalar field. In a three-dimensional density field, these include peaks, voids, filaments, and walls, which are local elements of the cosmic web [13]. Related classifications appear in the Zel'dovich approximation, where structures are associated with the eigenvalues of the deformation tensor [14]. Filaments trace anisotropic matter transport toward clusters [15–17]; voids provide sensitive probes of dark energy, massive neutrinos, and modified gravity [18–20]; and walls form sheet-like structures that often surround voids and connect to filaments [21, 22]. The Euler characteristic is closely related to these CPs because it can be expressed as an alternating sum of critical-point counts [23]. It also measures how the connectivity of excursion sets changes with threshold [24–27].

The goal of this thesis is to develop and test analytical and numerical tools for critical-point statistics in weakly non-Gaussian cosmological fields. The analytical side follows the random-field approach: the number density of CPs is written in terms of the joint statistics of the field and its derivatives, and weak non-Gaussian corrections are organized by cumulants or polyspectra. This connects naturally to Gram–Charlier and Wiener–Hermite expansions [24, 28]. Building on the generalized peak formalism of Ref. [28] and its extension to two-dimensional critical-point statistics [29], this thesis considers critical sets beyond peaks and compares the resulting predictions with direct numerical counts. It is worth noting that fully analytic formulae for Betti numbers and persistent homology are not yet available in the same form, and their numerical estimation can be computationally expensive. Similarly, next-to-leading-order corrections to critical-point statistics are not yet fully developed and still have important limitations. The thesis gave an attempt in Appendix. B for the kurtosis-order correction.

The numerical side is designed to test how far the leading non-Gaussian correction can be trusted; we do not develop an emulator for it. We use toy two-dimensional weak-lensing-like maps with controlled skewness, as well as three-dimensional matter density fields from N-body simulations. CPs are identified with a Newton-based refinement algorithm [30], and the resulting cumulative and differential number densities are compared with Gaussian and weakly non-Gaussian theoretical predictions. This comparison clarifies when leading-order non-Gaussian corrections improve on the Gaussian baseline and when stronger nonlinear evolution requires higher-order information.

As a practical extension, this thesis also connects CPs to forward modeling and parameter inference, exploring how these statistics may contribute to cosmological constraints. We first study the sensitivity of CPs to cosmological parameters. We then construct a reasonable data vector and use Bayesian analysis to test the stability and usefulness of the resulting constraints. Finally, we combine interpolation with nested sampling to perform standard cosmological inference. We find that the sensitivity of CPs is comparable to the Betti-number results of Ref. [31]. We also show that σ_8 can help break degeneracies with other parameters because of non-Gaussian effects.

The thesis is organized as follows:

- Chapter 2 reviews the physical background from inflationary initial conditions to late-time large-scale structure, including spectra from slow-roll inflation, linear and nonlinear perturbation theory in both Eulerian and Lagrangian space, and probes based on weak lensing and the cosmic web. This chapter provides the cosmological background needed to connect critical points to additional probes of structure.
- Chapter 3 develops the Gaussian and weakly non-Gaussian theory of critical-point number densities. We briefly discuss probability theory for random fields and the geometric description based on the Kac–Rice formula. Then, using the Gaussian theory pioneered by BBKS [3], we develop the critical-point theory based on Matsubara's work [9]. To validate the theory, we compare the predictions with a numerical critical-point-finding method in two-dimensional and three-dimensional weakly non-Gaussian test fields.
- Chapter 4 treats critical points as cosmological summary statistics and discusses forward modeling, covariance estimation, likelihood construction, and parameter constraints.
- Chapter 5 summarizes the main conclusions and future directions, while the appendices discuss numerical methods and higher-order non-Gaussian terms in the kurtosis contribution.

Chapter 2

Primordial Perturbations and Structure Formation

This chapter explains how primordial Gaussian fields are generated during inflation and how subsequent matter growth and galaxy formation lead to the cosmic web. It provides the physical background and motivations for the main content of this thesis of critical-point statistics. Section 2.1 introduces scalar-field fluctuations during inflation and shows how quantized fluctuations generate the primordial power spectrum. Section 2.2 then describes the evolution of matter perturbations with Eulerian and Lagrangian dynamics, together with the resulting power spectrum and higher-order statistics. Section 2.3 reviews two important observational probes of structure, with emphasis on weak lensing and cosmic web. The latter motivates the use of critical-point statistics as a complementary probe of cosmic fields.

Before discussing the initial conditions, we first review some basic cosmological background. The Einstein equation is

$$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}.$$

Here, $R_{\mu\nu}$ is the Ricci tensor, $g_{\mu\nu}$ is the metric, $R = R_{\mu\nu} g^{\mu\nu}$ is the Ricci scalar, and Λ is the cosmological constant. The energy-momentum tensor of a perfect fluid is

$$T^{\mu}{}_{\nu} = (\rho + p) u^{\mu} u_{\nu} + p \delta^{\mu}{}_{\nu}, \tag{2.1}$$

and it satisfies the conservation condition

$$\nabla_{\mu} T^{\mu\nu} = 0.$$

Here, u^{μ} is the fluid four-velocity, ρ is the energy density, and p is the fluid pressure. For a perfect fluid, the conservation equation gives

$$\begin{aligned} \dot{\rho} &= -3H(\rho + p), \qquad \text{(continuity)} \\ (\rho + p) a_i &= -\partial_i p, \qquad \text{(Euler)}, \end{aligned}$$

where the dot denotes a derivative with respect to cosmic time t, $H \equiv \dot{a}/a$ is the Hubble parameter, and $a^{\mu} \equiv u^{\nu} \nabla_{\nu} u^{\mu}$ is the four-acceleration. In a homogeneous and spatially flat universe, with the cosmological constant absorbed into the total energy density and pressure if present, Eq. (2.1) reduce to the Friedmann equations

$$\begin{aligned} \left(\frac{\dot{a}}{a}\right)^2 &= \frac{8\pi G}{3} \rho, \\ \frac{\ddot{a}}{a} &= -\frac{4\pi G}{3} (\rho + 3p). \end{aligned}$$

For scalar perturbations in the conformal-Newtonian gauge, assuming vanishing anisotropic stress, the line element is

$$ds^2 = -[1 + 2\Phi(\boldsymbol{x}, t)]dt^2 + a^2(t)\delta_{ij}[1 - 2\Phi(\boldsymbol{x}, t)]dx^i dx^j,$$

where Φ is the Newtonian potential.

These equations provide the simplest cosmological background for the discussion below.

2.1 Inflation as initial conditions

This section discusses primordial density fluctuations from inflation and why they have Gaussian statistics. We assume linear perturbation theory in this section and explain why linear perturbations from inflation are Gaussian.

Here, we adopt the simplest model, in which inflation is driven by a slowly rolling scalar field, the inflaton. We consider single-field slow-roll inflation as the standard paradigm for producing nearly Gaussian random fields in the early Universe, and split the inflaton field as

$$\phi(\boldsymbol{x}, t) = \phi_0(t) + \delta\phi(\boldsymbol{x}, t).$$

$\phi_0(t)$ is the classical, effectively homogeneous background field, and $\delta\phi$ is the quantum fluctuation. We focus on how Gaussian and non-Gaussian features arise in the early Universe.

See [5] and [32] for more detailed review.

2.1.1 Slow-roll dynamics in classical and quantum field theory

Inflation is a period of accelerated expansion of the Universe that lasts long enough to solve the horizon and flatness problems of standard hot Big Bang cosmology. Primordial perturbations generated during this stage provide the observable initial conditions for later structure formation. This era is characterized by $\ddot{a} > 0$, where a is the scale factor. Equivalently, $d(aH)^{-1}/dt < 0$, so the comoving Hubble horizon, $(aH)^{-1}$, decreases with time. The condition can also be written as $-\dot{H}/H^2 < 1$. In slow-roll inflation, H varies slowly on a Hubble timescale. If $|\dot{H}| \ll H^2$, then the Hubble radius $H^{-1} = a/\dot{a}$ remains nearly constant over many Hubble times, and the expansion is approximately exponential, $a \propto e^{Ht}$. Neglecting dark energy, or absorbing it into ρ and p, Eq. (2.3) implies that the pressure must satisfy $p < -\rho/3$, where ρ is the field energy density.

Classical field theory of Einstein gravity

During inflation, the dynamics are dominated by canonically normalized scalar fields, and the action is

$$S = \frac{1}{2}\int d^4x\sqrt{-g}M_{\rm Pl}^2 R,$$

where $M_{\rm Pl} \equiv 1/\sqrt{8\pi G}$ is the reduced Planck mass in natural units with $\hbar = c = 1$, and $g \equiv \det(g_{\mu\nu})$ is the determinant of the metric tensor. The equation of motion of a homogeneous scalar field $\phi = \phi_0(t)$ in an expanding, unperturbed spacetime is

$$\ddot{\phi} + 3H\dot{\phi} + V'(\phi) = 0. \tag{2.4}$$

Eq. (2.3) reads

$$3M_{\rm Pl}^2 H^2 = \frac{1}{2}\dot{\phi}^2 + V(\phi).$$

Differentiating this equation and combining it with Eq. (2.4) gives

$$2M_{\rm Pl}^2 \dot{H} = -\dot{\phi}^2.$$

To formulate the slow-roll approximation, we assume that inflation is approximately exponential:

$$\frac{|\dot{H}|}{H^2} \ll 1. \tag{2.5}$$

This is a strong condition and is equivalent to

$$3M_{\rm Pl}^2 H^2 \simeq V, \quad \Rightarrow \quad 3H\dot{\phi} \simeq -V'(\phi), \tag{2.6}$$

which is equivalent to $|\ddot{\phi}| \ll 3H|\dot{\phi}|$. This states that $\dot{\phi}$ does not change much within one Hubble time. These relations motivate the slow-roll parameter

$$\epsilon(\phi) \ll 1, \quad \text{where} \quad \epsilon(\phi) \equiv \frac{M_{\rm Pl}^2}{2}\left(\frac{V'}{V}\right)^2. \tag{2.7}$$

Taking the derivative of both sides in Eq. (2.6) gives

$$\ddot{\phi} \simeq -\frac{\dot{H}}{H}\dot{\phi} - \frac{V''\dot{\phi}}{3H}.$$

Since $\ddot{\phi} \simeq 0$, we obtain

$$|\eta(\phi)| \ll 1, \qquad \text{where} \qquad \eta \equiv M_{\rm Pl}^2 \frac{V''}{V} \simeq \frac{V''}{3H^2}. \tag{2.8}$$

The strong inequality in Eq. (2.5), together with the approximations above, constitutes the slow-roll approximation. It implies the conditions in Eqs. (2.7) and (2.8) on the potential, which we call the flatness conditions. Sometimes, we also define a such quantity that show how slowly the field is rolling:

$$\delta_{\rm sr} \equiv \frac{1}{H}\frac{\ddot{\phi}}{\dot{\phi}} \simeq \epsilon - \eta.$$

After inflation ends, the background inflaton field rolls quickly along its potential, oscillates around the minimum, and decays into Standard Model particles, reheating the Universe.

Quantum theory of a massless free scalar field during inflation

Besides the background dynamics, vacuum (quantum) fluctuations act as seeds of primordial energy-density perturbations. They survive after inflation and are the origin of all structure in the Universe. Quantum fluctuations of light fields are generated on sub-Hubble scales. Inflation stretches these fluctuations to super-Hubble scales, where they freeze. Inflation therefore converts microscopic quantum fluctuations into macroscopic classical perturbations.

For a given wavenumber before horizon exit, spacetime curvature is negligible and the field can be treated in flat-space theory. The particle number is assumed to be negligible, so each field is in the vacuum state. The vacuum state is taken to be the Bunch–Davies vacuum. Starting from the equation of motion in the unperturbed spacetime,

$$\ddot{\phi} + 3H\dot{\phi} - a^{-2}\nabla^2\phi + \frac{\partial V}{\partial \phi} = 0,$$

the first-order perturbation satisfies

$$\ddot{\delta\phi}_{\boldsymbol{k}} + 3H\dot{\delta\phi}_{\boldsymbol{k}} + \left(\frac{k}{a}\right)^2 \delta\phi_{\boldsymbol{k}} = 0.$$

Over a few Hubble times around horizon exit, we take the Hubble parameter $H = H_k = H_*$ to be constant [1]. Converting to conformal time, $d\eta = dt/a$, gives

$$\frac{d^2\delta\phi}{d\eta^2} + 2\frac{da}{ad\eta}\frac{d\delta\phi}{d\eta} + k^2\delta\phi = 0.$$

[1] this gives $a(t) = e^{H_* t}$ and then $\eta = \int dt/a(t) = -1/(aH_*)$.

Letting $\varphi = a\delta\phi$, we obtain the oscillator equation

$$\frac{d^2\varphi_k}{d\eta^2} + \omega_k^2\varphi_k = 0,$$

where

$$\omega_k^2 = k^2 - \frac{2}{\eta^2} = k^2 - 2(aH_k)^2,$$

for this equation, consider a small spacetime $k^{-1} \leqslant \Delta\eta \leqslant (aH_k)^{-1}$, and

$$(2\pi)^3\hat{\varphi}_{\boldsymbol{k}}(\eta) = \varphi_k(\eta)\hat{a}(\boldsymbol{k}) + \varphi_k^*(\eta)\hat{a}^\dagger(-\boldsymbol{k}), \tag{2.9}$$

here we quantize it using harmonic oscillator

$$(2\pi)^{-3}\big[\hat{a}(\boldsymbol{k}), \hat{a}^\dagger(\boldsymbol{k'})\big] = \delta^3(\boldsymbol{k} - \boldsymbol{k'}), \qquad \big[\hat{a}(\boldsymbol{k}), \hat{a}(\boldsymbol{k'})\big] = 0. \tag{2.10}$$

We impose

$$\varphi_k(\eta) = \frac{1}{\sqrt{2k}}e^{-ik\eta}, \tag{2.11}$$

at early times because, for $k \gg aH$, spacetime locally reduces to Minkowski space and each mode behaves as a simple harmonic oscillator. The full mode equation in inflation has an extra curvature term, $-2/\eta^2$. Its solution is a combination of Hankel functions and satisfies the initial behavior

$$\varphi_k(\eta) = \frac{e^{-ik\eta}}{\sqrt{2k}}\frac{(k\eta - i)}{k\eta}. \tag{2.12}$$

Here is the derivation. The general solution of

$$\frac{d^2\varphi_k}{d\eta^2} + \left(k^2 - \frac{\nu^2 - \frac{1}{4}}{\eta^2}\right)\varphi_k = 0,$$

is the Bessel equation of order $\nu = 3/2$, with solution

$$\varphi_k(\eta) = \sqrt{-\eta}\left[A_k H_\nu^{(1)}(-k\eta) + B_k H_\nu^{(2)}(-k\eta)\right],$$

where $H_\nu^{(1,2)}$ are Hankel functions. To fix the vacuum, we set $B_k = 0$, since the mode $e^{-ik\eta}$ has positive frequency. For large $x = -k\eta$, corresponding to the initial condition,

$$H_\nu^{(1)}(x) \sim \sqrt{\frac{2}{\pi x}}e^{i(x-\pi\nu/2-\pi/4)}, \qquad H_\nu^{(2)}(x) \sim \sqrt{\frac{2}{\pi x}}e^{-i(x-\pi\nu/2-\pi/4)},$$

so $H_\nu^{(1)}(-k\eta)$ corresponds to positive frequency, while $H_\nu^{(2)}(-k\eta)$ corresponds to negative frequency, exactly as for plane waves. Setting $B_k = 0$ defines the Bunch–Davies vacuum, which ensures that the mode reduces to the Minkowski vacuum at short distances. Put the equation when $\nu = 3/2$,

$$H_{3/2}^{(1)}(x) = -i\sqrt{\frac{2}{\pi x}}\left(1 + \frac{i}{x}\right)e^{ix},$$

and substituting these expressions gives back to Eq. (2.12). After horizon exit it approches $\varphi_k(\eta) = -i/(\sqrt{2k}k\eta)$. Given by the quantization relation Eq. (2.9) and Eq. (2.10), we have

$$(2\pi)^6\left\langle\varphi_{\boldsymbol{k}}\varphi_{\boldsymbol{k'}}\right\rangle = \left\langle\varphi_k\varphi_{k'}\hat{a}(\boldsymbol{k})\hat{a}(\boldsymbol{k'}) + \varphi_k\varphi_{k'}^*\hat{a}(\boldsymbol{k})\hat{a}(-\boldsymbol{k'}) + \varphi_k^*\varphi_{k'}\hat{a}^\dagger(-\boldsymbol{k})\hat{a}(\boldsymbol{k'}) + \varphi_k^*\varphi_{k'}^*\hat{a}^\dagger(-\boldsymbol{k})\hat{a}^\dagger(-\boldsymbol{k'})\right\rangle.$$

Thus, in dimensionless form [2]

$$(2\pi)^3 \langle \varphi_{\boldsymbol{k}} \varphi_{\boldsymbol{k}'} \rangle = \frac{2\pi^2}{k^3} \Delta_\varphi^2(k) \delta^3(\boldsymbol{k} + \boldsymbol{k}'), \qquad \Delta_\varphi^2(k,\eta) = \frac{k^3}{2\pi^2} |\varphi(\eta)|^2.$$

Dividing by a^2 and evaluating the result a few Hubble times after horizon exit, ($k < aH_k$), we find the time-independent quantity

$$P_{\delta\phi}(k) = \frac{H_k^2}{2k^3}, \qquad \Delta_{\delta\phi}^2(k) = \left(\frac{H_k}{2\pi}\right)^2.$$

Free quantum fields have Gaussian statistics because the modes are not coupled. Quantum fluctuations generate primordial density perturbations, which later evolve into the large-scale structure of the Universe. The classical background field drives inflation and sets the stage for these quantum fluctuations to be stretched to cosmological scales. Next, we see how linear evolution preserves Gaussianity and how interactions generate non-Gaussianity.

2.1.2 Perturbations and non-Gaussian corrections

Linear correction

Linear evolution preserves Gaussianity. Focusing on the simplest case of one light field and writing $\varphi = a\delta\phi$, we first include the effect of the potential:

$$\ddot{\delta\phi}_{\boldsymbol{k}} + 3H\dot{\delta\phi}_{\boldsymbol{k}} + \left(\frac{k}{a}\right)^2 \delta\phi_{\boldsymbol{k}} + m^2(\phi)\delta\phi = 0.$$

Here $m^2(\phi) \equiv \partial^2 V/\partial\phi^2$ is the effective mass-squared of the perturbation $\delta\phi$. We obtain

$$\frac{d^2\varphi_k}{d\eta^2} + \left[(am_k)^2 + k^2 - \frac{2}{\eta}\right]\varphi_k(\eta) = 0,$$

where m_k is approximately constant during the few Hubble times around horizon exit. With the initial condition in Eq. (2.11),

$$\varphi_k(\eta) = e^{i(\pi\nu/2 - \pi/4)} \sqrt{\frac{\pi}{4k}} \sqrt{-k\eta}\, H_\nu^{(1)}(-k\eta),$$

where

$$\nu = \sqrt{\frac{9}{4} - \frac{m_k^2}{H_k^2}} \simeq \frac{3}{2} - \frac{m_k^2}{3H_k^2}, \qquad (m_k^2 \ll H_k^2).$$

Well after horizon exit, the solution becomes

$$\varphi_k(\eta) = e^{i(\pi\nu/2 - \pi/4)} \frac{2^\nu \Gamma(\nu)}{2^{3/2}\Gamma\left(\frac{3}{2}\right)} \frac{1}{\sqrt{2k}} (-k\eta)^{\frac{1}{2} - \nu}, \qquad (|k\eta| \ll 1).$$

In a quasi-de Sitter universe, where $\dot{H} = -\epsilon H$, the dimensionless power spectrum becomes

$$\Delta_{\delta\phi}^2(k,\eta) \simeq \left(\frac{H_k}{2\pi}\right)^2 \left(\frac{k}{aH_k}\right)^{2m_k^2/3H_k^2}.$$

At leading order, the inflaton behaves like a free (or weakly interacting) scalar field in a quasi–de Sitter background.

More generally, one can consider a general potential, multiple fields, and metric perturbations in a generic spacetime. The gauge-invariant curvature perturbation $\mathcal{R}$ is conserved on super-horizon scales. Since $\mathcal{R}$ is linearly related to the inflaton fluctuation at first order, Gaussianity is preserved during its evolution. We discuss this point in Sec. (2.1.3).

[2] Define like $\sigma^2(\boldsymbol{x}) \equiv \langle f^2(\boldsymbol{x})\rangle = \int d^3k/(2\pi)^3\, P_f(k) = \int_0^\infty dk/k\, \Delta_f^2(k)$

Non-Gaussianity

Small departures from Gaussianity are expected from interactions and gravitational nonlinearities, although they are typically suppressed. In standard single-field slow-roll inflation, primordial perturbations are nearly Gaussian. The resulting primordial non-Gaussianity can seed additional late-time non-Gaussianity through gravitational instability. We will consider a small additional contribution $\hat{H}_I$ to the Hamiltonian, analogous to an interaction term in ordinary quantum field theory.

In the Heisenberg picture, the time dependence is

$$\hat{\varphi}_{\rm hp}(\eta) = \hat{\varphi}(\eta) + i\int_{-\infty}^{\eta} d\eta' \left[\hat{H}_I(\eta'), \hat{\varphi}(\eta)\right].$$

The connected part of the N-point correlation function is

$$\langle\hat{\varphi}_{\boldsymbol{k}_1}(\eta)\cdots\hat{\varphi}_{\boldsymbol{k}_N}(\eta)\rangle = \langle 0\,|\hat{\varphi}_{\boldsymbol{k}_1}(\eta)\cdots\hat{\varphi}_{\boldsymbol{k}_N}(\eta)|\,0\rangle - i\int_{-\infty}^{\eta} d\eta' \left\langle 0\left|\left[\hat{\varphi}_{\boldsymbol{k}_1}(\eta)\cdots\hat{\varphi}_{\boldsymbol{k}_N}(\eta),\, \hat{H}_I(\eta')\right]\right|0\right\rangle + \cdots .$$

The first term is the free, or Gaussian, contribution. The integral term is the leading interaction, or non-Gaussian, correction. The lower limit is often written as $-\infty(1-i\epsilon)$ to project onto the interacting vacuum. We define the Green function using the perturbative solution of a massless minimally coupled scalar in de Sitter spacetime:

$$\langle 0|\hat{\varphi}(\eta,\boldsymbol{k})\hat{\varphi}(\eta',\boldsymbol{k}')|0\rangle = \delta^{(3)}(\boldsymbol{k}+\boldsymbol{k}')\,\varphi_k(\eta)\,\varphi_k^*(\eta') = \delta^{(3)}(\boldsymbol{k}+\boldsymbol{k}')G_\varphi(k,\eta,\eta').$$

Thus,

$$G_\varphi(k,\eta,\eta') = \varphi_k(\eta)\varphi_k^*(\eta') = \frac{1}{2k}\left(1-\frac{i}{k\eta}\right)\left(1+\frac{i}{k\eta'}\right)e^{-ik(\eta-\eta')}.$$

Ignoring metric perturbations, we now keep the quadratic term in the perturbed field equation:

$$\ddot{\delta\phi} + 2H_*\dot{\delta\phi} - a^{-2}\nabla^2\delta\phi + \frac{1}{2}V_*'''(\delta\phi)^2 = 0.$$

Here the star denotes the value during inflation, which is taken to be constant, and $V_*''' = \partial^3 V/\partial\phi^3$. The interaction Hamiltonian is

$$\hat{H}_I(\eta) = \frac{V_*'''}{3!}\int d^3x\, a^4(\eta)\hat{\varphi}^3(\boldsymbol{x},\eta).$$

We calculate the bispectrum to first order in H_I. Defining $\lambda(\eta) = V_*'''a^4(\eta)/6$, we obtain

$$\begin{aligned}
\langle\hat{\varphi}_{\boldsymbol{k}_1}\hat{\varphi}_{\boldsymbol{k}_2}\hat{\varphi}_{\boldsymbol{k}_3}\rangle &= -i\int_{-\infty}^{\eta} d\eta' \left\langle 0\left|\left[\hat{\varphi}_{\boldsymbol{k}_1}\hat{\varphi}_{\boldsymbol{k}_2}\hat{\varphi}_{\boldsymbol{k}_3}, \hat{H}_I(\eta')\right]\right|0\right\rangle \\
&= -i\int_{-\infty}^{\eta} d\eta'\,\lambda(\eta')\int [d^3p]^3\,(2\pi)^3\delta(\Sigma\boldsymbol{p})\,\langle 0|[\hat{\varphi}_{\boldsymbol{k}_1}\hat{\varphi}_{\boldsymbol{k}_2}\hat{\varphi}_{\boldsymbol{k}_3}, \hat{\varphi}_{\boldsymbol{p}_1}\hat{\varphi}_{\boldsymbol{p}_2}\hat{\varphi}_{\boldsymbol{p}_3}]|0\rangle \\
&= 2\int_{-\infty}^{\eta} d\eta'\,\lambda(\eta')\mathrm{Im}\left(\prod_{i=1}^{3} G_{k_i}\right)\int [d^3p]^3\,(2\pi)^3\delta(\Sigma\boldsymbol{p})\sum_{\pi\in S_3}\prod_{i=1}^{3}\delta^{(3)}(\boldsymbol{k}_i+\boldsymbol{p}_{\pi(i)}) \\
&= 12(2\pi)^3\delta^{(3)}(\boldsymbol{k}_1+\boldsymbol{k}_2+\boldsymbol{k}_3)\int_{-\infty}^{\eta} d\eta'\,\lambda(\eta')\,\mathrm{Im}\Big[\prod_{i=1}^{3} G_{k_i}(\eta,\eta')\Big] \\
&= 12\,\mathrm{Im}\left[\varphi_{k_1}(\eta)\varphi_{k_2}(\eta)\varphi_{k_3}(\eta)\int_{-\infty}^{\eta} d\eta'\,\lambda(\eta')\,\varphi_{k_1}^*(\eta')\varphi_{k_2}^*(\eta')\varphi_{k_3}^*(\eta')\right](2\pi)^3\delta^{(3)}(\Sigma\boldsymbol{k}),
\end{aligned}$$

where the third line uses

$$\langle 0|\hat{\varphi}_{\boldsymbol{k}_1}\hat{\varphi}_{\boldsymbol{k}_2}\hat{\varphi}_{\boldsymbol{k}_3}\hat{\varphi}_{\boldsymbol{p}_1}\hat{\varphi}_{\boldsymbol{p}_2}\hat{\varphi}_{\boldsymbol{p}_3}|0\rangle_{\text{conn}} = \sum_{\pi\in S_3}\prod_{i=1}^{3}\langle 0|\hat{\varphi}_{\boldsymbol{k}_i}\hat{\varphi}_{\boldsymbol{p}_{\pi(i)}}|0\rangle$$
$$= \sum_{\pi\in S_3}\prod_{i=1}^{3}\delta^{(3)}(\boldsymbol{k}_i+\boldsymbol{p}_{\pi(i)})G_\varphi(k_i,\eta,\eta').$$

The $\boldsymbol{k}_i$ terms do not contract with one another because the bispectrum requires $\delta^{(3)}(\Sigma\boldsymbol{k})$. We also use that $G_\varphi^*(k,\eta,\eta')$ is the complex conjugate of $G_\varphi(k,\eta,\eta')$, so $\prod_i G_{k_i} - (\prod_i G_{k_i})^* = 2i\,\mathrm{Im}(\prod_i G_{k_i})$. Transforming to $\delta\phi$ and using $a(\eta) = -1/(H_*\eta)$ gives

$$\delta\phi_k(\eta) = \frac{H_*}{\sqrt{2k^3}}(1+ik\eta)e^{-ik\eta}.$$

The contribution to the bispectrum of $\delta\phi$ from self-interaction is therefore

$$B_{\delta\phi} = \frac{H_*^2 V_*'''}{4\prod_i k_i^3} I_3,$$

where

$$I_3 \equiv \mathrm{Re}\left(-ie^{-k_t\eta}\right)\left[\prod_j(1+ik_j\eta)\right]\int_{-\infty}^{\eta}\frac{d\eta'}{\eta'^4}e^{ik_t\eta'}\prod_i\left(1-ik_i\eta'\right),$$

where $k_t \equiv k_1+k_2+k_3$.

2.1.3 Primordial curvature perturbation in the slow-roll model

Primordial curvature perturbation

We now discuss the quantity $\mathcal{R}$, called the primordial curvature perturbation. On super-horizon scales, $\mathcal{R}$ is conserved under adiabatic evolution and later determines the total energy-density perturbation. Below, we discuss how it can be generated within the inflationary model. Here, we define it in a gauge with comoving threading. Neglecting tensor perturbations, the spatial metric on each slice is

$$g_{ij} = a^2(\boldsymbol{x},t)\delta_{ij}, \qquad a(\boldsymbol{x},t) \equiv a(t)e^{\psi(\boldsymbol{x},t)}.$$

The perturbation to the local number of e-folds between two generic slices is

$$\delta N_{12}(\boldsymbol{x}) = \delta\int_{t_1}^{t_2}\frac{1}{a}\frac{da}{dt}dt = \psi(\boldsymbol{x},t_2)-\psi(\boldsymbol{x},t_1). \tag{2.13}$$

When $\psi = 0$, the slicing is called flat slicing. If we define $N(\boldsymbol{x},t)$ as the local number of e-folds from a fixed flat slice to a uniform-density slice, then

$$\mathcal{R}(\boldsymbol{x},t) = \delta N(\boldsymbol{x},t).$$

This δN formula is useful for understanding how the curvature perturbation can originate in the early Universe.

Linear density perturbations

Starting from a uniform-density slice, we transform to a generic slicing with $\tilde{t}(\boldsymbol{x},t) = t+\delta t(\boldsymbol{x},t)$. To first order, $\tilde{t}(\boldsymbol{x},t-\delta t) = t$; that is, the time shift is $-\delta t$ at time $\tilde{t}$. Using Eq. (2.13), the first-order curvature perturbation on the generic slicing is given by

$$\psi = \mathcal{R} - H\delta t.$$

In the old and new gauges, the same physical density can be written as

$$\rho(t) + \delta\rho(\boldsymbol{x}, t) = \rho(\tilde{t}) + \widetilde{\delta\rho}(\boldsymbol{x}, \tilde{t}).$$

Hence, the gauge transformation for the perturbation is $\widetilde{\delta\rho}(\boldsymbol{x}, \tilde{t}) - \delta\rho(\boldsymbol{x}, t) = \rho(\tilde{t}) - \rho(t) = -\dot{\rho}(t)\delta t(\boldsymbol{x}, t)$. Dropping the tilde on the time coordinate of the generic slicing gives

$$\delta\rho(\boldsymbol{x}, t) = -\dot{\rho}(t)\delta t(\boldsymbol{x}, t).$$

This gives the gauge-invariant definition of $\mathcal{R}$:

$$\mathcal{R} = -H\frac{\delta\rho}{\dot{\rho}} = \frac{1}{3}\frac{\delta\rho}{\rho + p}. \tag{2.14}$$

Here, we impose the flat slicing with $\psi = 0$. Suppose that the cosmic fluid has separate components that do not exchange energy. Then

$$\mathcal{R}(t) = \frac{\sum(\rho_a + p_a)\mathcal{R}_a}{\rho + p}, \qquad \mathcal{R}_a \equiv \frac{1}{3}\frac{\delta\rho_a}{\rho_a + p_a}.$$

The continuity equation $a d\rho/da = -3(\rho + p)$, together with the adiabatic conditions $\rho_{\rm m} \propto a^{-3}$ and $\rho_{\rm r} \propto a^{-4}$, implies

$$\frac{1}{3}\delta_{\rm b} = \frac{1}{3}\delta_{\rm c} = \frac{1}{4}\delta_\gamma = \frac{1}{4}\delta_\nu,$$

where $\delta_a = \delta\rho_a/\rho_a$ are the density contrasts.

Within the slow-roll model

We assume that the curvature perturbation reaches its final value a few Hubble times after horizon exit. This corresponds to the scenario in which $\mathcal{R}$ remains constant throughout the era when the smoothing scale is outside the horizon, so that the locally defined pressure is a unique function of the locally defined energy density. Considering Eq. (2.14) with ϕ instead of ρ and using a flat slicing with $\psi = 0$, the curvature perturbation is

$$\mathcal{R} = -H\frac{\delta\phi}{\dot{\phi}}.$$

A few Hubble times after horizon exit, $\delta\phi(\boldsymbol{k})$ has the power spectrum $H^2/(2k^3)$. Using the slow-roll approximation $\dot{\phi} = -V'/3H$ and $3M_{\rm Pl}^2H^2 \simeq V$, we obtain

$$\mathcal{R}(\boldsymbol{x}) = \frac{1}{M_{\rm Pl}^2}\frac{V}{V'}\delta\phi(\boldsymbol{x}), \qquad P_{\mathcal{R}}(k) = \left(\frac{H}{\dot{\phi}}\right)^2 P_{\delta\phi}(k) = \frac{H^2}{4M_{\rm Pl}^2k^3\epsilon}\bigg|_{k=aH}. \tag{2.15}$$

We can see that this relation is linear in the field fluctuation and the coefficient $V/(M_{\rm Pl}^2V')$ is not a random field. Therefore the power spectrum simply rescales and the Gaussinity preserved, the higher-order extension like $\sim (\delta\phi)^2$ gives the non-Gaussinity.

Under the slow-roll approximation, we can work out the rates of change of H and ϵ in terms of the potential and its derivatives. We can define a similar quantity that gives the fractional change in the Hubble rate:

$$\epsilon_{\rm sr} \equiv \frac{d}{dt}\left(\frac{1}{H}\right) \backsimeq \frac{-\dot{\phi}^2/(2M_{\rm Pl}^2)}{V/(3M_{\rm Pl}^2)} \backsimeq \epsilon(\phi).$$

If we define $dN \equiv -Hdt$, we find

$$-\frac{d(\ln H)}{dN} \backsimeq -\epsilon, \qquad -\frac{d(\ln\epsilon)}{dN} \backsimeq 4\epsilon - 2\eta, \qquad -\frac{d\eta}{dN} \backsimeq 2\epsilon\eta - \xi,$$

where

$$\xi \equiv M_{\rm Pl}^4\frac{V'(d^3V/d\phi^3)}{V^2}.$$

Using $d\ln(aH) \backsimeq H dt$, this gives the spectral tilt

$$n(k) - 1 \equiv \frac{d\ln P_{\mathcal{R}}(k)}{d\ln k} = -6\epsilon + 2\eta,$$

with all quantities again evaluated at $k = aH$. Differentiating again gives the running,

$$n' \equiv \frac{dn}{d\ln k} = 16\epsilon\eta - 24\epsilon^2 - 2\xi.$$

2.2 Evolution to large-scale structure

Given these seeds as initial conditions for curvature perturbations, we can calculate the nonlinear evolution of the resulting matter fields. This section introduces standard perturbation theory (SPT) in the Eulerian framework, restricting the discussion to non-relativistic matter. It also provides the minimum background needed to discuss the information carried by the power spectrum and bispectrum, which are used in the numerical implementation in later chapters. We then briefly review Lagrangian dynamics, which provide insight into the structures observed today, although they are not central to the main results of this thesis. Metric perturbations and gravitational instability, combined with inflationary initial conditions, generate non-Gaussianity during structure formation.

For matter density perturbations, we define the density contrast as

$$\delta_{\rm m}(\boldsymbol{x}) = \frac{\rho_{\rm m}(\boldsymbol{x})}{\bar{\rho}_{\rm m}} - 1.$$

Here, the subscript m denotes matter, including both baryons and cold dark matter (CDM), and $\bar{\rho}_{\rm m}$ is the mean matter density. The physical particle velocity is $\boldsymbol{v} = d\boldsymbol{x}/dt$. The four-velocity is defined as $u^\mu = dx^\mu/d\tau$, with $x^\mu = (t, \boldsymbol{x})$, $d\tau = dt\sqrt{1-v^2} = dt/\gamma$, and $v = |\boldsymbol{v}|$, so $u^\mu = \gamma(1, \boldsymbol{v})$. The four-momentum is $p^\mu = mu^\mu = (E, \boldsymbol{p})$, with $E = \sqrt{p^2 + m^2}$, $p = |\boldsymbol{p}|$, and $\boldsymbol{p} = \gamma m \boldsymbol{v}$. For CDM or galaxies, we usually write the peculiar velocity as $\boldsymbol{u} \equiv a\boldsymbol{v}_{\rm pec}$. In the fluid description, we define

$$n_{\rm m} = \frac{\rho_{\rm m}}{m} = \int \frac{d^3p}{(2\pi)^3} f_{\rm m}, \qquad u_{\rm c}^i \equiv \frac{1}{n_{\rm m}} \int \frac{d^3p}{(2\pi)^3} f_{\rm m} \frac{p\,\hat{p}^i}{E(p)}.$$

Here, $n_{\rm m}$ is the matter number density and $f_{\rm m}$ is the phase-space distribution function. The fluid, or bulk, velocity can be much smaller than the individual particle velocities.

See Refs. [6, 33] for detailed background.

2.2.1 Eulerian dynamics

Eulerian linear theory

In linear perturbation theory, the fluid equations lead to the standard growing-mode solution. In the large-k, radiation-free limit, Poisson's equation gives

$$k^2\Phi(\boldsymbol{k}, a) = 4\pi G\rho_{\rm m}(a)a^2\delta_{\rm m}(\boldsymbol{k}, a).$$

Together with the background matter density $\rho_{\rm m} = \Omega_{\rm m}\rho_{\rm cr}/a^3$, where $\Omega_{\rm m} = \rho_{{\rm m},0}/\rho_{\rm cr}$ and $\rho_{\rm cr} = 3H_0^2/(8\pi G)$, the matter density perturbation generated by the primordial curvature perturbation is

$$\delta_{\rm m}(\boldsymbol{k}, a) = \frac{2}{5}\frac{k^2}{\Omega_{\rm m}H_0^2}\mathcal{R}(\boldsymbol{k})T(k)D_+(a) \qquad (a > a_{\rm late}, k \gg aH),$$

where $T(k)$ is the transfer function describing the evolution of perturbations through horizon crossing and the radiation–matter transition, and $D_+(a)$ is the growth factor describing wavelength-independent growth at late times. We can rewrite Eq. (2.15) as

$$P_{\mathcal{R}}(k) \equiv 2\pi^2\mathcal{A}_s k^{-3}\left(\frac{k}{k_{\rm p}}\right)^{n_s-1},$$

where $\mathcal{A}_s$ is the variance of curvature perturbations in a logarithmic wavenumber interval centered on the pivot scale $k_\mathrm{p} = 0.05\,\mathrm{Mpc}^{-1}$, and n_s is the spectral index. The linear matter power spectrum at late times is therefore

$$P_\mathrm{L}(k,a) = \frac{8\pi^2}{25}\frac{\mathcal{A}_s}{\Omega_\mathrm{m}^2}T^2(k)D_+^2(a)\frac{k^{n_s}}{H_0^4 k_\mathrm{p}^{n_s-1}}.$$

Eulerian nonlinear theory

The starting point of perturbative approaches to nonlinear growth is to take moments of the 6 + 1-dimensional Vlasov equation,

$$\frac{df_\mathrm{m}}{dt} = \frac{\partial f_\mathrm{m}}{\partial t} + \frac{\partial f_\mathrm{m}}{\partial x^j}\frac{p^j}{ma} - \frac{\partial f_\mathrm{m}}{\partial x^j}\left[Hp^j + \frac{m}{a}\frac{\partial \Phi}{\partial x^j}\right] = 0.$$

We omit the detailed calculation here and keep only the final 3 + 1 dimensional system. In conformal time, the continuity, Euler, and Poisson equations are

$$\begin{aligned}
{\delta_\mathrm{m}}' + \frac{\partial}{\partial x^j}\left[(1+\delta_\mathrm{m})u_\mathrm{m}^j\right] &= 0,\\
{u_\mathrm{m}^i}' + u_\mathrm{m}^j\frac{\partial}{\partial x^j}u_\mathrm{m}^i + aHu_\mathrm{m}^i + \frac{\partial \Phi}{\partial x^i} + \frac{1}{\rho_\mathrm{m}}\frac{\partial}{\partial x^j}\sigma_\mathrm{m}^{ij} &= 0,\\
\nabla^2\Phi &= \frac{3}{2}\Omega_\mathrm{m}(\eta)(aH)^2\delta_\mathrm{m}.
\end{aligned}$$

Here, σ_m^{ij} is the stress tensor defined by $\langle p^i p^j\rangle_{f_\mathrm{m}}/m = \rho_\mathrm{m}u_\mathrm{m}^i u_\mathrm{m}^j + \sigma_\mathrm{m}^{ij}$, which will be neglected, and $\Omega_\mathrm{m}(\eta)$ is the time-dependent density parameter. Introducing the velocity divergence $\theta_\mathrm{m} \equiv \partial_i u_\mathrm{m}^i$, these equations can be rewritten as

$$\begin{aligned}
{\delta_\mathrm{m}}' + \theta_\mathrm{m} &= -\delta_\mathrm{m}\theta_\mathrm{m} - u_\mathrm{m}^j\frac{\partial}{\partial x^j}\delta_\mathrm{m},\\
{\theta_\mathrm{m}}' + aH\theta_\mathrm{m} + \frac{3}{2}\Omega_\mathrm{m}(\eta)(aH)^2\delta_\mathrm{m} &= -u_\mathrm{m}^j\frac{\partial}{\partial x^j}\theta_\mathrm{m} - (\partial_i u_\mathrm{m}^j)(\partial_j u_\mathrm{m}^i).
\end{aligned}$$

If we set the right-hand side to zero, these equations reduce to the linear equations. We can expand the fields as

$$\begin{aligned}
\delta_\mathrm{m}(\boldsymbol{x},\eta) &= \delta^{(1)}(\boldsymbol{x},\eta) + \delta^{(2)}(\boldsymbol{x},\eta) + \cdots + \delta^{(n)}(\boldsymbol{x},\eta),\\
\theta_\mathrm{m}(\boldsymbol{x},\eta) &= \theta^{(1)}(\boldsymbol{x},\eta) + \theta^{(2)}(\boldsymbol{x},\eta) + \cdots + \theta^{(n)}(\boldsymbol{x},\eta).
\end{aligned}$$

Let the reference linear field be $\delta_0(\boldsymbol{x}) = \delta^{(1)}(\boldsymbol{x},\eta_\mathrm{ref})/D_+(\eta_\mathrm{ref})$. Then $\delta^{(1)}(\boldsymbol{x},\eta) = D_+(\eta)\delta_0(\boldsymbol{x})$, and the linear continuity equation yields

$$\theta^{(1)}(\boldsymbol{x},\eta) = -\delta^{(1)\prime}(\boldsymbol{x},\eta) = -aHf(\eta)\delta^{(1)}(\boldsymbol{x},\eta),$$

where $f = d\ln D_+/d\ln a$ is the growth rate. The second-order density perturbation can be written as

$$\delta^{(2)}(\boldsymbol{k},\eta) = D_+^2(\eta)\int\frac{d^2k_1}{(2\pi)^3}\int\frac{d^2k_2}{(2\pi)^3}(2\pi)^3\delta_\mathrm{D}^{(3)}(\boldsymbol{k}-\boldsymbol{k}_1-\boldsymbol{k}_2)F_2(\boldsymbol{k}_1,\boldsymbol{k}_2)\delta_0(\boldsymbol{k}_1)\delta_0(\boldsymbol{k}_2),$$

where the kernel is

$$F_2(\boldsymbol{k}_1,\boldsymbol{k}_2) = \frac{5}{7} + \frac{1}{2}\frac{\boldsymbol{k}_1\cdot\boldsymbol{k}_2}{k_1k_2}\left(\frac{k_1}{k_2}+\frac{k_2}{k_1}\right) + \frac{2}{7}\left(\frac{\boldsymbol{k}_1\cdot\boldsymbol{k}_2}{k_1k_2}\right)^2.$$

This expression will be used in the numerical implementation.

In SPT, the power spectrum receives loop corrections. Since we use a halo-model prescription to correct the linear-order result later, we do not write these corrections explicitly here. The tree-level bispectrum is

$$
\begin{aligned}
&\langle \delta_{\rm m}(\boldsymbol{k}_1,\eta)\delta_{\rm m}(\boldsymbol{k}_2,\eta)\delta_{\rm m}(\boldsymbol{k}_3,\eta)\rangle = \langle \delta_1^{(2)}\delta_2^{(1)}\delta_3^{(1)}\rangle + \langle \delta_1^{(1)}\delta_2^{(2)}\delta_3^{(1)}\rangle \, \langle \delta_1^{(1)}\delta_2^{(1)}\delta_3^{(2)}\rangle \\
=&D_+^4\int_{\boldsymbol{q}_1,\boldsymbol{q}_2}(2\pi)^3\delta_{\rm D}^{(3)}(\boldsymbol{k}-\boldsymbol{k}_1-\boldsymbol{k}_2)F_2(\boldsymbol{k}_1,\boldsymbol{k}_2)\,\langle\delta_0(\boldsymbol{q}_1)\delta_0(\boldsymbol{q}_2)\delta_0(\boldsymbol{k}_1)\delta_0(\boldsymbol{k}_2)\rangle + \text{cyc.} \\
=&(2\pi)^3\delta_{\rm D}^3(\boldsymbol{k}_1,\boldsymbol{k}_2,\boldsymbol{k}_3)[2F(\boldsymbol{k}_1,\boldsymbol{k}_2)P_{\rm L}(k_1,\eta)P_{\rm L}(k_2,\eta)+\text{cyc.}].
\end{aligned}
$$

At third order and beyond, one usually applies recursion relations, which we do not write explicitly here. The corresponding vertices ν_n, defined by the spherical averages of the kernels,

$$\nu_n \equiv n!\int\frac{d\Omega_1}{4\pi}\cdots\frac{d\Omega_n}{4\pi}\,F_n(\boldsymbol{k}_1,\ldots\boldsymbol{k}_n),$$

are useful for deriving cumulant parameters and other perturbative results. For instance,

$$\nu_1=1,\qquad \nu_2=\frac{34}{21},\qquad \nu_3=\frac{682}{189}. \tag{2.16}$$

Halo model

Beyond SPT, in the numerical implementation, we directly use the nonlinear power spectrum modeled with Halofit fitting formulae [34, 35]. It can be written as the sum of a two-halo term and a one-halo term:

$$
\begin{aligned}
P_{\rm 2H}(k,z) &= P_{\rm L}(k,z)\left[\int_0^\infty dM\, b(M,z)W(M,k,z)n(M,z)\right]^2, \\
P_{\rm 1H}(k,z) &= \int_0^\infty dM\, W^2(M,k,z)n(M,z).
\end{aligned} \tag{2.17}
$$

Here, M is the halo mass, $b(M,z)$ is the linear halo bias, and $n(M,z)$ is the halo mass function that is number density of halos of M. The function $W(M,k,z)$ is the spherical Fourier transform of the halo matter density profile:

$$W(M,k,z)=\frac{1}{\bar\rho_{\rm m}}\int_0^{r_{\rm v}}dr\,4\pi r^2\frac{\sin(kr)}{kr}\rho_{\rm m}(M,r,z).$$

Here, $\rho_{\rm m}(M,r,z)$ is the radial matter density profile in a host halo of mass M, and $\bar\rho_{\rm m}$ is the mean matter density. The halo mass is related to the virial radius $r_{\rm v}$ by

$$M=\frac{4}{3}\pi r_{\rm v}^3\Delta_{\rm v}(z)\bar\rho_{\rm m},$$

where $\Delta_{\rm v}(z)$ is the virial halo overdensity.

This brief review clarifies the relationship between the linear power spectrum and the nonlinear power spectrum used later.

2.2.2 Lagrangian dynamics

Unlike the Eulerian framework, the Lagrangian framework follows the trajectories of particles or fluid elements. In the Lagrange–Newton system, the initial particle positions $\boldsymbol{q}$ are mapped to the final Eulerian particle positions $\boldsymbol{x}$ through

$$\boldsymbol{x}(\eta)=\boldsymbol{q}+\boldsymbol{\Psi}(\boldsymbol{q},\eta).$$

Using $1+\delta_{\rm m}(\boldsymbol{x})=\int d^3q\,\delta_{\rm D}(\boldsymbol{x}-\boldsymbol{q}-\boldsymbol{\Psi})$, we obtain, in Fourier space,

$$\delta_{\rm m}(\boldsymbol{k})=\int d^3q\,e^{-i\boldsymbol{k}\cdot\boldsymbol{q}}\Big(e^{-i\boldsymbol{k}\cdot\boldsymbol{\Psi}}-1\Big).$$

Because of mass conservation, the density field obeys $\bar{\rho}_{\rm m}(1+\delta_{\rm m}(\boldsymbol{x}))d^3x = \bar{\rho}_{\rm m}d^3q$. Thus the Jacobian of the transformation between Eulerian and Lagrangian space is

$$1+\delta_{\rm m} = \frac{1}{J}, \qquad J = \det\left(\frac{\partial x_i}{\partial q_j}\right) = \det\left(\delta_{ij} + \Psi_{i,j}\right),$$

where $\Psi_{i,j} \equiv \partial \Psi_i/\partial q_j$. The equation of motion for particle trajectories is then

$$\frac{d^2\boldsymbol{x}}{d\eta^2} + aH(\eta)\frac{d\boldsymbol{x}}{d\eta} = -\nabla\Phi.$$

The total, or convective, time derivative along $\boldsymbol{u}$ is $d/d\eta \equiv \partial/\partial\eta + \boldsymbol{u}\cdot\nabla$. Taking the divergence of this equation, we obtain a closed equation for the displacement field:

$$J(\boldsymbol{q},\eta)\nabla\cdot\left[\frac{d^2\boldsymbol{\Psi}}{d\eta^2} + aH(\eta)\frac{d\boldsymbol{\Psi}}{d\eta}\right] = \frac{3}{2}\Omega_{\rm m}(\eta)(aH)^2(J-1). \tag{2.19}$$

Using $\partial_{x_i} = (\delta_{ij}+\Psi_{i,j})^{-1}\partial_{q_j}$, we solve the equation in a perturbative expansion, similar to SPT. We first write the displacement field as

$$\boldsymbol{\Psi}(\boldsymbol{q},\eta) = \boldsymbol{\Psi}^{(1)}(\boldsymbol{q},\eta) + \boldsymbol{\Psi}^{(2)}(\boldsymbol{q},\eta) + \cdots.$$

Here, $\Psi^{(1)}(\boldsymbol{q},\eta) = D_+(\eta)\Psi^{(1)}(\boldsymbol{q})$ and $\Psi^{(2)}(\boldsymbol{q},\eta) = D_2(\eta)\Psi^{(2)}(\boldsymbol{q})$, where $D_2(\eta)$ denotes the second-order growth factor. For any matrix A,

$$\det(I+A) = 1 + \operatorname{Tr}A + \frac{1}{2}\big[(\operatorname{Tr}A)^2 - \operatorname{Tr}\big(A^2\big)\big] + \frac{1}{6}\big[(\operatorname{Tr}A)^3 - 3\operatorname{Tr}A\operatorname{Tr}\big(A^2\big) + 2\operatorname{Tr}\big(A^3\big)\big],$$

and therefore

$$\begin{aligned} J(\boldsymbol{q}) = 1 + \nabla_{\boldsymbol{q}}\cdot\boldsymbol{\Psi} &+ \frac{1}{2}\Big[(\nabla_{\boldsymbol{q}}\cdot\boldsymbol{\Psi})^2 - \sum_{i,j}\Psi_{i,j}\Psi_{j,i}\Big] \\ &+ \frac{1}{6}\left[(\nabla_{\boldsymbol{q}}\cdot\boldsymbol{\Psi})^3 - 3\nabla_{\boldsymbol{q}}\cdot\boldsymbol{\Psi}\sum_{i,j}\Psi_{i,j}\Psi_{j,i} + 2\sum_{i,j,k}\Psi_{i,j}\Psi_{j,k}\Psi_{k,i}\right] + O(\Psi^4). \end{aligned}$$

Substituting this expansion gives Eq. (2.19) order by order.

Linear solutions and the Zel'dovich approximation

Using $1+\delta_{\rm m} \simeq 1 - \nabla_{\boldsymbol{q}}\cdot\boldsymbol{\Psi}$, the linear solution of Eq. (2.19) is

$$\nabla_{\boldsymbol{q}}\cdot\boldsymbol{\Psi}^{(1)} = -D_+(\eta)\delta_0(\boldsymbol{q}), \qquad \boldsymbol{\Psi}^{(1)}(\boldsymbol{q},\eta) = -D_+(\eta)\nabla_{\boldsymbol{q}}\Phi^{(1)}(\boldsymbol{q}).$$

Thus, the tidal tensor $\Psi_{i,j}$ can be written as the Hessian of the gravitational potential. This implies that the velocities of particles initially at $\boldsymbol{q}$ are given by

$$\boldsymbol{u} = a\frac{d\boldsymbol{\Psi}}{dt} \approx -aD_+(\eta)H(\eta)f\,\nabla_{\boldsymbol{q}}\Phi^{(1)}(\boldsymbol{q}),$$

where we use $\dot{D} = HDf$.

The Zel'dovich approximation uses the linear displacement field as an approximate solution to the dynamical equations. The local density field then reads

$$1+\delta_{\rm m}(\boldsymbol{x},\eta) = \frac{1}{[1-\lambda_1 D_+(\eta)][1-\lambda_2 D_+(\eta)][1-\lambda_3 D_+(\eta)]}.$$

Here, λ_i are the local eigenvalues of $\Psi_{i,j}$. One positive eigenvalue corresponds to planar collapse, two positive eigenvalues correspond to filamentary collapse, and three positive eigenvalues correspond to spherical collapse. If all eigenvalues are negative, the evolution describes an underdense region that can asymptotically approach $\delta_{\rm m} = -1$. This physical interpretation is useful for explaining critical points in the cosmic web.

Second-order Lagrangian PT

In practice, second-order Lagrangian perturbation theory is used to generate more accurate initial particle positions and velocities for cosmological simulations. The second-order solution describes the correction to the linear-order displacement due to gravitational tidal effects and reads

$$\nabla_q \cdot \boldsymbol{\Psi}^{(2)} = \frac{1}{2} D_2(\eta) \sum_{i \neq j} \left(\Psi^{(1)}_{i,i} \Psi^{(1)}_{j,j} - \Psi^{(1)}_{i,j} \Psi^{(1)}_{j,i} \right).$$

This expression is curl-free and can be written as $\Psi^{(2)} = -D_2(\eta) \nabla^2_{\boldsymbol{q}} \Phi^{(2)}$. Thus, in 2LPT,

$$\begin{aligned} \boldsymbol{x} &= \boldsymbol{q} - D_+ \nabla_{\boldsymbol{q}} \Phi^{(1)} - D_2 \nabla_{\boldsymbol{q}} \Phi^{(2)}, \\ \boldsymbol{u} &= -a D_+ H f \nabla_{\boldsymbol{q}} \Phi^{(1)} - a D_2 H f_2 \nabla_{\boldsymbol{q}} \Phi^{(2)}. \end{aligned}$$

Here, $f_2 \equiv d \ln D_2 / d \ln a$.

Lagrangian perturbation theory can naturally incorporate Lagrangian bias and redshift-space distortions through resummation techniques [36]. Unlike the effective field theory of large-scale structure (EFTofLSS), it does not require nuisance parameters to fit the data, while still giving good predictions.

2.2.3 Statistics: skewness and kurtosis

This subsection emphasizes geometric one-point statistics and field-strength analysis, and discusses the relationship between N-point functions and cumulants. We are interested in geometric statistics of the LSS because they contain information complementary to that in N-point functions and are more directly related to field morphology. In particular, they can capture non-Gaussian features that may be missed by traditional polyspectra, especially in the nonlinear regime of structure formation.

One-point normalized cumulant statistics for the density and velocity-divergence fields are

$$S_p(R) \equiv \frac{\langle \delta^p(R) \rangle_{\mathrm{c}}}{\langle \delta^2(R) \rangle^{p-1}}, \qquad T_p(R) \equiv \frac{\langle \theta^p(R) \rangle_{\mathrm{c}}}{\langle \theta^2(R) \rangle^{p-1}}.$$

where p is the cumulant order and R is the smoothing scale. When $p = 2$, the normalized cumulant reduces to the unit variance; $p = 3$ gives the skewness, and $p = 4$ gives the kurtosis. We also define the logarithmic derivatives of the variance:

$$\gamma_p = \frac{d^p \ln \sigma^2(R)}{d \ln^p R},$$

where $\sigma^2(R)$ denotes either $\langle \delta^2(R) \rangle$ or $\langle \theta^2(R) \rangle$. We do not discuss the velocity-divergence field in detail here.

Skewness

For $p = 3$, the reduced cumulant describes the skewness of the unsmoothed density field at leading order in σ:

$$\frac{\langle \delta^3 \rangle}{\langle \delta^2 \rangle^2} \simeq \frac{3 \langle (\delta^{(1)})^2 \delta^{(2)} \rangle}{\langle \delta^2 \rangle^2} = 6 \int_{-1}^{1} d\mu \, F_2(k_1, k_2, \mu) = 3\nu_2 = \frac{34}{7}.$$

In practice, the fields are defined with a smoothing window function $W(kR)$, so

$$\begin{aligned} \sigma^2(R) &= \int_{\boldsymbol{k}} P_{\mathrm{L}}(k) W^2(kR), \\ \langle \delta_R^3 \rangle &= 6 \int_{\boldsymbol{k}_1, \boldsymbol{k}_2} F_2 P_{\mathrm{L}1} P_{\mathrm{L}2} W(k_1 R) W(k_2 R) W(|\boldsymbol{k}_1 + \boldsymbol{k}_2| R). \end{aligned}$$

Using a top-hat smoothing filter,

$$W(kR) = \frac{3}{(kR)^3}[\sin(kR) - kR\cos(kR)],$$

one obtains

$$S_3 = \frac{34}{7} + \frac{d\ln\sigma^2(R)}{d\ln R} = \frac{34}{7} + \gamma_1.$$

Thus, we sometimes define

$$\bar{\nu}_2 = \nu_2 + \frac{\gamma_1}{3},$$

for the modified vertex. For a power-law spectrum, $P_{\rm L}(k) \propto k^n$, we can directly calculate

$$\sigma^2(R) \propto \int_0^\infty dk\, k^{n+2} W^2(kR) = R^{-(n+3)} \int_0^\infty dx\, W^2(x) \propto R^{-(n+3)}.$$

Therefore, $\gamma_1 = -(n+3)$ in this case. Skewness measures the asymmetry of the local density distribution. It appears because underdense regions evolve less rapidly than overdense regions once nonlinearities become important. It is expected to be smaller for power spectra with more small-scale fluctuations. In the leading-order critical-point statistics, we will see that the contributions all come from generalized skewness parameters.

Kurtosis

For $p = 4$, the corresponding reduced cumulant describes the kurtosis, which measures the connected four-point contribution to the one-point distribution. Although it is not part of the main structure of this thesis, it is useful to review because it illustrates the type of statistical quantity that appears in cosmology. The connected fourth-order cumulant of the local density is given by

$$\langle\delta^4\rangle_{\rm c} = \langle\delta^4\rangle - 3\langle\delta^2\rangle^2 = 6\langle(\delta^{(1)})^2(\delta^{(2)})^2\rangle_{\rm c} + 4\langle(\delta^{(1)})^3\delta^{(3)}\rangle_{\rm c}.$$

The terms of order σ^4 are disconnected parts, so the connected contribution starts at order σ^6. Applying the top-hat window function gives

$$S_4 = \frac{\langle\delta^4\rangle}{\sigma^6} = 4\nu_3 + 12\nu_2^2 + (14\nu_2 - 2)\gamma_1 + \frac{7}{3}\gamma_1^2 + \frac{2}{3}\gamma_2.$$

Substituting Eq. (2.16) gives

$$S_4 = \frac{60712}{1323} + \frac{62\gamma_1}{3} + \frac{7\gamma_1^2}{3} + \frac{2\gamma_2}{3}.$$

For a power-law spectrum, $\gamma_i = 0$ for $i \geqslant 2$.

Similar results can also be derived for the velocity-divergence field.

2.3 Probes of structure

After reviewing inflationary initial conditions and perturbation theory for large-scale structure, this section discusses two cosmological fields to which geometric statistics will be applied in later chapters: weak gravitational lensing fields and the cosmic web, both treated in terms of the matter density field. We explain why peak statistics are useful for matter fields and then extend the discussion to other critical points, such as filaments, walls, and voids, which are central topics of this thesis. Peaks correspond to high-density regions where galaxies are more likely to form. They can be used to identify and study large-scale structures, such as galaxy clusters and filaments, and they provide insight into the underlying matter distribution and the processes that drive galaxy formation and evolution. Together with other critical points, they offer an additional probe of the Universe beyond N-point correlation-function statistics.

2.3.1 Weak gravitational lensing

Lensing is the gravitational effect by which inhomogeneities in the Universe distort light paths from distant sources. Examples include time-delay cosmography for measuring H_0, microlensing for constraining the contribution of massive compact halo objects to dark matter, substructure lensing, and cluster lensing, such as in the Bullet Cluster. Here, we consider weak lensing, which traces the evolution of the initial density field through tidal forces and gravitational instability and is now measured precisely in both CMB and LSS surveys. This probe is sensitive to the full mass distribution, not only to galaxies or baryonic matter.

Neglecting scattering and absorption in the late Universe, there is no collision term. The Boltzmann equation then states that the photon distribution function is conserved: $df(\boldsymbol{x}, \boldsymbol{p})/dt = 0$. We introduce the specific intensity I_ν, defined as the energy incident on a detector per solid angle, per unit area, per unit time, and per unit frequency. Integrating the intensity gives the total energy on the detector,

$$dE = I_\nu d\Omega\, dA_\perp\, dt\, d\nu.$$

All information is contained in $f(\boldsymbol{x}, \boldsymbol{p}, t)$, and one can show that $I_\nu = 4\pi h \nu^3 f(\boldsymbol{x}, p = 2\pi\nu, \hat{\boldsymbol{p}}, t)$. We start from the fundamental property of weak lensing: it conserves surface brightness, or intensity, $I_{\rm obs}(\boldsymbol{\theta}) = I_{\rm true}(\boldsymbol{\theta}_{\rm s})$, where θ is the observed angular position and θ_s is the true angular position of the source (the sky position in the absence of lensing). We assume a small deflection angle. Then the true and observed source positions are located at

$$\boldsymbol{x}_{\rm obs} = (\boldsymbol{\theta}_{\rm s}, 1)\chi, \qquad \boldsymbol{x} = (\boldsymbol{\theta}, 1)\chi.$$

Here, the radial distance χ is the comoving distance to the source, and the transverse distance is $\chi_{\rm s} = \chi\boldsymbol{\theta}_{\rm s}$ for the true source. The convergence $\kappa(\boldsymbol{\theta})$ is

$$\kappa(\boldsymbol{\theta}) = \int d\chi\, q(\chi)\delta_{\rm m}(\chi\boldsymbol{\theta}, \chi), \qquad q(\chi) \equiv \frac{3H_0^2\Omega_{\rm m,0}}{2a(\chi)c^2}\frac{\chi_{\rm s} - \chi}{\chi\chi_{\rm s}}.$$

The Fourier transform of κ is

$$\begin{aligned}\tilde{\kappa}(\boldsymbol{\ell}) = \int d^2\theta\, e^{-i\boldsymbol{\theta}\cdot\boldsymbol{\ell}}\kappa(\boldsymbol{\theta}) &= \int d\chi\, q(\chi) \int d^2\theta\, e^{-i\boldsymbol{\theta}\cdot\boldsymbol{\ell}}\delta_{\rm m}(\chi\boldsymbol{\theta}, \chi) \\ &= \int d\chi \frac{q(\chi)}{\chi^2} \int d^2x_\perp\, e^{-i\boldsymbol{x}_\perp\cdot(\boldsymbol{\ell}/\chi)}\delta(\boldsymbol{x}_\perp, \chi) = \int d\chi \frac{q(\chi)}{\chi^2}\tilde{\delta}_{\rm m}\left(\frac{\boldsymbol{\ell}}{\chi}, \chi\right).\end{aligned}$$

In the second line, we used $\tilde{\delta}_{\rm m}(\boldsymbol{k}_\perp, \chi) = \int d^2x_\perp\, e^{-i\boldsymbol{k}_\perp\cdot\boldsymbol{x}_\perp}\delta(\boldsymbol{x}_\perp, \chi)$, where $\boldsymbol{k}_\perp$ is the transverse mode and $\boldsymbol{k}_\parallel = 0$. We define the convergence-field power spectrum by

$$\langle\tilde{\kappa}(\boldsymbol{\ell})\tilde{\kappa}(\boldsymbol{\ell}')\rangle \equiv (2\pi)^2\delta_{\rm D}^{(2)}(\boldsymbol{\ell} + \boldsymbol{\ell}')P_\kappa(\ell).$$

Then,

$$\langle\tilde{\kappa}(\boldsymbol{\ell})\tilde{\kappa}(\boldsymbol{\ell}')\rangle = \int d\chi d\chi' \frac{q(\chi)q(\chi')}{\chi^2\chi'^2}\left\langle\tilde{\delta}_{\rm m}\left(\frac{\boldsymbol{\ell}}{\chi}, \chi\right)\tilde{\delta}_{\rm m}\left(\frac{\boldsymbol{\ell}'}{\chi'}, \chi'\right)\right\rangle.$$

In the Limber approximation, the correlation length of the density field is much smaller than the distance over which the lensing kernel varies significantly. Then,

$$\langle\tilde{\delta}_{\rm m}(\boldsymbol{k}_\perp, \chi)\tilde{\delta}_{\rm m}(\boldsymbol{k}'_\perp, \chi')\rangle = (2\pi)^2\delta_{\rm D}^{(2)}(\boldsymbol{k}_\perp + \boldsymbol{k}'_\perp)\delta_{\rm D}(\chi - \chi')P_\delta(k = |\boldsymbol{k}_\perp|, \chi).$$

Using $\delta_D^{(2)}(\boldsymbol{k}_\perp + \boldsymbol{k}'_\perp) = \chi^2\delta_D^{(2)}(\boldsymbol{\ell} + \boldsymbol{\ell}')$, we obtain

$$P_\kappa(\ell) = \int d\chi \frac{q^2(\chi)}{\chi^2}P_\delta\left(k = \frac{\ell}{\chi}, \chi\right).$$

If we add a smoothing window function to the convergence field,

$$\kappa_\vartheta(\boldsymbol{\theta}) = \int d^2\theta' W\left(\frac{|\boldsymbol{\theta}' - \boldsymbol{\theta}|}{\vartheta}\right)\kappa(\boldsymbol{\theta}'),$$

the Fourier transform is $\tilde{\kappa}_\vartheta(\ell) = W(\ell\vartheta)\tilde{\kappa}(\ell)$, and its power spectrum is $P_{\kappa,\vartheta}(\ell) = W^2(\ell\vartheta)P_\kappa(\ell)$. For the bispectrum, we define the convergence-field bispectrum as

$$\langle\tilde{\kappa}(\boldsymbol{\ell}_1)\tilde{\kappa}(\boldsymbol{\ell}_2)\tilde{\kappa}(\boldsymbol{\ell}_3)\rangle \equiv (2\pi)^2\delta_{\mathrm{D}}^{(2)}(\boldsymbol{\ell}_1 + \boldsymbol{\ell}_2 + \boldsymbol{\ell}_3)B_\kappa(\ell_1, \ell_2, \ell_3),$$

and the matter bispectrum as

$$\begin{aligned}&\langle\tilde{\delta}_{\mathrm{m}}(\boldsymbol{k}_{\perp 1}, \chi_1)\tilde{\delta}_{\mathrm{m}}(\boldsymbol{k}_{\perp 2}, \chi_2)\tilde{\delta}_{\mathrm{m}}(\boldsymbol{k}_{\perp 3}, \chi_3)\rangle\\ &\quad= (2\pi)^2\delta_{\mathrm{D}}^{(2)}(\boldsymbol{k}_{\perp 1} + \boldsymbol{k}_{\perp 2} + \boldsymbol{k}_{\perp 3})\delta_{\mathrm{D}}(\chi_1 - \chi_2)\delta_{\mathrm{D}}(\chi_1 - \chi_3)B_\delta(k_1, k_2, k_3, \chi_1).\end{aligned}$$

Thus, the convergence bispectrum derived from the matter bispectrum is

$$B_\kappa(\ell_1, \ell_2, \ell_3) = \int d\chi \frac{q^3(\chi)}{\chi^4} B_\delta\left(k_1 = \frac{\ell_1}{\chi}, k_2 = \frac{\ell_2}{\chi}, k_3 = \frac{\ell_3}{\chi}, \chi\right).$$

Here, we have introduced the basic statistics of the weak-lensing field. In later chapters, we will construct a toy model for a two-dimensional non-Gaussian field defined on angular coordinates to represent this application.

2.3.2 Cosmic web: peak, void, filament, wall

Peak statistics, pioneered by the BBKS framework [3], can be extended to all critical points of the cosmic density field, namely peaks, voids, filaments, and walls. These critical points are local building blocks of the cosmic web; see Ref. [13] for a short review and Fig. 2.1 for an illustration. In cosmic-web applications, the Zel'dovich approximation [14] classifies these structures by the eigenvalues of the deformation tensor, corresponding to three, two, one, and zero positive eigenvalues, respectively, as discussed in Sec. 2.2.2. Here, we briefly review these structures and their recent cosmological applications. In the next chapter, we will discuss the theory of their statistics in detail.

Peak

BBKS counted peaks statistically and derived the famous peak abundance $\bar{n}^{\mathrm{pk}}(\nu)$, where ν is the field value divided by its variance and is interpreted as a threshold. Instead of using standard peak theory directly for the evolved density field $\delta(\boldsymbol{x}, z)$ in Eulerian space, we can also describe peak positions in Lagrangian space, leading to the peak model of bias as a tracer. In Fourier space, the perturbative peak-bias expansion takes the form

$$\delta_{\mathrm{pk}}^L(\boldsymbol{q}) = \sum_{n=1}^{\infty}\frac{1}{n!}\int_{\boldsymbol{k}_1}\cdots\int_{\boldsymbol{k}_n} c_n^L(\boldsymbol{k}_1, \ldots, \boldsymbol{k}_n)\left[\delta^{(1)}(\boldsymbol{k}_1)\cdots\delta^{(1)}(\boldsymbol{k}_n) + \cdots\right]e^{i\boldsymbol{k}_{1\ldots n}\cdot\boldsymbol{q}}.$$

Here, the Lagrangian peak-bias functions $c_n^L(\boldsymbol{k}_1, \ldots, \boldsymbol{k}_n)$ are defined at the collapse time, and $\delta^{(1)}$ is the linear density contrast field. This theory can be embedded naturally into integrated perturbation theory (iPT) [36].

In peak theory, there is a cloud-in-cloud problem: a peak identified on a small smoothing scale may lie inside a larger collapsed region. This issue can be addressed by excursion-set theory. Excursion-set theory considers the density field smoothed on different scales. As the smoothing

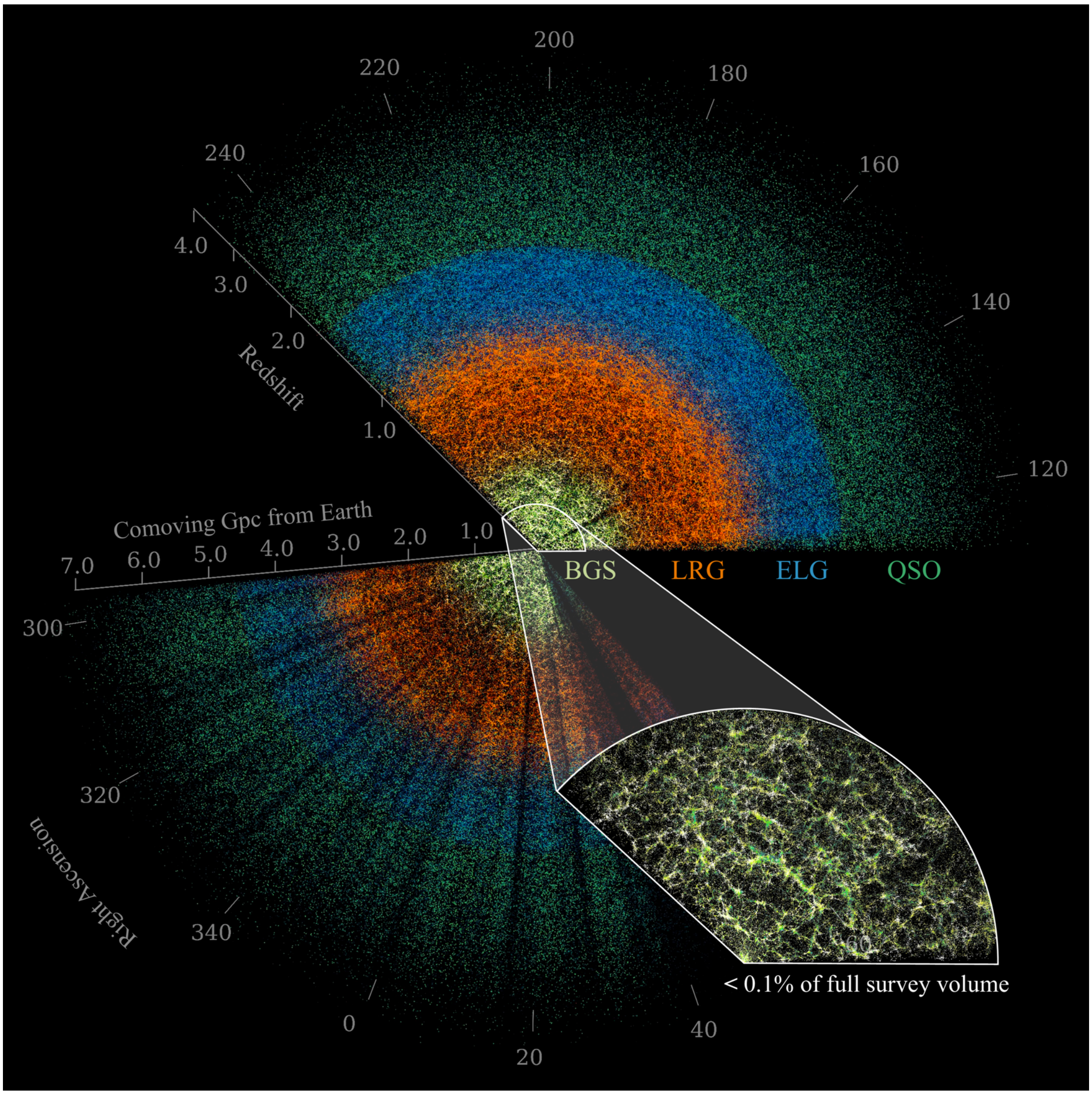

Figure 2.1: Cosmic web slice mapped by DR1 from Ref. [37]. The four major extragalactic samples are BGS galaxies, LRGs, ELGs, and QSOs, shown as yellow, orange, blue, and green points, respectively.

scale changes, the smoothed overdensity traces a random walk. The first crossing of the collapse threshold $\delta_c \simeq 1.686$ determines the scale on which a halo of mass M forms. Combining these two theories, the excursion-set peak (ESP) approach predicts the halo mass function as

$$\bar{n}_h(M) = \bar{\rho}_{\mathrm{m}} f_{\mathrm{ESP}}(\nu_c) \frac{d\nu_c}{dM},$$

where $\nu_c = \delta_c/\sigma_0(M)$, $\bar{\rho}_{\mathrm{m}}$ is the mean mass density, and $f_{\mathrm{ESP}}(\nu_c)$ is the ESP peak multiplicity function,

$$f_{\mathrm{ESP}}(\nu_c) = f_{\mathrm{pk}}(\nu_c) \frac{\langle x \rangle_{\mathrm{ESP}}}{\gamma\nu}, \qquad f_{\mathrm{pk}} = \frac{M}{\bar{\rho}_{\mathrm{m}}} \bar{n}^{\mathrm{pk}}(\nu_c).$$

Here, $\gamma = \sigma_1^2/(\sigma_0\sigma_2)$, and $M/\bar{\rho}_{\mathrm{m}}$ denotes the Lagrangian volume occupied by a halo of mass M. The threshold is $\nu = \delta_{\mathrm{m}}/\sigma_0$, and $x(R) = -\nabla^2\delta_R/\sigma_2$ depends on the smoothing scale R. The factor $\gamma\nu$ is a normalization because the covariance satisfies $\langle x\nu \rangle \approx \gamma\nu$, and $\bar{n}^{\mathrm{pk}}$ is the mean number density of peaks. The correction factor $\langle x \rangle_{\mathrm{ESP}}/(\gamma\nu)$ is one of the main results of Ref. [38]. In this thesis, we compute $\bar{n}^{\mathrm{pk}}$ at a given threshold in both Gaussian and non-Gaussian density fields.

Void

Voids are powerful probes because they are low-density, weakly nonlinear environments where cosmological effects, such as dark energy [18], massive neutrinos [19], and modified gravity [20], are amplified and easier to model. They are enormous underdense regions with typical sizes of about 20–50 h^{-1}Mpc and occupy a large fraction of the volume of the Universe. Previous studies have shown that void shapes, particularly their ellipticity, are sensitive to structure growth and therefore indirectly to the properties of dark energy. Different dark-energy models predict different growth histories and consequently different values of $\sigma_8(z)$, the rms amplitude of matter-density fluctuations smoothed on a scale of $8\,h^{-1}$Mpc. Because void ellipticity is strongly correlated with $\sigma_8(z)$, a degeneracy arises: changes in void shape may reflect differences in the clustering amplitude rather than a direct signature of dark energy itself. For neutrino applications, it has been found that cosmic voids in massive-neutrino cosmologies are less evolved than their counterparts in massless-neutrino cosmologies. As a result, massive-neutrino models contain a larger abundance of small voids and a smaller abundance of large voids. In addition, void properties provide a powerful way to break the degeneracy between the total neutrino mass Ω_ν (or $\sum m_\nu$) and the fluctuation amplitude σ_8. Voids are good environments for studying the so-called unscreened regime, $|f_R| \gtrsim |\Phi|$, where the modified fifth force is strong and deviations from standard gravity can be large. The force points toward the surrounding walls and filaments, producing emptier void centers and steeper density profiles. In Chap. 4, we explain why degeneracies involving standard cosmological parameters and σ_8 can be broken. Parameters such as dark energy, massive neutrinos, and $f(R)$ gravity are not considered in detail here.

Filaments and walls

Filaments [15–17] connect peaks and trace the preferred directions of anisotropic matter transport toward cluster-scale structures. The longest filaments can reach lengths of $60\,h^{-1}$Mpc. They contain 35–40 percent of the total galaxy luminosity and cover roughly 5–8 percent of the total volume. A filament can be approximated as an infinite cylinder with line mass λ (mass per unit length). Thus, the gravitational field at distance r from the filament axis is $g(r) = 2G\lambda/r$. If galaxies accelerate according to $d^2\boldsymbol{r}/dt^2 = \nabla\Phi$, then $d^2r/dt^2 = -2G\lambda/r$, showing why galaxies migrate toward filament centers and cluster nodes. The mass flux entering a galaxy for a filament of radius R_f is approximately $dM/dt \approx \rho v \pi R_f^2$. Thus, dark matter halos grow through filament accretion according to $dM_h/dt = \dot{M}_{\rm filament} + \dot{M}_{\rm merger}$, where $\dot{M}_{\rm merger}$ is the merger contribution from halos. Moreover, tidal torque theory explains how filaments influence galaxy spin: $L_i = a^2\dot{D}_+\epsilon_{ijk}T_{jl}I_{lk}$, where I is the inertia tensor, T is the tidal tensor, and ϵ_{ijk} is the Levi-Civita tensor.

Cosmic walls [21, 22] are sheet-like structures in the cosmic web that arise from anisotropic gravitational collapse along one direction. They often enclose voids and connect to filaments. They have received less attention than peaks, filaments, or voids, largely because they are harder to define physically. A wall may contain many clusters and filaments, and there is no universally accepted boundary. Walls are also too large to be single gravitationally bound structures. In the Zel'dovich picture, matter first collapses into walls, while filaments and peaks form through further collapse.

There are also many other statistical probes for investigating the evolutionary history of the Universe. Minkowski functionals and genus statistics [39] characterize geometrical information analytically. Betti numbers and persistent homology, often calibrated with simulations or emulators, measure how the connectivity of excursion sets changes with threshold [24–27]. Critical-point statistics are related to these probes, but they also provide complementary information from current cosmic observations.

Chapter 3

Non-Gaussian Critical-Point Statistics

This chapter develops baseline critical-point statistics for both Gaussian and non-Gaussian (NG) fields. We assume that the fields are statistically homogeneous and isotropic. We first introduce the geometry of random fields through probability theory [40, 41], the Kac–Rice formalism [42], and the Morse index [23], forming a self-contained theory for CPs. We then derive the one-point Gaussian critical-point statistics in 2D and 3D Gaussian fields, following the original BBKS result [3] but using the clearer formulation of Matsubara [9]. Next, we extend the leading-order non-Gaussian correction derived by Matsubara [9] for peak theory and compare the NG predictions with direct numerical counts in 2D and 3D fields. In Appendix A, we describe the FFT normalization and the critical-point finding algorithm [30]. In Appendix B, we present an attempt at the next-to-leading-order correction, although it is not successful.

3.1 Geometry of random fields

3.1.1 Field probability theory

Independence and randomness

Independence means an absence of correlation, a condition that simplifies many problems. Two events X and Y are independent if and only if $P(X \cap Y) = P(X)P(Y)$ or $P(Y|X) = P(Y)$. In addition, any functions f and g satisfy

$$\langle f(X)g(Y)\rangle = \langle f(X)\rangle \langle g(Y)\rangle .$$

The correlation function of X and Y, $C \equiv \langle XY\rangle - \langle X\rangle \langle Y\rangle = \langle (X - \langle X\rangle)(Y - \langle Y\rangle)\rangle$, is zero if X and Y are independent.

For a sequence of random variables $X_1, X_2, \ldots$, if all variables are mutually independent, then the sequence is random. Let $\langle f(X)\rangle \equiv \frac{1}{N}\sum_{i=k}^{N+k} f(X_i)$, and assume that this value is independent of N and k. If the sequence is much longer than N, let $X(r) = X_{i+r}$, with $i = 0, 1, 2, \ldots$. If

$$\frac{1}{N}\sum_{i=k}^{N+k} f(X_i)g(X_{i+r}) \equiv \langle f(X(0))g(X(r))\rangle = \langle f(X)\rangle \langle g(X)\rangle ,$$

then each $X(r)$ is independent and the sequence is random. In other words, the correlation function of $X(0)$ and $X(r)$

$$C(r) \equiv \langle X(0)X(r)\rangle - \langle X\rangle^2 ,$$

is zero for all $r \neq 0$.

For the two-point correlation function of a cosmic density field, we define $\xi(r) = \langle \delta(\boldsymbol{x})\delta(\boldsymbol{x} + \boldsymbol{r})\rangle$. Then $\xi(r)$ is nonzero only for $r < R$, where R is the correlation length. On length scales larger than R, $\delta(\boldsymbol{x})$ is random, i.e. the value of $\delta(\boldsymbol{x})$ at different points is independent.

Gaussian fields

A real-valued random variable X is said to be Gaussian (or normally distributed) if it has the density function

$$f(X) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(X-\mu)^2}{2\sigma^2}\right), \qquad X \in \mathbb{R},$$

where $\mu = \langle X \rangle$ and $\sigma^2 = \langle (X-\mu)^2 \rangle$; we write this as $X \sim \mathcal{N}(\mu, \sigma^2)$. An $\mathbb{R}^d$-valued random variable X is said to be multivariate Gaussian if, for every $\alpha = (\alpha_1, \cdots, \alpha_d) \in \mathbb{R}^d$, the real-valued variable $\alpha \cdot X = \sum_{i=1}^d \alpha_i X_i$ is Gaussian. Its probability density function is

$$f(X) = \frac{1}{(2\pi)^{d/2} \det(C)^{1/2}} \exp\left(-\frac{1}{2}(X-\mu)^T C^{-1} (X-\mu)\right),$$

where μ is the mean vector with $\mu_j = \langle X_j \rangle$ and C is a nonnegative definite $d \times d$ covariance matrix with elements $c_{ij} = \langle (X_i - \mu_i)(X_j - \mu_j) \rangle$. In a Gaussian field, all statistical information is determined by the mean function and the covariance function.

For a zero-mean Gaussian field, the moments are

$$\langle X^2 \rangle = \sigma^2, \qquad \langle X^{2n} \rangle = \frac{(2n)!}{2^n n!} \sigma^{2n}, \qquad \langle X^{2n+1} \rangle = 0, \tag{3.1}$$

with $n = 1, 2, \cdots$. The same results can be obtained from the characteristic function of $f(X)$:

$$\tilde{f}(k) = \langle e^{-ikX} \rangle = \int_{-\infty}^{\infty} dX\, e^{-ikX} f(X) = \exp\left(-\frac{1}{2}\sigma^2 k^2 - i\mu k\right). \tag{3.2}$$

Expanding this expression in powers of k gives Eq. (3.1). We can recover $f(X)$ by the inverse Fourier transform of $\tilde{f}(k)$:

$$f(X) = \frac{1}{2\pi} \int dk\, e^{ikX} \tilde{f}(k).$$

In cosmology, the initial density field is well approximated as a Gaussian random field, as described by the inflationary paradigm in the previous chapter. In density fields, the field values, gradients, and higher-order derivatives can be treated as Gaussian random variables following Gaussian probability density functions (PDFs).

The central limit theorem (CLT)

Independence is a strong condition that leads to the CLT. This theorem states that if N is large, then the distribution of the sum of N independent random variables approaches a Gaussian distribution, with the width proportional to $\sqrt{N}$. Here we give an intuitive proof of the CLT.

Let $X_1, X_2, \ldots, X_N$ be a sequence of independent random variables. Suppose $\langle X_i \rangle = 0$, and define $Y \equiv \frac{1}{\sqrt{N}}(X_1 + X_2 + \cdots + X_N)$. The characteristic function of Y is

$$\langle e^{-ikY} \rangle = \langle \prod_{j=1}^{N} e^{-ikX_j/\sqrt{N}} \rangle = \prod_{j=1}^{N} \langle e^{-ikX_j/\sqrt{N}} \rangle = \exp \sum_{j=1}^{N} A_j(k/\sqrt{N}),$$

where $A_j(k/\sqrt{N}) \equiv \ln \langle e^{-ikX_j/\sqrt{N}} \rangle$. For large N, we can expand $A_j(k/\sqrt{N})$ as

$$A_j(k/\sqrt{N}) \simeq A_j(0) + \frac{k^2}{2N} A_j''(0) + \frac{k^3}{N} \mathcal{O}(N^{-1/2}),$$

since $\langle X_j \rangle = 0$, and $A_j''(0) = -\langle X_j^2 \rangle$. Therefore,

$$\langle e^{-ikY} \rangle = \exp\left[-\frac{1}{2}k^2\sigma^2 + \mathcal{O}(\frac{1}{\sqrt{N}})\right],$$

where $\sigma^2 = \frac{1}{N}\sum_{j=1}^{N} \langle X_j^2 \rangle$. This is consistent with Eq. (3.2) and proves the CLT: Y is normally distributed with width σ. The CLT explains why Gaussian distributions arise so frequently in random-field applications.

Moments, cumulants and their generating function

The mean is $\mu \equiv \langle X \rangle$. The raw moments are $\langle X^n \rangle$, and the central moments are

$$\mu_n \equiv \langle (X-\mu)^n \rangle .$$

Thus, the variance is $\sigma_0^2 \equiv \sigma^2 = \mu_2$. The third and fourth central moments define the normalized skewness and excess kurtosis:

$$\gamma_1 = \frac{\mu_3}{\sigma_0^3}, \qquad \gamma_2 = \frac{\mu_4}{\sigma_0^4} - 3.$$

For a Gaussian variable, $\gamma_1 = \gamma_2 = 0$; more generally, departures from Gaussianity are measured by nonzero higher-order connected moments.

Equation (3.2) gives the characteristic function of a Gaussian variable. For any random field, the moment-generating function and characteristic function are

$$M_X(t) \equiv \langle e^{tX} \rangle, \qquad \tilde{f}(k) \equiv \langle e^{-ikX} \rangle = M_X(-ik).$$

Expanding $M_X(t)$ around $t = 0$ gives

$$M_X(t) = \sum_{n=0}^{\infty} \frac{t^n}{n!} \langle X^n \rangle, \qquad \langle X^n \rangle = \left. \frac{d^n M_X}{dt^n} \right|_{t=0} .$$

The cumulant-generating function is the logarithm of the moment-generating function,

$$K_X(t) \equiv \ln M_X(t) = \sum_{n=1}^{\infty} \frac{t^n}{n!} \kappa_n, \qquad \kappa_n = \left. \frac{d^n K_X}{dt^n} \right|_{t=0} .$$

The first few cumulants are

$$\begin{aligned}
\kappa_1 &= \mu, \\
\kappa_2 &= \langle X^2 \rangle - \langle X \rangle^2 = \sigma_0^2, \\
\kappa_3 &= \langle X^3 \rangle - 3 \langle X^2 \rangle \langle X \rangle + 2 \langle X \rangle^3 = \mu_3, \\
\kappa_4 &= \langle X^4 \rangle - 4 \langle X^3 \rangle \langle X \rangle - 3 \langle X^2 \rangle^2 + 12 \langle X^2 \rangle \langle X \rangle^2 - 6 \langle X \rangle^4 = \mu_4 - 3\sigma_0^4.
\end{aligned} \tag{3.3}$$

The normalized skewness and kurtosis satisfy $\gamma_1 = \kappa_3/\kappa_2^{3/2}$ and $\gamma_2 = \kappa_4/\kappa_2^2$. For several variables, the same definition gives connected correlation functions. If $\boldsymbol{X} = (X_1, \ldots, X_d)$ and $K(\boldsymbol{t}) = \ln \langle \exp(\sum_a t_a X_a) \rangle$, then

$$\langle X_{a_1} \cdots X_{a_n} \rangle_{\mathrm{c}} = \left. \frac{\partial^n K(\boldsymbol{t})}{\partial t_{a_1} \cdots \partial t_{a_n}} \right|_{\boldsymbol{t}=0} .$$

Therefore, the two-point correlation can be decomposed as

$$\langle X_i X_j \rangle = \langle X_i \rangle \langle X_j \rangle + \langle X_i X_j \rangle_{\mathrm{c}} ,$$

where $\langle X_i X_j \rangle_{\mathrm{c}}$ is the connected two-point correlation function, or cumulant. If X_i and X_j are mutually independent, then their correlation or cumulant is zero. For three variables,

$$\langle X_i X_j X_k \rangle = \langle X_i \rangle \langle X_j \rangle \langle X_k \rangle + \langle X_i X_j \rangle_{\mathrm{c}} \langle X_k \rangle + \langle X_i X_k \rangle_{\mathrm{c}} \langle X_j \rangle + \langle X_j X_k \rangle_{\mathrm{c}} \langle X_i \rangle + \langle X_i X_j X_k \rangle_{\mathrm{c}} .$$

The last term is the connected three-point correlation function, or third cumulant. If the variables are not mutually independent, then besides the pure correlation $\langle X_i X_j X_k \rangle_{\mathrm{c}}$, there are also pairwise contributions. The fourth cumulant can be similarly defined as

$$\begin{aligned}
\langle X_i X_j X_k X_l \rangle = {} & \langle X_i \rangle \langle X_j \rangle \langle X_k \rangle \langle X_l \rangle + \langle X_i X_j \rangle_{\mathrm{c}} \langle X_k \rangle \langle X_l \rangle + \cdots \\
& + \langle X_i X_j \rangle_{\mathrm{c}} \langle X_k X_l \rangle_{\mathrm{c}} + \cdots + \langle X_i X_j X_k \rangle_{\mathrm{c}} \langle X_l \rangle + \cdots + \langle X_i X_j X_k X_l \rangle_{\mathrm{c}} .
\end{aligned}$$

The special case when variables are all the same gives

$$
\begin{aligned}
\langle X^3\rangle &= \langle X\rangle^3 + 3\langle X^2\rangle_{\rm c}\langle X\rangle + \langle X^3\rangle_{\rm c}\,,\\
\langle X^4\rangle &= \langle X\rangle^4 + 6\langle X^2\rangle_{\rm c}\langle X\rangle^2 + 4\langle X^3\rangle_{\rm c}\langle X\rangle + 3\langle X^2\rangle_{\rm c}^2 + \langle X^4\rangle_{\rm c}\,.
\end{aligned}
$$

This is consistent with Eq. (3.3). In later sections, we will use the third- and fourth-order cumulants to extract field information beyond the power spectrum, i.e. the bispectrum and trispectrum.

The cumulant expansion theorem states that

$$
f_c(k) \equiv \ln\langle e^{-ikX}\rangle = \sum_{n=1}^{\infty}\frac{(-ik)^n}{n!}\langle X^n\rangle_{\rm c} = \langle (e^{-ikX}-1)\rangle_{\rm c}\,, \tag{3.4}
$$

which is an important result that allows us to express the characteristic function in terms of cumulants. We will use it later to derive the perturbative expansion of the non-Gaussian distribution.

Cauchy–Schwarz inequality

The Cauchy–Schwarz inequality states that for any two vectors u and v in an inner-product space, $|\langle u,v\rangle|^2 \leqslant \langle u,u\rangle\langle v,v\rangle$. In integral form with a positive measure $d\mu$,

$$
\left(\int d\mu\, fg\right)^2 \leqslant \left(\int d\mu\, f^2\right)\left(\int d\mu\, g^2\right).
$$

The spectral moments of a random field are defined as

$$
\sigma_j^2 = \int \frac{d^Nk}{(2\pi)^N}k^{2j}P(k) = \int d\mu(k)\,k^{2j}, \tag{3.5}
$$

in N-dimensional space. We can define a dimensionless spectral-shape parameter γ:

$$
\gamma \equiv \frac{\sigma_1^2}{\sigma_0\sigma_2}.
$$

It measures the correlation between the field f and its Laplacian $\nabla^2 f$. The Cauchy–Schwarz inequality gives

$$
(\sigma_1^2)^2 \leqslant \sigma_0^2\sigma_2^2,
$$

which guarantees that

$$
0 \leqslant \gamma \leqslant 1.
$$

This bound is useful for estimating the relative size of moments and cumulants. For cosmological fields, $\gamma = \mathcal{O}(1) \sim 0.4 - 0.8$, so that all spectral moments are of comparable order.

3.1.2 Kac–Rice formalism for CPs

Critical-point statistics are computed with the Kac–Rice framework, which expresses number densities in terms of the joint distribution of field derivatives. Morse-theoretic language then classifies critical points into maxima, minima, and saddles according to the Hessian eigenvalues. From now on, we normalize the field as follows: α is the normalized field value, η_i are normalized first derivatives, and ζ_{ij} are normalized second derivatives,

$$
\alpha \equiv \frac{f}{\sigma_0}, \qquad \eta_i \equiv \frac{\partial_i f}{\sigma_1}, \qquad \zeta_{ij} \equiv \frac{\partial_i\partial_j f}{\sigma_2}, \tag{3.6}
$$

where the spectral moments σ_0, σ_1, σ_2 are defined in Eq. (3.5).

Kac–Rice formula

Many geometric observables of a random field can be formulated as counting problems. The Kac–Rice formula provides a general framework for computing the expected number density of points satisfying a set of local constraints.

Consider a smooth map $\phi : \mathbb{R}^n \to \mathbb{R}^{n'}$ with $n \geqslant n'$, and a target value $u \in \mathbb{R}^{n'}$. The number of solutions of $\phi(\boldsymbol{x}) = u$ is

$$N_u(\phi) = \#\left\{\boldsymbol{x} \in \mathbb{R}^n : \phi(\boldsymbol{x}) = u\right\}.$$

The Kac–Rice formula states that the ensemble average of this quantity is

$$\langle N_u(\phi)\rangle = \int d^n x\, \mathcal{P}_\phi(u) \det\bigl(\partial\phi^i/\partial x_j\bigr), \qquad \phi(\boldsymbol{x}) = u,$$

where $\mathcal{P}_\phi(u)$ is the probability density function of ϕ. The determinant has a simple geometric interpretation. It measures how rapidly the constraint variables ϕ^i vary in the neighbourhood of a solution and therefore converts the volume element in configuration space into a density of roots. The larger the determinant, the more isolated the solutions are.

Critical points

Critical points are the locations where all first derivatives of the field vanish,

$$\nabla\alpha = 0.$$

The corresponding counting variable is

$$N^{\mathrm{cp}} = \#\left\{\boldsymbol{x} : \eta_1 = 0, \eta_2 = 0, \ldots, \eta_N = 0\right\},$$

where $\eta_i \equiv \partial_i\alpha$. Using the Kac–Rice formula,

$$\langle N^{\mathrm{cp}}\rangle = \int d\alpha\, d\boldsymbol{\eta}\, d\zeta\, \mathcal{P}(\alpha, \boldsymbol{\eta}, \zeta) \prod_{i=1}^{N} \delta_{\mathrm{D}}(\eta_i) J,$$

where

$$J = \det\left(\frac{\partial\eta_i}{\partial x_j}\right) = \det\zeta.$$

Therefore, the CP number density above a threshold can be written as

$$\langle n^{\mathrm{cp}}(\nu)\rangle = \left(\frac{\sigma_2}{\sigma_1}\right)^N \left\langle\Theta(\alpha - \nu)\, \delta_{\mathrm{D}}^N(\boldsymbol{\eta})\, \Theta(\lambda^{\mathrm{cp}})\, |\det\zeta|\right\rangle.$$

The $(\sigma_2/\sigma_1)^N$ factor comes from Eq. (3.6) for $\delta_{\mathrm{D}}^N(\boldsymbol{\eta})$ and $\det\zeta$. The factor $\Theta(\alpha-\nu)$ is the step function for excursion sets, since we want a threshold-dependent number density. We also include $\Theta(\lambda^{\mathrm{cp}})$ to select different CP species, and we take the absolute value of $\det\zeta$ to obtain a positive count. We order the eigenvalues of $-\zeta$ as $\lambda_1 \geqslant \cdots \geqslant \lambda_N$. The quantity λ^{cp} denotes the eigenvalue whose sign selects the critical-point species under consideration; equivalently, it is the boundary eigenvalue that enforces the required Hessian-sign pattern. In 2D, we consider $\mathrm{cp} \in \{\mathrm{peak}, \mathrm{saddle}, \mathrm{void}\}$, corresponding to 2, 1, and 0 negative eigenvalues, respectively. In 3D, we consider $\mathrm{cp} \in \{\mathrm{peak}, \mathrm{filament}, \mathrm{wall}, \mathrm{void}\}$, corresponding to 3, 2, 1, and 0 negative eigenvalues, respectively. For peaks, $\lambda^{\mathrm{cp}} = \lambda_N$; analogous boundary eigenvalues are used for the other critical points.

Euler characteristic

As another example, consider the Euler characteristic of the excursion set $\alpha > \nu_t$. Geometrically, this quantity represents the number density of upcrossing points along contour lines in 2D or along surfaces in 3D at a given threshold. Following the upcrossing formulation, the Euler characteristic in 3D can be expressed as the number of points satisfying

$$\alpha = \nu_t, \qquad \eta_1 = 0, \qquad \eta_2 = 0, \qquad \eta_3 > 0.$$

The corresponding counting variable is

$$N^{\rm Euler} = \#\left\{\boldsymbol{x} : \alpha = \nu_t, \eta_1 = 0, \eta_2 = 0, \eta_3 > 0\right\}.$$

Using the Kac–Rice formula,

$$\langle N^{\rm Euler}\rangle = \int d\alpha\, d\boldsymbol{\eta}\, d\zeta\, \mathcal{P}(\alpha, \boldsymbol{\eta}, \zeta)\, \delta_{\rm D}(\alpha - \nu_t)\delta_{\rm D}(\eta_1)\delta_{\rm D}(\eta_2)\Theta(\eta_3)J,$$

where $\zeta_{ij} = \partial_i\partial_j\alpha$ is the Hessian matrix. Under the imposed constraints,

$$J = \det\left(\frac{\partial(\alpha, \eta_1, \eta_2)}{\partial(x_1, x_2, x_3)}\right) = \det\begin{pmatrix} 0 & 0 & \eta_3 \\ \zeta_{11} & \zeta_{12} & \zeta_{13} \\ \zeta_{21} & \zeta_{22} & \zeta_{23}\end{pmatrix} = \eta_3\left(\zeta_{11}\zeta_{22} - \zeta_{12}^2\right).$$

More generally, for a normalized field α in N-dimensional space, using the same normalization conventions as for CPs, we can express the density as

$$\langle n\rangle^{\rm Euler}(\nu) = (-1)^{N+1}\frac{\sigma_2^{N-1}}{\sigma_0\sigma_1^{N-2}}\left\langle\delta_{\rm D}(\alpha - \nu)\prod_{i=1}^{N-1}\delta_{\rm D}(\eta_i)\,|\eta_N|\det\hat{\zeta}\right\rangle,$$

where η_N is the only component along which the gradient does not vanish, and $\hat{\zeta}$ is the minor matrix of ζ obtained by deleting the Nth row and column. The next subsection explains in more detail why this quantity is a topological invariant.

3.1.3 Morse index

The Euler characteristic is a topological invariant. For the present application, it is convenient to write it as an alternating sum over the Morse indices [23]:

$$\bar{n}^{\chi}(\nu) \equiv \sum_{\mu^{\rm cp}=0}^{N}(-1)^{\mu^{\rm cp}}\bar{n}^{\rm cp}(\nu) = \left(\frac{\sigma_2}{\sigma_1}\right)^N\left\langle\Theta(\alpha - \nu)\,\delta_{\rm D}^N(\boldsymbol{\eta})\det\zeta\right\rangle, \tag{3.7}$$

where $\mu^{\rm cp}$ is the number of positive Hessian eigenvalues. In 2D this reduces to

$$\bar{n}^{\chi} = \bar{n}^{\rm peak} - \bar{n}^{\rm saddle} + \bar{n}^{\rm void}.$$

In 3D it becomes

$$\bar{n}^{\chi} = \bar{n}^{\rm peak} - \bar{n}^{\rm filament} + \bar{n}^{\rm wall} - \bar{n}^{\rm void}.$$

Identity proof

Here we prove[1] that $\bar{n}^{\chi} = 2(-1)^N\bar{n}^{\rm Euler}$, i.e.,

$$(-1)^{N+1}\frac{\sigma_1^2}{\sigma_0\sigma_2}\left\langle\delta_{\rm D}(\alpha - \nu)\prod_{i=1}^{N-1}\delta_{\rm D}(\eta_i)\,|\eta_N|\det\hat{\zeta}\right\rangle = 2(-1)^N\left\langle\Theta(\alpha - \nu)\,\delta_{\rm D}^N(\boldsymbol{\eta})\det\zeta\right\rangle, \tag{3.8}$$

[1] Mostly by ChatGPT, but checked by hand.

where

$$\zeta = \begin{pmatrix} \hat{\zeta} & \boldsymbol{v} \\ \boldsymbol{v}^T & \zeta_{NN} \end{pmatrix}, \qquad \boldsymbol{v} = (\zeta_{1N}, \zeta_{2N}, \ldots, \zeta_{N-1\,N}).$$

The factor $(-1)^N$ on the right-hand side is chosen so that the final expression is positive. Recall that $\sigma_1 \eta_i = \sigma_0 \partial_i \alpha$ and $\sigma_2 \zeta_{ij} = \sigma_0 \partial_i \partial_j \alpha$, hence $\sigma_1 \partial_j \eta_i = \sigma_2 \zeta_{ij}$. We also use $\delta(x) = d\Theta(x)/dx$ and $\mathrm{sgn}(x) = x/|x| = 2\Theta(x) - 1$. Then

$$\delta_{\mathrm{D}}(\alpha - \nu)|\eta_N| = \mathrm{sgn}(\eta_N)\, \eta_N\, \delta(\alpha - \nu) = \frac{\sigma_0}{\sigma_1}\, \mathrm{sgn}(\eta_N)\, \partial_N \Theta(\alpha - \nu).$$

Using integration by parts, and dropping total derivatives in the distributional sense, we obtain

$$\begin{aligned} \mathrm{RHS} &= (-1)^{N+1} \frac{\sigma_1}{\sigma_2} \left\langle \prod_{i=1}^{N-1} \delta_{\mathrm{D}}(\eta_i) \det\hat{\zeta} \left[\partial_N \big(\Theta(\alpha - \nu)\, \mathrm{sgn}(\eta_N) \big) - \Theta(\alpha - \nu)\, \partial_N \mathrm{sgn}(\eta_N) \right] \right\rangle \\ &= 2(-1)^N \left\langle \Theta(\alpha - \nu)\, \delta_{\mathrm{D}}^N(\boldsymbol{\eta}) \det\hat{\zeta}\, \zeta_{NN} \right\rangle, \end{aligned}$$

where in the second line we used $\partial_N \mathrm{sgn}(\eta_N) = 2\,\delta(\eta_N)\,\partial_N \eta_N$. The boundary terms vanish because the expectation value is an integral over field derivatives weighted by a rapidly decaying probability density, so the integrand goes to zero at large field values. It remains to justify that $\zeta_{NN}\det\hat{\zeta}$ may be replaced by $\det\zeta$ in the sense of distributions. Expanding $\det\zeta$ along the last row gives

$$\begin{aligned} \det\zeta &= \sum_{j=1}^{N} (-1)^{N+j} \zeta_{Nj} M_{Nj} = \zeta_{NN}\det\hat{\zeta} + \sum_{j=1}^{N-1} (-1)^{N+j} \zeta_{Nj} M_{Nj} \\ &= \zeta_{NN}\det\hat{\zeta} + \sum_{i,j=1}^{N-1} (-1)^{i+j}\, \zeta_{Nj} \zeta_{iN} \det\big(\hat{\zeta} \text{ with row } i \text{ and column } j \text{ deleted}\big). \end{aligned}$$

When multiplied by $\prod_{i=1}^{N-1} \delta_{\mathrm{D}}(\eta_i)$, the second term vanishes distributionally: indeed $\delta(\eta_i)\,\partial_N \eta_i = \partial_N \Theta(\eta_i)$ is a total derivative, hence it integrates to zero against test functions. Therefore $\zeta_{NN}\det\hat{\zeta}$ can be replaced by $\det\zeta$ under the expectation, completing the proof.

Morse theory

In the previous subsections, we introduced the Euler characteristic as a topological invariant. Here we briefly discuss why it is topological and how it is related to Betti numbers, following Sec. 8.1 of Ref. [23]. Some basic concepts from algebraic topology are useful for this discussion.

In Eq. (3.7), we introduced the signs $(-1)^{\mu^{\mathrm{cp}}}$ to obtain cancellations between the contributions from individual critical points. For example, in Fig. 3.1, we consider a compact manifold X given by the 2-sphere S^2, and we study smooth functions on this manifold. The function f_1 has a maximum p_1 and a minimum p_2, whereas f_2 has two maxima p_1, p_2, a saddle p_3, and a minimum p_4. Using the Morse indices, we can calculate χ and find that it is 2 in both cases. We introduce the boundary operator ∂. Consider pairs (p, q) of CPs with $\mu^{\mathrm{cp}}(q) = \mu^{\mathrm{cp}}(p) - 1$, i.e. with index difference 1. We then count the number of trajectories from p to q modulo 2:

$$\partial p = \sum_{\substack{q \text{ CPs of } f \\ \mu^{\mathrm{cp}}(q) = \mu^{\mathrm{cp}}(p) - 1}} (\#\{\text{flow lines from } p \text{ to } q\} \mod 2)\, q.$$

In this way, we obtain an operator from the chain group $C_*(f, \mathbb{Z}_2)$, the vector space over $\mathbb{Z}_2$ generated by the CPs of f, to itself. Then we can define the homology groups

$$H_k(X, f, \mathbb{Z}_2) \equiv \text{kernel of } \partial \text{ on } C_k(f, \mathbb{Z}_2)/ \text{ image of } \partial \text{ from } C_{k+1}(f, \mathbb{Z}_2).$$

Here $C_k(f, \mathbb{Z}_2)$ is generated by the critical points of Morse index k. Since $\partial \circ \partial = 0$, the image of ∂ from $C_{k+1}(f, \mathbb{Z}_2)$ is always contained in the kernel of ∂ on $C_k(f, \mathbb{Z}_2)$. Denoting the restriction of ∂ to $C_k(f, \mathbb{Z}_2)$ by ∂_k, we obtain the chain complex

$$0 \to C_2 \xrightarrow{\partial_2} C_1 \xrightarrow{\partial_1} C_0 \to 0.$$

For f_1, there are no pairs of critical points with index difference 1. Then $C_2 = \{p_1\}$, $C_1 = 0$, and $C_0 = \{p_2\}$, so $\partial_2 = 0$ and $\partial_1 = 0$. Therefore,

$$\ker \partial_2 = C_2 = \{p_1\}, \qquad \ker \partial_0 = C_0 = \{p_2\}.$$

In addition, $\operatorname{img} \partial_2 = \{\partial_2(p) \mid p \in C_2\} = 0$, and the other images are also trivial. Thus,

$$H_2(X, f_1, \mathbb{Z}_2) = \mathbb{Z}_2, \qquad H_1(X, f_1, \mathbb{Z}_2) = 0, \qquad H_0(X, f_1, \mathbb{Z}_2) = \mathbb{Z}_2.$$

The Betti numbers are defined as

$$b_k \equiv \dim_{\mathbb{Z}_2} H_k(X, f, \mathbb{Z}_2).$$

The Euler characteristic is

$$\chi(X) = \sum_j (-1)^j b_j.$$

For f_1, this gives $\chi = 1 - 0 + 1 = 2$. Next, consider f_2, for which $C_2 = \{p_1, p_2\}$, $C_1 = \{p_3\}$, and $C_0 = \{p_4\}$. Since $\partial_2 p_1 = \partial_2 p_2 = p_3$, we have $\partial_2(p_1 + p_2) = 2p_3 = 0$ and $\partial_1 p_3 = 2p_4 = 0$, because we compute modulo 2. Finally, $\partial_0 p_4 = 0$. Therefore,

$$\ker \partial_2 = \{p_1 + p_2\}, \qquad \ker \partial_1 = \{p_3\}, \qquad \ker \partial_0 = \{p_4\}.$$

The nontrivial images are

$$\operatorname{img} \partial_2 = \{p_3\}, \qquad \operatorname{img} \partial_1 = 0.$$

Thus,

$$H_2(X, f_2, \mathbb{Z}_2) = \mathbb{Z}_2, \qquad H_1(X, f_2, \mathbb{Z}_2) = 0, \qquad H_0(X, f_2, \mathbb{Z}_2) = \mathbb{Z}_2.$$

The homology groups, and therefore also the Betti numbers, are the same for both functions. This is a basic fact of Morse theory. The equality arises from cancellations between critical points induced by the boundary operator.

It is worth noting that the field is defined on a periodic domain. Consequently, the underlying manifold is T^2 in 2D and T^3 in 3D, both of which have vanishing Euler characteristic. Therefore, the Euler characteristic of the full field satisfies

$$\chi(T^2) = \chi(T^3) = 0.$$

3.2 One-point Gaussian critical-point statistics

In Sec. 3.1, we introduced the statistical description of a random field and the geometric properties of critical points and the Euler characteristic. In this section, we derive the general formula for these quantities in N dimensions and then specialize it to the 2D and 3D cases. We follow the method of Refs. [24, 28], which is based mainly on the framework pioneered by BBKS [3].

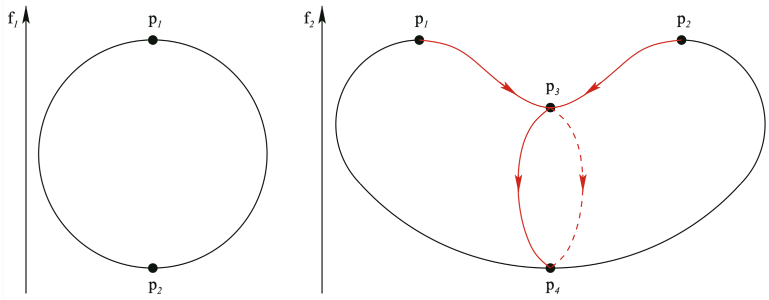

Figure 3.1: Graphs of two functions and their critical points. The graph of f_2 also shows flow lines between CPs with index difference 1. (From Ref.[23])

3.2.1 Statistics of field derivatives

The following covariances will be useful later:

$$\begin{aligned}
&\langle \alpha^2 \rangle_{\mathrm{G}} = 1, \qquad \langle \eta_i \eta_j \rangle_{\mathrm{G}} = \frac{1}{N}\delta_{ij}, \qquad \langle \alpha \zeta_{ij} \rangle_{\mathrm{G}} = -\frac{\gamma}{N}\delta_{ij}, \\
&\langle \zeta_{ij}\zeta_{kl} \rangle_{\mathrm{G}} = \frac{1}{N(N+2)}(\delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}).
\end{aligned} \tag{3.9}$$

$\langle \cdots \rangle_{\mathrm{G}}$ denotes an average over the Gaussian probability distribution; these covariances follow from isotropy and the spectral moments. Other cross-correlations, such as $\langle \alpha \eta_i \rangle$ and $\langle \eta_i \zeta_{jk} \rangle$, vanish. Since ζ_{ij} is a symmetric tensor, only components with $i \leqslant j$ are independent. We collect the independent variables as

$$\boldsymbol{Y} = (\alpha, \eta_1, \ldots, \eta_N, \zeta_{11}, \zeta_{12}, \ldots, \zeta_{N-1,N}, \zeta_{NN}).$$

There are $N_0 = (N+1)(N+2)/2$ variables in total. The multivariate Gaussian distribution for these variables at a single point is

$$\mathcal{P}_{\mathrm{G}}(\boldsymbol{Y}) = \frac{1}{\sqrt{(2\pi)^{N_0} \det M}} \exp\left(-\frac{1}{2} \boldsymbol{Y}^{\mathrm{T}} M^{-1} \boldsymbol{Y} \right), \tag{3.10}$$

where $M_{ab} \equiv \langle X_a X_b \rangle$ is the $N_0 \times N_0$ covariance matrix given by Eq. (3.9). To reduce the number of variables, we introduce the following rotationally invariant quantities as variables of the probability distribution:

$$\begin{aligned}
&\eta^2 \equiv \boldsymbol{\eta} \cdot \boldsymbol{\eta}, \qquad J_1 \equiv -\zeta_{ii}, \qquad J_2 \equiv \frac{N}{N-1}\tilde{\zeta}_{ij}\tilde{\zeta}_{ji} \quad (N \geqslant 2), \\
&J_3 \equiv \frac{N^2}{(N-1)(N-2)}\tilde{\zeta}_{ij}\tilde{\zeta}_{jk}\tilde{\zeta}_{ki} \quad (N \geqslant 3).
\end{aligned} \tag{3.11}$$

Here repeated indices are summed over, and the traceless part of ζ is defined as

$$\tilde{\zeta}_{ij} \equiv \zeta_{ij} + \frac{1}{N}\delta_{ij} J_1.$$

The normalization factors in J_2 and J_3 are explained in the next subsubsection.

In terms of the rotationally invariant variables, the multivariate Gaussian PDF in Eq. (3.10) becomes

$$\mathcal{P}_{\mathrm{G}}(\boldsymbol{Y}) \propto \mathcal{N}(\alpha, J_1) \exp\left[-\frac{N}{2}\eta^2 - \frac{(N-1)(N+2)}{4} J_2 \right], \tag{3.12}$$

where $\mathcal{N}$ is the bivariate normal distribution with unit variance,

$$\mathcal{N}(\alpha, J_1) = \frac{1}{2\pi\sqrt{1-\gamma^2}} \exp\left[-\frac{\alpha^2 + J_1^2 - 2\gamma\alpha J_1}{2(1-\gamma^2)} \right]. \tag{3.13}$$

The form of this expression is fixed by the Gaussian distribution and the chosen normalization conventions. The details are discussed later in this section.

Rotationally invariant variables

Here we explain why J_2 and J_3 are defined as in Eq. (3.11). Consider a matrix X. For $R \in SO(3)$ with $RR^{\mathrm{T}} = 1$, the transformation $X \to X' = RXR^T$ leaves the trace invariants $\mathrm{Tr}\big(X^N\big)$ unchanged. For example,

$$\begin{aligned} \mathrm{Tr}(X) &= \sum_i X_{ii} = \mathrm{Tr}\big(RXR^T\big) = \mathrm{Tr}\big(X'\big), \\ \mathrm{Tr}\big(X^2\big) &= \sum_{ij} X_{ij}X_{ji} = \mathrm{Tr}\big((RXR^T)(RXR^T)\big) = \mathrm{Tr}\big(X'^2\big). \end{aligned}$$

The characteristic equation is

$$\det(X - \lambda \mathbb{1}) = 0,$$

and the corresponding characteristic polynomial is

$$\prod_{a=1}^{N}(t-\lambda_a) = t^N - I_1 t^{N-1} + I_2 t^{N-2} + \cdots + (-1)^N I_N.$$

The coefficients I_s are elementary symmetric polynomials in the eigenvalues, e.g., $I_1 = \sum_i \lambda_i$, $I_2 = \sum_{i<j} \lambda_i\lambda_j$, $I_3 = \sum_{i<j<k} \lambda_i\lambda_j\lambda_k$, and $I_N = \prod_i \lambda_i$. In general,

$$I_s(\lambda_1, \ldots, \lambda_N) = \sum_{1 \leqslant i_1 < \cdots < i_s \leqslant N} \lambda_{i_1} \cdots \lambda_{i_s}.$$

We define the traceless matrix

$$\tilde{X} = X_{ij} - \frac{1}{N}\delta_{ij}\,\mathrm{Tr}(X),$$

and the invariant functions p_k and $\tilde{p}_k$, where k is the order of the polynomial:

$$p_k = \mathrm{Tr}\Big(X^k\Big), \qquad \tilde{p}_k = \mathrm{Tr}\Big(\tilde{X}^k\Big).$$

With $p_0 = N$ and $p_1 = \mathrm{Tr}(X)$, we have

$$\tilde{p}_k = \mathrm{Tr}\left(\left(X - \frac{p_1}{N}\mathbb{1}\right)^k\right) = \sum_{s=0}^{k} C_k^s \Big(-\frac{p_1}{N}\Big)^{k-s} p_s = \Big(-\frac{p_1}{N}\Big)^k N + \sum_{s=1}^{k} C_k^s \Big(-\frac{p_1}{N}\Big)^{k-s} p_s.$$

For instance,

$$\tilde{p}_2 = -\frac{1}{N}p_1^2 + p_2, \qquad \tilde{p}_3 = \frac{2p_1^3}{N^2} - \frac{3p_1p_2}{N} + p_3.$$

Thus $\tilde{p}_k$ can be expressed in terms of p_k. Returning to the Hessian variables, the matrix invariants of $\tilde{\zeta}_{ij}$ are proportional to $\tilde{p}_k$, so they can be represented by p_k. For instance,

$$\begin{aligned} I_1 &= p_1, \qquad I_2 = \frac{1}{2}(p_1^2 - p_2), \\ I_3 &= \frac{1}{3}\sum_{i=1}^{3}(-1)^{i-1}p_i I_{3-i} = \frac{1}{6}(p_1^3 - 3p_1p_2 + 2p_3), \\ I_4 &= \frac{1}{4}(p_1I_3 - p_2I_2 + p_3I_1 - p_4) = \frac{1}{24}(p_1^4 - 5p_1^2p_2 + 3p_2^2 + 8p_1p_3 - 6p_4). \end{aligned} \tag{3.14}$$

In general, these are Newton's identities:

$$I_k = \frac{1}{k}\sum_{i=1}^{k}(-1)^{i-1}p_i I_{k-i}.$$

Following Ref. [24], the variables J_k are defined as normalized coefficients of the characteristic polynomial of the traceless matrix $\tilde{X}$:

$$J_{k\geqslant 2} = I_1^k - \sum_{s=2}^{k} \frac{(-N)^s C_k^s}{(k-1)C_N^s} I_1^{k-s} I_s.$$

This equation can also be derived directly. Let $\tilde{x}_{ij} = N x_{ij} - I_1 \mathbb{1}$ be the traceless part of H. The characteristic polynomial of $\tilde{H}$ has coefficients

$$J_s = \sum_{1\leqslant i_1<\cdots<i_s\leqslant N} \prod_{r=1}^{s} (N\lambda_{i_r} - I_1).$$

Expanding the product gives

$$\begin{aligned} J_s &= C_N^s(-I_1)^s + \sum_{1\leqslant i_1<\cdots<i_s\leqslant N} \sum_{p=1}^{s} \binom{s-p}{N-p} (N\lambda_{i_1})\cdots(N\lambda_{i_p})(-I_1)^{s-p} \\ &= C_N^s(-I_1)^s + C_{N-1}^{s-1}(-N)(-I_1)^s + \sum_{p=2}^{s} C_{N-p}^{s-p} N^p I_p (-I_1)^{s-p}. \end{aligned}$$

After applying the chosen normalization, we obtain the expression for J_s.

For a traceless matrix, $I_1 = 0$. Equation (3.14) then gives $2I_2 = -\tilde{p}_2$, $3I_3 = \tilde{p}_3$, and $8I_4 = \tilde{p}_2^2 - 2\tilde{p}_4$. By analogy with I_s, $J_2 \propto \tilde{p}_2$ and $J_3 \propto \tilde{p}_3$, so

$$J_2 = \frac{N}{N-1}\tilde{p}_2, \qquad J_3 = \frac{N^2}{(N-1)(N-2)}\tilde{p}_3.$$

The normalization factor $N/(N-1)$ in J_2 is chosen so that $\langle J_2 \rangle = 1$, analogous to $\langle J_1^2 \rangle = \langle \zeta_{ii}\zeta_{jj} \rangle = 1$. To see this, compute

$$\begin{aligned} \langle \tilde{\zeta}_{ij}\tilde{\zeta}_{kl} \rangle &= \langle \zeta_{ij}\zeta_{kl} \rangle + \langle \frac{1}{N}\delta_{ij} J_1 \zeta_{kl} \rangle + \langle \frac{1}{N}\delta_{kl} J_1 \zeta_{ij} \rangle + \frac{1}{N^2}\delta_{ij}\delta_{kl} \langle J_1^2 \rangle \\ &= \frac{1}{N(N+2)} \left(-\frac{2}{N}\delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk} \right). \end{aligned}$$

Then

$$\langle J_2 \rangle = \frac{N}{N-1} \langle \tilde{\zeta}_{ij}\tilde{\zeta}_{ji} \rangle = \frac{N}{N-1}\frac{1}{N(N+2)}\left(N^2 + N - 2\right) = 1.$$

Similarly, $\langle J_3 \rangle = 1$ can be obtained from the rank-6 tensor calculation shown in Appendix B.

Distribution conventions

The pair (α, J_1) follows the bivariate normal distribution above. Using $\langle \alpha^2 \rangle = 1$, $\langle J_1^2 \rangle = 1$, and $\langle \alpha J_1 \rangle = \gamma$, the covariance matrix is $\begin{pmatrix} 1 & \gamma \\ \gamma & 1 \end{pmatrix}$, which gives $\mathcal{N}(\alpha, J_1)$ in Eq. (3.13).

We now explain why the distribution has this form for (J_2, J_3). $\mathcal{P}(J_2)$ follows a χ^2 distribution. For a traceless matrix, let $d = N(N+1)/2 - 1$ be the number of independent variables. If $X = \sum_{a=1}^{d} x_a^2$ and $x_a \sim \mathcal{N}(0, \sigma_t^2)$, then $X/\sigma_t^2 \sim \chi_d^2$. Thus $J_2 = \frac{N}{N-1}\sum_{a=1}^{d} x_a^2$ gives

$$\sigma_t^2 \cdot \left(\frac{N(N+1)}{2} - 1 \right) = \frac{N-1}{N}.$$

This fixes the variance assigned to each independent component. Thus

$$\mathcal{P}(J_2) = \frac{1}{\Gamma(d/2)} \frac{1}{[2(N/(N-1))\sigma_t^2]^{d/2}} J_2^{d/2-1} \exp\left[\frac{-J_2}{2(N/N-1)\sigma_t^2} \right]. \tag{3.15}$$

This reproduces Eq. (3.12) when $N = 2$. When $N = 3$, let $\tilde{\lambda}_1$, $\tilde{\lambda}_2$, and $\tilde{\lambda}_3 = -\tilde{\lambda}_1 - \tilde{\lambda}_2$ be the eigenvalues of $\tilde{\zeta}$. Therefore,

$$J_2 = 3(\tilde{\lambda}_1^2 + \tilde{\lambda}_2^2 + \tilde{\lambda}_1\tilde{\lambda}_2), \qquad J_3 = -\frac{27}{2}\tilde{\lambda}_1\tilde{\lambda}_2(\tilde{\lambda}_1 + \tilde{\lambda}_2).$$

The Jacobian is

$$\frac{\partial(J_2, J_3)}{\partial(\tilde{\lambda}_1, \tilde{\lambda}_2)} = -\frac{81}{2}(\tilde{\lambda}_1 - \tilde{\lambda}_2)(2\tilde{\lambda}_1 + \tilde{\lambda}_2)(\tilde{\lambda}_1 + 2\tilde{\lambda}_2).$$

Meanwhile, the Vandermonde factor for the traceless matrix, explained in the next subsubsection, gives the same measure:

$$\prod_{i<j}\left|\tilde{\lambda}_i - \tilde{\lambda}_j\right| d\tilde{\lambda}_1 d\tilde{\lambda}_2 \propto dJ_2 dJ_3.$$

We can also use

$$\tilde{\lambda}_i = A\cos\left(\theta + \frac{2\pi i}{3}\right), \qquad i = 1, 2, 3,$$

to satisfy the traceless condition. Then

$$\sum_i \cos\left(\theta + \frac{2\pi i}{3}\right) = 0, \qquad \sum_i \cos^2\left(\theta + \frac{2\pi i}{3}\right) = \frac{3}{2}, \qquad \sum_i \cos^3\left(\theta + \frac{2\pi i}{3}\right) = \frac{3}{4}\cos 3\theta.$$

It follows that $J_2 = \frac{9}{4}A^2$ and $J_3 = \frac{27}{8}A^3\cos 3\theta$. This gives $J_3 = J_2^{3/2}\cos 3\theta$ and $-J_2^{3/2} \leqslant J_3 \leqslant J_2^{3/2}$.

$$dJ_3 = -3J_2^{3/2}\sin 3\theta d\theta.$$

Integrating out θ returns Eq. (3.15). The marginal distribution of J_2 is reproduced by the joint form $\mathcal{P}(J_2, J_3) \propto e^{5J_2/2}$.

The volume element in the space of symmetric matrices

We now prove that the volume element in the six-dimensional space of real symmetric matrices is given by

$$d\,\mathrm{vol} = \prod_{A=1}^{6} d\zeta_A = |(\lambda_1 - \lambda_2)(\lambda_2 - \lambda_3)(\lambda_1 - \lambda_3)| d\lambda_1 d\lambda_2 d\lambda_3 d\,\mathrm{vol}[SO(3)],$$

where $d\,\mathrm{vol}[SO(3)] = \sin\beta d\beta d\alpha d\gamma$ is the volume element on $SO(3)$, which is parametrized by the Euler angles ($\alpha, \gamma \in [0, 2\pi], \beta \in [0, \pi]$). $|(\lambda_1 - \lambda_2)(\lambda_2 - \lambda_3)(\lambda_1 - \lambda_3)|$ is called the Vandermonde factor. Define the inner product of two symmetric matrices as $\mathrm{Tr}(S_1 S_2)$. Then the metric on the space of symmetric matrices is $ds^2 = \mathrm{Tr}\big[(dS)^2\big]$. The matrix S can be diagonalized by a rotation, $\lambda = \mathrm{diag}(\lambda_1, \lambda_2, \lambda_3)$, so that $S = R^\dagger \lambda R$. Since $d(R^\dagger)R + R^\dagger dR = 0$, we have

$$\begin{aligned} dS &= d(R^\dagger)\lambda R + R^\dagger d\lambda R + R^\dagger \lambda dR = R^\dagger d\lambda R + R^\dagger \lambda R R^\dagger dR - R^\dagger dR R^\dagger \lambda R \\ &= R^\dagger d\lambda R + [R^\dagger \lambda R, R^\dagger dR] = R^\dagger (d\lambda + [\lambda, R^\dagger dR])R, \end{aligned}$$

where $[,]$ denotes the commutator. Thus,

$$ds^2 = \mathrm{Tr}\big[(d\lambda)^2\big] + \mathrm{Tr}\big[\lambda, R^\dagger dR\big]^2.$$

The cross terms disappear since they involve the trace of a product of symmetric and antisymmetric matrices, i.e. $\mathrm{Tr}\big(d\lambda[\lambda, R^\dagger dR]\big) = 0$. Since $(\lambda R^\dagger dR)_{ij} = \sum_k \lambda_{ik}(R^\dagger dR)_{kj}$ and $(R^\dagger dR\lambda)_{ij} = \sum_k (R^\dagger dR)_{ik}\lambda_{kj}$, it follows that

$$\begin{aligned} [\lambda, R^\dagger dR]_{ij} &= \sum_k \big(\lambda_i \delta_{ik}(R^\dagger dR)_{kj} - (R^\dagger dR)_{ik}\lambda_j \delta_{jk}\big) = \lambda_i (R^\dagger dR)_{ij} - (R^\dagger dR)_{ij}\lambda_j \\ &= \lambda_i \epsilon_{ijk}\omega_k - \epsilon_{ijk}\omega_k \lambda_j = (\lambda_i - \lambda_j)\epsilon_{ijk}\omega_k, \end{aligned}$$

since $(R^\dagger dR)^\dagger = -R^\dagger dR$. Here we define $\omega_i = \frac{1}{2}\epsilon_{ijk}(R^\dagger dR)_{jk}$. Thus,

$$ds^2 = \sum_i (d\lambda_i)^2 + (\lambda_2 - \lambda_3)^2\omega_1^2 + (\lambda_3 - \lambda_1)^2\omega_2^2 + (\lambda_1 - \lambda_2)^2\omega_3^2,$$

with the orthonormal basis

$$\{d\lambda_1, d\lambda_2, d\lambda_3, |\lambda_2 - \lambda_3|\omega_1, |\lambda_3 - \lambda_1|\omega_2, |\lambda_1 - \lambda_2|\omega_3\}.$$

Since the volume element is the wedge product of the orthonormal basis elements, we have

$$d\,\mathrm{vol} = |(\lambda_1 - \lambda_2)(\lambda_2 - \lambda_3)(\lambda_1 - \lambda_3)|\lambda_1 \wedge \lambda_2 \wedge \lambda_3 \wedge \omega_1 \wedge \omega_2 \wedge \omega_3.$$

The matrix $R^\dagger dR = \begin{pmatrix} 0 & \omega_3 & -\omega_2 \\ -\omega_3 & 0 & \omega_1 \\ \omega_2 & -\omega_1 & 0 \end{pmatrix}$ is an element of the Lie algebra of $SO(3)$. The ω_i form an orthonormal basis, so the volume element of $SO(3)$ is given by

$$d\mathrm{vol}[SO(3)] = \omega_1 \wedge \omega_2 \wedge \omega_3.$$

Similarly, in the 2D case, $R^\dagger dR = \begin{pmatrix} 0 & \omega \\ -\omega & 0 \end{pmatrix}$ and $(R^\dagger dR)_{ij} = \epsilon_{ij}\omega$, so that

$$[\lambda, R^\dagger dR]_{ij} = (\lambda_i - \lambda_j)\epsilon_{ij}\omega,$$

and the line element is

$$ds^2 = (d\lambda_1)^2 + (d\lambda_2)^2 + (\lambda_1 - \lambda_2)^2\omega^2.$$

Thus,

$$d\,\mathrm{vol}_{\mathrm{2D}} = |(\lambda_1 - \lambda_2)|\lambda_1 \wedge \lambda_2 \wedge \omega,$$

and $d\mathrm{vol}[SO(2)] = \omega$, with angular range $[0, 2\pi]$.

For example, $\Omega \equiv R^\dagger dR = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} -\sin d\theta & -\cos\theta d\theta \\ -\cos\theta d\theta & -\sin\theta d\theta \end{pmatrix} = \begin{pmatrix} 0 & -d\theta \\ d\theta & 0 \end{pmatrix}$.

This is important when transforming the variables from the elements of ζ to its eigenvalues and then to other orthogonal linear variables.

3.2.2 Number density in Gaussian fields

After specifying the Gaussian probability density in Eq. (3.10) and the Kac–Rice formula for critical points in Sec. 3.1.2, we can combine them to calculate the number density in Gaussian fields analytically:

$$\begin{aligned}
\bar{n}^{\mathrm{cp}}(\nu) &= \int_{-\infty}^{\infty} d\alpha d\boldsymbol{\eta} d\prod_{i\leqslant j}\zeta_{ij} \left(\frac{\sigma_2}{\sigma_1}\right)^N \Theta(\alpha - \nu)\delta_{\mathrm{D}}^N(\boldsymbol{\eta})\Theta(\lambda^{\mathrm{cp}})|\det\zeta|\,\mathcal{P}_{\mathrm{G}}(\boldsymbol{Y}) \\
&= N_1 \left(\frac{\sigma_2}{\sigma_1}\right)^N \left(\frac{N}{2\pi}\right)^{N/2} \int_\nu^\infty d\alpha \int d\prod_{i\leqslant j}\zeta_{ij}\, \Theta(\lambda^{\mathrm{cp}})|\det\zeta|\mathcal{N}(\alpha, J_1)\exp\left[-\frac{(N-1)(N+2)}{4}J_2\right].
\end{aligned} \tag{3.16}$$

The factor $(N/2\pi)^{N/2}$ comes from the normalized distribution of η^2, and the normalization factor N_1 is given by

$$N_1^{-1} = \frac{1}{\sqrt{2\pi}} \int d\prod_{i\leqslant j}\zeta_{ij}\, \exp\left[-\frac{J_1^2}{2} - \frac{(N-1)(N+2)}{4}J_2\right],$$

where we have used $\int d\alpha\, \mathcal{N}(\alpha, J_1) = e^{-J_1^2/2}/\sqrt{2\pi}$. Next, we change the integration variables as

$$N_1 \prod_{i\leqslant j} d\zeta_{ij} = \frac{1}{\Omega_N} dx\, d^{N-1}W\, d\Omega_N,$$

where $x = J_1 = \lambda_1 + \cdots + \lambda_N$. Here $d^{N-1}W$ represents the volume element of the traceless rotational invariants, and $d\Omega_N$ represents the volume element of the rotational angular components.

$$\Omega_N = \int d\Omega_N = \mathrm{Vol}_{\mathrm{SO}(N)} = \frac{2^{N-1}\pi^{(N-1)(N+2)/4}}{\prod_{n=1}^{N-1}\Gamma(n/2)}.$$

In principle, $d^{N-1}W$ is obtained by Gram–Schmidt orthogonalization of the $N(N-1)/2$ independent components of ζ_{ij} so that the eigenvalues take the form $-(\lambda_1, \lambda_2, \ldots, \lambda_N)$, ordered by $\lambda_1 \geqslant \lambda_2 \geqslant \cdots \geqslant \lambda_N$, and satisfy

$$\int_D d^{N-1}W\, \exp\left[-\frac{(N-1)(N+2)}{4} J_2\right] = 1, \tag{3.17}$$

where D is the integration domain defined by $\lambda_1 \geqslant \lambda_2 \geqslant \cdots \geqslant \lambda_N$. Therefore, Eq. (3.16) reduces to

$$\bar{n}^{\mathrm{cp}}(\nu) = \frac{1}{(2\pi)^{N/2}} \left(\frac{\sigma_2}{\sqrt{N}\sigma_1}\right)^N \int_0^\infty dx\, f_{00}^{\mathrm{cp}}(x) \int_\nu^\infty d\alpha\, \mathcal{N}(\alpha, x), \tag{3.18}$$

where

$$f_{00}^{\mathrm{cp}}(x) \equiv N^N \int_D d^{N-1}W\, \Theta(\lambda^{\mathrm{cp}})|\lambda_1 \cdots \lambda_N| \exp\left[-\frac{(N-1)(N+2)}{4} J_2\right]. \tag{3.19}$$

The differential number density can also be written as

$$-\frac{d\bar{n}^{\mathrm{cp}}(\nu)}{d\nu} = \frac{1}{(2\pi)^{N/2}} \left(\frac{\sigma_2}{\sqrt{N}\sigma_1}\right)^N \int_{a^{\mathrm{cp}}}^{b^{\mathrm{cp}}} dx\, f_{00}^{\mathrm{cp}}(x) \mathcal{N}(\nu, x),$$

by differentiating Eq. (3.18). In the section on non-Gaussian corrections, we explain why this function is denoted by $f_{00}(x)$. In Eq. (3.8), we proved the identity between critical points and the Euler characteristic; the only difference is replacing $\Theta(\lambda^{\mathrm{cp}})|\lambda_1 \cdots \lambda_N|$ with $\lambda_1 \cdots \lambda_N$ in Eq. (3.19). Therefore, we will use $-d\bar{n}^\star/d\nu$ to denote these quantities collectively, because the differential number density is the main quantity used in the theory.

2D space

In the 2D case, this reduces to

$$-\frac{d\bar{n}^\star(\nu)}{d\nu} = \frac{1}{4\pi}\left(\frac{\sigma_2}{\sigma_1}\right)^2 \int_{a^\star}^{b^\star} dx\, f_{00}^\star(x) \mathcal{N}(\nu, x). \tag{3.20}$$

We change variables from λ_1, λ_2 to

$$x = \lambda_1 + \lambda_2, \qquad y = \frac{\lambda_1 - \lambda_2}{2},$$

so that $J_1 = x$, $J_2 = 4y^2$, and $\det\zeta = x^2/4 - y^2$. The Vandermonde factor gives $(\lambda_1 - \lambda_2) \propto y dy$, so $dW \propto y$. The normalization condition in Eq. (3.17) gives the factor $1/\int dy\, y \exp(-4y^2)$ when $\lambda_1 \geqslant \lambda_2$ so that $dW = 8y dy$. Thus, for different CPs, we have

$$f_{00}^\star(x) = 8\int_{a^{\mathrm{cp}}}^{b^{\mathrm{cp}}} dy\, y|x^2 - 4y^2| e^{-y^2} = \begin{cases} e^{-x^2} + x^2 - 1, & \text{peak,void} \\ -e^{-x^2}, & \text{saddle} \\ x^2 - 1, & \chi \end{cases}.$$

The peak/void domain is given by $\lambda_2 = x/2 - y \geqslant 0 \Rightarrow y \leqslant x/2$, together with $\lambda_1 \geqslant \lambda_2 \Rightarrow y > 0$. For a saddle, the domain of y is given by $\lambda_1 \geqslant 0 \geqslant \lambda_2$, so $y \in (|x|/2, \infty)$; for the Euler characteristic, $y \in (0, \infty)$. Substituting $f_{00}^{\mathrm{cp}}(x)$ into Eq. (3.20) makes the result straightforward to evaluate numerically. Since $x = -\lambda_1 - \lambda_2$, peaks have $x \in [0, \infty)$, voids have $x \in (-\infty, 0]$, and saddles and χ have $x \in (-\infty, \infty)$.

3D space

In the 3D case, this reduces to

$$-\frac{d\bar{n}^\star(\nu)}{d\nu} = \frac{1}{(2\pi)^{3/2}}\left(\frac{\sigma_2}{\sqrt{3}\sigma_1}\right)^3 \int_{a^\star}^{b^\star} dx\, f_{00}^\star(x)\mathcal{N}(\nu, x).$$

We also introduce an orthogonal, linearly independent set of variables,

$$\begin{aligned}
& x = \lambda_1 + \lambda_2 + \lambda_3, \qquad && y = \frac{\lambda_1 - \lambda_3}{2}, \qquad && z = \frac{\lambda_1 - 2\lambda_2 + \lambda_3}{2} \\
\Rightarrow \quad & \lambda_1 = \frac{x}{3} + y + \frac{z}{3}, && \lambda_2 = \frac{x}{3} - \frac{2z}{3}, && \lambda_3 = \frac{x}{3} - y + \frac{z}{3} \\
\Rightarrow \quad & J_1 = x, && J_2 = 3y^2 + z^2, && J_3 = z^3 - 9y^2 z.
\end{aligned}$$

The determinant is

$$|\det \zeta| = |\lambda_1\lambda_2\lambda_3| = \frac{1}{27}\left|(x-2z)\left[(x+z)^2 - (3y)^2\right]\right|.$$

Using the Vandermonde factor, the volume element should satisfy $d^2W \propto y(y^2 - z^2)dydz$. Imposing the normalization condition and the ordering $\lambda_1 \geqslant \lambda_2 \geqslant \lambda_3$ gives $d^2W = (2\pi)^{-1/2}3^2 5^{5/2} y(y^2 - z^2)dydz$. For peaks and voids, since $\lambda_1 \geqslant \lambda_2 \geqslant \lambda_3 \geqslant 0$, f_{00} is given by

$$f_{00}^{\text{peak/void}}(x) = \frac{3^2 5^{5/2}}{\sqrt{2\pi}}\left(\int_0^{x/4} dy \int_{-y}^{y} dz + \int_{x/4}^{x/2} dy \int_{3y-x}^{y} dz\right) e^{-5(3y^2+z^2)/2}(x-2z)\left[(x+z)^2 - (3y)^2\right] y(y^2 - z^2).$$

For filaments, $\lambda_1 \geqslant \lambda_2 \geqslant 0 \geqslant \lambda_3$. The integrand is the same, but the domain is

$$\int_0^\infty dx \int_{x/4}^{x/2} dy \int_{-y}^{3y-x} dz + \int_0^\infty dx \int_{x/2}^{\infty} dy \int_{-y}^{x/2} dz + \int_{-\infty}^0 dx \int_{-x/2}^{\infty} dy \int_{-y}^{x/2} dz.$$

For walls, the domain is

$$\int_0^\infty dx \int_{x/2}^{\infty} dy \int_{x/2}^{y} dz + \int_{-\infty}^0 dx \int_{-x/2}^{\infty} dy \int_{x/2}^{y} dz + \int_{-\infty}^0 dx \int_{-x/4}^{-x/2} dy \int_{-x-3y}^{y} dz.$$

The corresponding final expressions for filaments, walls, and the Euler characteristic are given in Appendix A.4.

3.3 Non-Gaussian leading-order correction

In the previous section, we discussed critical-point statistics in Gaussian fields. In this section, we consider weakly non-Gaussian fields. The main change is that the probability density function (PDF) is now non-Gaussian and can be expanded around the Gaussian PDF using generalized Wiener–Hermite functionals.

3.3.1 Generalized Wiener–Hermite functionals expansion

We derive the weakly non-Gaussian (NG) field by expanding around the Gaussian field with Wiener–Hermite functionals. Reference [43] gives the configuration-space derivation; here we follow Ref. [9] and work in Fourier space. The Fourier transform of a field $f(\boldsymbol{x})$ is

$$\tilde{f}(\boldsymbol{k}) = \int d^N x\, f(\boldsymbol{x}) e^{-i\boldsymbol{k}\cdot\boldsymbol{x}}, \qquad f(\boldsymbol{x}) = \int \frac{d^N k}{(2\pi)^N} \tilde{f}(\boldsymbol{k}) e^{i\boldsymbol{k}\cdot\boldsymbol{x}}.$$

The moment-generating functional, or partition function, is

$$\mathcal{Z}[J] = \int \mathcal{D}\tilde{f} \exp\left(i \int \frac{d^N k}{(2\pi)^N} J(\boldsymbol{k})\tilde{f}(\boldsymbol{k})\right)\mathcal{P}[\tilde{f}],$$

where $\mathcal{D}\tilde{f}$ is the volume element of the functional integral over $\tilde{f}(\boldsymbol{k})$, with the appropriate measure. The cumulant-expansion theorem in Eq. (3.4) then gives[2]

$$\ln \mathcal{Z}[J] = \sum_{n=1}^{\infty} \frac{i^n}{n!} \int_{\boldsymbol{k}_1 \cdots \boldsymbol{k}_n} \langle \tilde{f}(\boldsymbol{k}_1) \cdots \tilde{f}(\boldsymbol{k}_n) \rangle_{\mathrm{c}} J(\boldsymbol{k}_1) \cdots J(\boldsymbol{k}_n).$$

By functional differentiation, the nth-order cumulant in Fourier space is

$$\mu^{(n)}(\boldsymbol{k}_1, \ldots, \boldsymbol{k}_n) = \langle \tilde{f}(\boldsymbol{k}_1) \cdots \tilde{f}(\boldsymbol{k}_n) \rangle_{\mathrm{c}} = (-i)^n \left. \frac{\delta^n \ln \mathcal{Z}[J]}{\delta J(\boldsymbol{k}_1) \cdots \delta J(\boldsymbol{k}_n)} \right|_{J=0}.$$

Carrying out the $n = 2$ term, we have

$$Z[J] = \exp\left[-\frac{1}{2} \int_{\boldsymbol{k}} P(k) J(\boldsymbol{k}) J(-\boldsymbol{k})\right] \exp\left[\sum_{n=3}^{\infty} \frac{i^n}{n!} \int_{\boldsymbol{k}_1 \cdots \boldsymbol{k}_n} \mu^{(n)}(\boldsymbol{k}_1, \ldots, \boldsymbol{k}_n) J(\boldsymbol{k}_1) \cdots J(\boldsymbol{k}_n)\right],$$

where $P(k)$ is the power spectrum. By inverting the functional integral, we obtain the probability distribution

$$\begin{aligned} \mathcal{P}[\tilde{f}] &= \int \bar{\mathcal{D}} J Z[J] \exp\left[-i \int_{\boldsymbol{k}} J(\boldsymbol{k}) \tilde{f}(\boldsymbol{k})\right] \\ &= \exp\left[\sum_{n=3}^{\infty} \frac{(-1)^n}{n!} \int_{\boldsymbol{k}_1 \cdots \boldsymbol{k}_n} \frac{\delta}{\delta \tilde{f}(\boldsymbol{k}_1)} \cdots \frac{\delta}{\delta \tilde{f}(\boldsymbol{k}_n)}\right] \mathcal{P}_{\mathrm{G}}[\tilde{f}]. \end{aligned} \tag{3.21}$$

The Gaussian functional is

$$\begin{aligned} \mathcal{P}_{\mathrm{G}}[\tilde{f}] &= \int \bar{\mathcal{D}} J \exp\left[-\frac{1}{2} \int \frac{d^N k}{(2\pi)^N} P(k) J(\boldsymbol{k}) J(-\boldsymbol{k}) - i \int \frac{d^N k}{(2\pi)^N} J(\boldsymbol{k}) \tilde{f}(\boldsymbol{k})\right] \\ &\propto \exp\left[-\frac{1}{2} \int \frac{d^N k}{(2\pi)^N} \frac{\tilde{f}(\boldsymbol{k}) \tilde{f}(-\boldsymbol{k})}{P(k)}\right]. \end{aligned}$$

We define the Wiener–Hermite polynomials by

$$\mathcal{H}_n(\boldsymbol{k}_1, \ldots, \boldsymbol{k}_n) \equiv \frac{(-1)^n}{\mathcal{P}_{\mathrm{G}}[\tilde{f}]} \frac{\delta^n \mathcal{P}_{\mathrm{G}}[\tilde{f}]}{\delta \tilde{f}(\boldsymbol{k}_1) \cdots \delta \tilde{f}(\boldsymbol{k}_n)}, \tag{3.22}$$

with $\mathcal{H}_0 = 1$. We then define the dual functionals by

$$\mathcal{H}_n^*(\boldsymbol{k}_1, \ldots, \boldsymbol{k}_n) \equiv (2\pi)^{Nn} P(k_1) \cdots P(k_n) \mathcal{H}_n(-\boldsymbol{k}_1, \ldots, -\boldsymbol{k}_n).$$

Some low-order examples follow from $\delta \tilde{f}(\boldsymbol{k}_1) / \delta \tilde{f}(\boldsymbol{k}_2) = \delta^N(\boldsymbol{k}_1 - \boldsymbol{k}_2)$:

$$\begin{aligned} \mathcal{H}_1(\boldsymbol{k}_1) &= \frac{1}{2} \frac{\delta}{\delta \tilde{f}(\boldsymbol{k}_1)} \int \frac{d^N k}{(2\pi)^N} \frac{\tilde{f}(\boldsymbol{k}) \tilde{f}(-\boldsymbol{k})}{P(k)} \\ &= \frac{1}{2} \int \frac{d^N k}{(2\pi)^N} \frac{1}{P(k)} \left[\tilde{f}(-\boldsymbol{k}) \delta^N(\boldsymbol{k} - \boldsymbol{k}_1) + \tilde{f}(\boldsymbol{k}) \delta^N(-\boldsymbol{k} - \boldsymbol{k}_1)\right] = \frac{\tilde{f}(-\boldsymbol{k}_1)}{(2\pi)^N P(k_1)}. \end{aligned}$$

We denote $\frac{1}{P} \frac{\delta P}{\delta f_i} = (\ln P)_{,i} = -\mathcal{H}_1(\boldsymbol{k}_i)$. Then, for $n = 2$, $\frac{1}{P} \frac{\delta^2 P}{\delta f_i \delta f_j} = (\ln P)_{,i} (\ln P)_{,j} + (\ln P)_{,ij} = \mathcal{H}_2(\boldsymbol{k}_i, \boldsymbol{k}_j)$, so

$$\mathcal{H}_2(\boldsymbol{k}_1, \boldsymbol{k}_2) = \frac{\tilde{f}(-\boldsymbol{k}_1) \tilde{f}(-\boldsymbol{k}_2)}{(2\pi)^{2N} P(k_1) P(k_2)} - \frac{\delta^N(\boldsymbol{k}_1 + \boldsymbol{k}_2)}{(2\pi)^N P(k_1)}.$$

For $n = 3$,

$$\mathcal{H}_3(\boldsymbol{k}_1, \boldsymbol{k}_2, \boldsymbol{k}_3) = \frac{\tilde{f}(-\boldsymbol{k}_1) \tilde{f}(-\boldsymbol{k}_2) \tilde{f}(-\boldsymbol{k}_3)}{(2\pi)^{3N} P(k_1) P(k_2) P(k_3)} - \left[\frac{\tilde{f}(-\boldsymbol{k}_1) \delta^N(\boldsymbol{k}_2 + \boldsymbol{k}_3)}{(2\pi)^{2N} P(k_1) P(k_2)} + \text{cyc.}\right].$$

[2]We use $\int \frac{d^N \boldsymbol{k}_n}{(2\pi)^N} \equiv \int_{\boldsymbol{k}_n}$ below.

Therefore,

$$
\begin{aligned}
&\mathcal{H}_0^* = 1,\\
&\mathcal{H}_1^*(\boldsymbol{k}) = \tilde{f}(\boldsymbol{k}),\\
&\mathcal{H}_2^*(\boldsymbol{k}_1, \boldsymbol{k}_2) = \tilde{f}(\boldsymbol{k}_1)\tilde{f}(\boldsymbol{k}_2) - (2\pi)^N \delta^N(\boldsymbol{k}_1 + \boldsymbol{k}_2) P(k_1),\\
&\mathcal{H}_3^*(\boldsymbol{k}_1, \boldsymbol{k}_2, \boldsymbol{k}_3) = \tilde{f}(\boldsymbol{k}_1)\tilde{f}(\boldsymbol{k}_2)\tilde{f}(\boldsymbol{k}_3) - \Big[(2\pi)^N \delta^N(\boldsymbol{k}_2 + \boldsymbol{k}_3)\tilde{f}(\boldsymbol{k}_1) P(k_2) + \text{cyc.}\Big].
\end{aligned}
$$

We then have the orthogonality relation

$$
\langle \mathcal{H}_n^*(\boldsymbol{k}_1, \ldots, \boldsymbol{k}_n)\mathcal{H}_m(\boldsymbol{k}'_1, \ldots, \boldsymbol{k}'_m)\rangle_{\mathrm{G}} = \delta_{nm}[\delta^N(\boldsymbol{k}_1 - \boldsymbol{k}'_1)\cdots\delta^N(\boldsymbol{k}_n - \boldsymbol{k}'_n) + \text{permutations}]. \tag{3.23}
$$

Here "permutations" means all possible permutations of $\boldsymbol{k}'_1, \ldots, \boldsymbol{k}'_m$; the proof is omitted here. Using Eq. (3.21) and comparing with the Edgeworth expansion, the σ expansion to second order gives

$$
\begin{aligned}
\mathcal{P} =& \mathcal{H}_0\mathcal{P}_{\mathrm{G}} + \frac{1}{6}\int d^N k_1 d^N k_2 d^N k_3\, \mu^{(3)}(\boldsymbol{k}_1, \boldsymbol{k}_2, \boldsymbol{k}_3)\mathcal{H}_3(\boldsymbol{k}_1, \boldsymbol{k}_2, \boldsymbol{k}_3)\mathcal{P}_{\mathrm{G}}\\
&+ \frac{1}{72}\int d^N k_1 \cdots d^N k_6\, \mu^{(3)}(\boldsymbol{k}_1, \boldsymbol{k}_2, \boldsymbol{k}_3)\mu^{(3)}(\boldsymbol{k}_4, \boldsymbol{k}_5, \boldsymbol{k}_6)\mathcal{H}_6(\boldsymbol{k}_1, \ldots, \boldsymbol{k}_6)\mathcal{P}_{\mathrm{G}}\\
&+ \frac{1}{24}\int d^N k_1 \cdots d^N k_4\, \mu^{(4)}(\boldsymbol{k}_1, \boldsymbol{k}_2, \boldsymbol{k}_3, \boldsymbol{k}_4)\mathcal{H}_4(\boldsymbol{k}_1, \boldsymbol{k}_2, \boldsymbol{k}_3, \boldsymbol{k}_4)\mathcal{P}_{\mathrm{G}}.
\end{aligned}
$$

Equation (3.23) gives the orthogonality of the generalized Wiener–Hermite functionals. Therefore, any functional $\mathcal{F}(\boldsymbol{x})$ of the random field f can be expanded in this basis in Fourier space:

$$
\mathcal{F}(\boldsymbol{k}) = \sum_{n=0}^{\infty}\frac{1}{n!}\int d^N k_1 \cdots d^N k_n\, \mathcal{F}^{(n)}(\boldsymbol{k}_1, \ldots, \boldsymbol{k}_n, \boldsymbol{k})\mathcal{H}_n^*(\boldsymbol{k}_1, \ldots, \boldsymbol{k}_n).
$$

The Gaussian measure defines the orthogonality. Once a complete orthogonal decomposition is available, the coefficients follow by projection:

$$
\begin{aligned}
\langle \mathcal{F}(\boldsymbol{k})\mathcal{H}_m(\boldsymbol{k}_1, \ldots, \boldsymbol{k}_m)\rangle_{\mathrm{G}} &= \sum_{n=0}^{\infty}\frac{1}{n!}\int d^N k'_1 \cdots d^N k'_n\, \mathcal{F}^{(n)}(\boldsymbol{k}'_1, \ldots, \boldsymbol{k}'_n, \boldsymbol{k})\\
&\qquad\qquad \times \langle \mathcal{H}_n^*(\boldsymbol{k}'_1, \ldots, \boldsymbol{k}'_n)\mathcal{H}_m(\boldsymbol{k}_1, \ldots, \boldsymbol{k}_m)\rangle_{\mathrm{G}}\\
&= \mathcal{F}^{(m)}(\boldsymbol{k}_1, \ldots, \boldsymbol{k}_m, \boldsymbol{k}).
\end{aligned}
$$

This can be rewritten as

$$
\mathcal{F}(\boldsymbol{k}) = \sum_{n=0}^{\infty}\frac{1}{n!}\int_{\boldsymbol{k}_1,\ldots,\boldsymbol{k}_n}(2\pi)^N\delta^N(\boldsymbol{k}_1 + \cdots + \boldsymbol{k}_n - \boldsymbol{k})\mathcal{G}^{(n)}(\boldsymbol{k}_1, \ldots, \boldsymbol{k}_n)\mathcal{H}_n^*(\boldsymbol{k}_1, \ldots, \boldsymbol{k}_n),
$$

where the delta function is a consequence of translational invariance. Using the orthogonality relation in Eq. (3.23), the coefficient functions $\mathcal{G}_n$ are then given by

$$
\begin{aligned}
(2\pi)^N\delta^N(\boldsymbol{k}_1 + \cdots + \boldsymbol{k}_n - \boldsymbol{k})\mathcal{G}^{(n)}(\boldsymbol{k}_1, \ldots, \boldsymbol{k}_n) &= (2\pi)^{Nn}\langle \mathcal{F}(\boldsymbol{k})\mathcal{H}_n(\boldsymbol{k}_1, \ldots, \boldsymbol{k}_n)\rangle_{\mathrm{G}}\\
&= (2\pi)^{Nn}\langle \frac{\delta^n\tilde{\mathcal{F}}(\boldsymbol{k})}{\delta\tilde{f}(\boldsymbol{k}_1)\cdots\delta\tilde{f}(\boldsymbol{k}_n)}\rangle_{\mathrm{G}}.
\end{aligned}
$$

The last equality uses integration by parts in the functional integral. Fourier transforming back to real space gives

$$
\mathcal{G}_n(\boldsymbol{k}_1, \ldots, \boldsymbol{k}_n) = (2\pi)^{Nn}e^{i(\boldsymbol{k}_1+\cdots+\boldsymbol{k}_n)\cdot\boldsymbol{x}}\langle \frac{\delta^n\tilde{\mathcal{F}}(\boldsymbol{x})}{\delta\tilde{f}(\boldsymbol{k}_1)\cdots\delta f(\boldsymbol{k}_n)}\rangle_{\mathrm{G}} = (2\pi)^{Nn}\langle \frac{\delta^n\mathcal{F}(\boldsymbol{x})}{\delta f(\boldsymbol{x}_1)\cdots\delta f(\boldsymbol{x}_n)}\rangle_{\mathrm{G}}.
$$

Since the right-hand side is independent of $\boldsymbol{x}$, we can set $\boldsymbol{x} = 0$ for convenience. The expectation value of $\mathcal{F}$ is

$$
\begin{aligned}
\langle \mathcal{F} \rangle = \int \mathcal{D}\mathcal{F}[f]\mathcal{P} = & \mathcal{G}_0 + \frac{1}{6}\int_{\boldsymbol{k}_1,\boldsymbol{k}_2,\boldsymbol{k}_3} \mu^{(3)}(\boldsymbol{k}_1,\boldsymbol{k}_2,\boldsymbol{k}_3)\mathcal{G}_3(\boldsymbol{k}_1,\boldsymbol{k}_2,\boldsymbol{k}_3) \\
& + \frac{1}{24}\int_{\boldsymbol{k}_1,\boldsymbol{k}_2,\boldsymbol{k}_3,\boldsymbol{k}_4} \mu^{(4)}(\boldsymbol{k}_1,\boldsymbol{k}_2,\boldsymbol{k}_3,\boldsymbol{k}_4)\mathcal{G}_4(\boldsymbol{k}_1,\boldsymbol{k}_2,\boldsymbol{k}_3,\boldsymbol{k}_4) \\
& + \frac{1}{72}\int_{\boldsymbol{k}_1,\boldsymbol{k}_2,\boldsymbol{k}_3,\boldsymbol{k}_4,\boldsymbol{k}_5,\boldsymbol{k}_6} \mu^{(3)}(\boldsymbol{k}_1,\boldsymbol{k}_2,\boldsymbol{k}_3)\mu^{(3)}(\boldsymbol{k}_4,\boldsymbol{k}_5,\boldsymbol{k}_6)\mathcal{G}_6(\boldsymbol{k}_1,\ldots,\boldsymbol{k}_6).
\end{aligned}
$$

In this thesis, we keep only the first line as the leading-order correction. In Appendix B, we discuss an attempt to include the second-line contribution.

3.3.2 Gaussian response functions and generalized skewness parameters

We develop the Mathematica code to compute the Gaussian response functions $\mathcal{G}_n$ for the critical-point counts in 2D and 3D.

The response function is defined by

$$
\mathcal{G}_n(\boldsymbol{k}_1,\ldots,\boldsymbol{k}_n) = (2\pi)^{Nn}\langle \frac{\delta^n n^\star}{\delta \tilde{f}(\boldsymbol{k}_1)\cdots\delta \tilde{f}(\boldsymbol{k}_n)}\rangle_{\mathrm{G}}.
$$

The functional derivative can be calculated by the chain rule:

$$
(2\pi)^N \frac{\delta}{\delta \tilde{f}(\boldsymbol{k})} \to (2\pi)^N \left[\frac{\delta\alpha}{\delta \tilde{f}(\boldsymbol{k})}\frac{\partial}{\partial\alpha} + \frac{\delta\eta_i}{\delta \tilde{f}(\boldsymbol{k})}\frac{\partial}{\partial\eta_i} + \frac{\delta\zeta_{ij}}{\delta \tilde{f}(\boldsymbol{k})}\frac{\partial}{\partial\zeta_{ij}}\right].
$$

For example,

$$
(2\pi)^N \frac{\delta\alpha}{\delta \tilde{f}(\boldsymbol{k})} = (2\pi)^N \frac{1}{\sigma_0}\frac{\delta \int_{\boldsymbol{k}'} e^{i\boldsymbol{k}'\cdot\boldsymbol{x}}\tilde{f}(\boldsymbol{k}')}{\delta \tilde{f}(\boldsymbol{k})} = \frac{1}{\sigma_0}e^{i\boldsymbol{k}\cdot\boldsymbol{x}}.
$$

Similarly, we can set $\boldsymbol{x} = 0$. Define

$$
\hat{\mathcal{D}}(\boldsymbol{k}) \equiv \frac{1}{\sigma_0}\frac{\partial}{\partial\alpha} + \frac{ik_i}{\sigma_1}\frac{\partial}{\partial\eta_i} - \frac{k_i k_j}{\sigma_2}\frac{\partial}{\partial\zeta_{ij}}.
$$

Thus,

$$
\mathcal{G}_n(\boldsymbol{k}_1,\ldots,\boldsymbol{k}_n) = (-1)^n \int d^{N_Y}Y n^\star \hat{\mathcal{D}}(\boldsymbol{k}_1)\cdots\hat{\mathcal{D}}(\boldsymbol{k}_n)\mathcal{P}_{\mathrm{G}}(\mathbf{Y}).
$$

To calculate the derivatives, we need the following relations:

$$
\begin{aligned}
&\frac{\partial\eta^2}{\partial\eta_i} = 2\eta_i, \qquad \frac{\partial J_1}{\partial\zeta_{ij}} = -\delta_{ij}, \qquad \frac{\partial\tilde{\zeta}_{kl}}{\partial\zeta_{ij}} = \delta_{ik}\delta_{jl} - \frac{1}{N}\delta_{ij}\delta_{kl}, \\
&\frac{\partial J_2}{\partial\zeta_{ij}} = \frac{N}{N-1}\frac{\partial(\tilde{\zeta}_{kl}\tilde{\zeta}_{lk})}{\partial\zeta_{ij}} = \frac{N}{N-1}\frac{\partial\tilde{\zeta}_{ij}}{\partial\zeta_{ij}}\frac{\partial(\tilde{\zeta}_{kl}\tilde{\zeta}_{lk})}{\partial\tilde{\zeta}_{ij}} = \frac{2N}{N-1}\tilde{\zeta}_{ji}.
\end{aligned}
$$

Note that $\partial\tilde{\zeta}_{ij}\big/\partial\zeta_{ij} = 1$ $(i \neq j)$. Then we can calculate the derivatives of the Gaussian PDF:

$$
\begin{aligned}
&\frac{\partial}{\partial\eta_i}\mathcal{P}_{\mathrm{G}} = 2\eta_i\frac{\partial}{\partial(\eta^2)}\mathcal{P}_{\mathrm{G}}, \qquad \frac{\partial^2}{\partial\eta_i\partial\eta_j}\mathcal{P}_{\mathrm{G}} = 2\left[\delta_{ij}\frac{\partial}{\partial(\eta^2)} + 2\eta_i\eta_j\frac{\partial^2}{\partial(\eta^2)^2}\right]\mathcal{P}_{\mathrm{G}}, \\
&\frac{\partial}{\partial\zeta_{ij}}\mathcal{P}_{\mathrm{G}} = \left[-\delta_{ij}\frac{\partial}{\partial J_1} + \frac{2N}{N-1}\tilde{\zeta}_{ji}\frac{\partial}{\partial J_2}\right]\mathcal{P}_{\mathrm{G}}, \\
&\frac{\partial^2}{\partial\zeta_{ij}\partial\zeta_{kl}}\mathcal{P}_{\mathrm{G}} = \left[\delta_{ij}\delta_{kl}\frac{\partial^2}{\partial J_1^2} - \frac{2N}{N-1}(\delta_{ij}\tilde{\zeta}_{lk} + \delta_{kl}\tilde{\zeta}_{ji})\frac{\partial^2}{\partial J_1\partial J_2} + \frac{4N^2}{(N-1)^2}\tilde{\zeta}_{ji}\tilde{\zeta}_{kl}\frac{\partial^2}{\partial J_2^2}\right. \\
&\qquad\qquad\qquad \left. + \frac{2N}{N-1}(\delta_{jk}\delta_{il} - \frac{1}{N}\delta_{ij}\delta_{kl})\frac{\partial}{\partial J_2}\right]\mathcal{P}_{\mathrm{G}}.
\end{aligned}
$$

For angular averages of second moments,

$$\langle \eta_i \eta_j \rangle_\Omega = \frac{1}{N}\delta_{ij}\eta^2, \quad \langle \tilde{\zeta}_{ij}\tilde{\zeta}_{kl}\rangle_\Omega = \frac{J_2}{N(N+2)}\left(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk} - \frac{2}{N}\delta_{ij}\delta_{kl}\right).$$

Together with $\langle \eta_i \rangle_\Omega = 0$ and $\langle \tilde{\zeta}_{ij}\rangle_\Omega = 0$, these averages give

$$\begin{aligned}\langle \mathcal{D}(\boldsymbol{k})\mathcal{P}_\mathrm{G}\rangle_\Omega =& \langle \frac{1}{\sigma_0}\frac{\partial}{\partial\alpha} + \frac{ik_i}{\sigma_1}2\eta_i\frac{\partial}{\partial(\eta^2)} - \frac{k_ik_j}{\sigma_2}(-\delta_{ij}\frac{\partial}{\partial J_1} + \frac{2N}{N-1}\tilde{\zeta}_{ji}\frac{\partial}{\partial J_2})\rangle_\Omega \\ =& \left(\frac{1}{\sigma_0}\frac{\partial}{\partial\alpha} + \frac{k^2}{\sigma_2}\frac{\partial}{\partial J_1}\right)\mathcal{P}_\mathrm{G}.\end{aligned}$$

$$\begin{aligned}\langle \mathcal{D}(\boldsymbol{k}_1)\mathcal{D}(\boldsymbol{k}_2)\mathcal{P}_\mathrm{G}\rangle_\Omega =& \left(\frac{1}{\sigma_0}\frac{\partial}{\partial\alpha} + \frac{ik_{1i}}{\sigma_1}\frac{\partial}{\partial\eta_i} - \frac{k_{1i}k_{1j}}{\sigma_2}\frac{\partial}{\partial\zeta_{ij}}\right)\left(\frac{1}{\sigma_0}\frac{\partial}{\partial\alpha} + \frac{ik_{2k}}{\sigma_1}\frac{\partial}{\partial\eta_i} - \frac{k_{2k}k_{2l}}{\sigma_2}\frac{\partial}{\partial\zeta_{kl}}\right)\mathcal{P}_\mathrm{G} \\ =& \{(\frac{1}{\sigma_0}\frac{\partial}{\partial\alpha} + \frac{k_1^2}{\sigma_2}\frac{\partial}{\partial J_1})(\frac{1}{\sigma_0}\frac{\partial}{\partial\alpha} + \frac{k_2^2}{\sigma_2}\frac{\partial}{\partial J_1}) - \frac{2(\boldsymbol{k}_1\cdot\boldsymbol{k}_2)}{\sigma_1^2}\left[1 + \frac{2}{N}\eta^2\frac{\partial}{\partial(\eta^2)}\right]\frac{\partial}{\partial(\eta^2)} \\ &+ \frac{2N}{(N-1)\sigma_2^2}\left[(\boldsymbol{k}_1\cdot\boldsymbol{k}_2)^2 - \frac{1}{N}k_1^2k_2^2\right]\left[1 + \frac{4J_2}{(N-1)(N+2)}\frac{\partial}{\partial J_2}\right]\frac{\partial}{\partial J_2}\}\mathcal{P}_\mathrm{G}.\end{aligned}$$

Since $\mathcal{P}_\mathrm{G}$ includes $\exp\left(-\frac{N}{2}\eta^2\right)$ and $\exp\left(-\frac{(N-1)(N+2)}{4}J_2\right)$, the differential operators $\partial/\partial(\eta^2)$ and $\partial/\partial J_2$ can be evaluated directly.

$$\left[1 + \frac{2}{N}\eta^2\frac{\partial}{\partial(\eta^2)}\right]\frac{\partial}{\partial(\eta^2)}\mathcal{P}_\mathrm{G} = \frac{N}{2}(\eta^2 - 1)\mathcal{P}_\mathrm{G} = -L_1^{(\frac{N}{2}-1)}\left(\frac{N}{2}\eta^2\right)\mathcal{P}_\mathrm{G}.$$

Here the generalized Laguerre polynomials are defined by

$$L_k^{(a)}(x) = \frac{x^{-a}e^x}{k!}\frac{d^k}{dx^k}(e^{-x}x^{k+a}).$$

For instance,

$$L_0^a(x) = 1, \qquad L_1^{(a)}(x) = -x + a + 1, \qquad L_2^{(a)}(x) = \frac{1}{2}(x^2 - 2(a+2)x + (a+1)(a+2)).$$

Similarly,

$$\left[1 + \frac{4J_2}{(N-1)(N+2)}\frac{\partial}{\partial J_2}\right]\frac{\partial}{\partial J_2}\mathcal{P}_\mathrm{G} = \frac{(N-1)(N+2)}{4}(J_2 - 1)\mathcal{P}_\mathrm{G} = F_{10}(J_2, J_3)\mathcal{P}_\mathrm{G}.$$

We define

$$F_{lm}(J_2, J_3) \equiv (-1)^l J_2^{3m/2} L_l^{(3m+(N-2)(N+3)/4)}\left(\frac{(N-1)(N+2)}{4}J_2\right)P_m\left(\frac{J_3}{J_2^{3/2}}\right).$$

Here the Legendre polynomials are defined by

$$P_m(x) \equiv \frac{1}{2^m m!}\frac{d^m}{dx^m}(x^2-1)^m.$$

For instance, if $N = 3$,

$$F_{00}(J_2, J_3) = 1, \qquad F_{10}(J_2, J_3) = \frac{5}{2}J_2 - \frac{5}{2}, \qquad F_{01}(J_2, J_3) = J_3.$$

We need to calculate the new $\mathcal{G}$ after angular averaging:

$$\mathcal{G}_n(\boldsymbol{k}_1, \ldots, \boldsymbol{k}_n) = (-1)^n \int d^{Nz}Z\, n^\star \langle \mathcal{D}(\boldsymbol{k}_1)\cdots\mathcal{D}(\boldsymbol{k}_n)\mathcal{P}_\mathrm{G}\rangle_\Omega\,.$$

The new variables are $\mathbf{Z} = (\alpha, \eta^2, J_1, J_2, J_3)$. Define

$$G_{ijklm}(\nu) \equiv (-1)^k \left\langle n^\star H_{ij}(\alpha, J_1) L_k^{(N/2-1)}\left(\frac{N}{2}\eta^2\right) F_{lm}(J_2, J_3)\right\rangle_{\rm G}.$$

The bivariate Hermite polynomials are

$$H_{ij}(\alpha, J_1) = \frac{1}{\mathcal{N}(\alpha, J_1)}\left(-\frac{\partial}{\partial\alpha}\right)^i\left(-\frac{\partial}{\partial J_1}\right)^j \mathcal{N}(\alpha, J_1).$$

This gives

$$\begin{aligned}
&\mathcal{G}_0 = \int d^{Nz} Z\, n^\star \mathcal{P}_{\rm G} = G_{00000}(\nu),\\
&\mathcal{G}_1(\boldsymbol{k}) = \int d^{Nz} Z\, n^\star \left(\frac{1}{\sigma_0}\frac{\partial}{\partial\alpha} + \frac{k^2}{\sigma_2}\frac{\partial}{\partial J_1}\right)\mathcal{P}_{\rm G} = \frac{G_{10000}}{\sigma_0} + \frac{G_{01000}}{\sigma_2}k^2,\\
&\mathcal{G}_2(\boldsymbol{k}_1, \boldsymbol{k}_2) = \frac{G_{20000}}{\sigma_0^2} + \frac{G_{11000}}{\sigma_0\sigma_2}({k_1}^2 + {k_2}^2) + \frac{G_{02000}}{\sigma_2^2}{k_1}^2{k_2}^2 - \frac{2G_{00100}}{\sigma_1^2}(\boldsymbol{k}_1\cdot\boldsymbol{k}_2) + \frac{2NG_{00010}}{(N-1)\sigma_2^2}\left[(\boldsymbol{k}_1\cdot\boldsymbol{k}_2)^2 - \frac{1}{N}{k_1}^2{k_2}^2\right].
\end{aligned}$$

We have shown the expressions only up to second order. For the leading-order correction, the expansion to $\mathcal{G}_3$ is necessary. We give the final result here; see the Mathematica notebook for the detailed calculation.

$$\begin{aligned}
\mathcal{G}_3(\boldsymbol{k}_1, \boldsymbol{k}_2, \boldsymbol{k}_3) = {}& \frac{G_{30000}}{\sigma_0^3} + \frac{G_{21000}}{\sigma_0^2\sigma_2}({k_1}^2 + {k_2}^2 + {k_3}^2) + \frac{G_{12000}}{\sigma_0\sigma_2^2}({k_1}^2{k_2}^2 + {k_2}^2{k_3}^2 + {k_3}^2{k_1}^2) + \frac{G_{03000}}{\sigma_2^3}{k_1}^2{k_2}^2{k_3}^2\\
&- \frac{2G_{10100}}{\sigma_0\sigma_1^2}[(\boldsymbol{k}_1\cdot\boldsymbol{k}_2) + (\boldsymbol{k}_2\cdot\boldsymbol{k}_3) + (\boldsymbol{k}_3\cdot\boldsymbol{k}_1)] - \frac{2G_{01100}}{\sigma_1^2\sigma_2}[(\boldsymbol{k}_1\cdot\boldsymbol{k}_2){k_3}^2 + (\boldsymbol{k}_2\cdot\boldsymbol{k}_3){k_1}^2 + (\boldsymbol{k}_3\cdot\boldsymbol{k}_1){k_2}^2]\\
&+ \frac{2NG_{10010}}{(N-1)\sigma_0\sigma_2^2}\left[[(\boldsymbol{k}_1\cdot\boldsymbol{k}_2)^2 - \frac{1}{N}{k_1}^2{k_2}^2] + \text{cyc.}\right] + \frac{2NG_{01010}}{(N-1)\sigma_2^3}\left[(\boldsymbol{k}_1\cdot\boldsymbol{k}_2)^2{k_3}^2 + \text{cyc.} - \frac{3}{N}{k_1}^2{k_2}^2k_3^3\right]\\
&- \frac{N^2(N+2)^2G_{00001}}{(N+4)\sigma_2^3}\left[(\boldsymbol{k}_1\cdot\boldsymbol{k}_2)(\boldsymbol{k}_2\cdot\boldsymbol{k}_3)(\boldsymbol{k}_3\cdot\boldsymbol{k}_1) - \frac{(\boldsymbol{k}_1\cdot\boldsymbol{k}_2)^2{k_3}^2 + \text{cyc.}}{N} + \frac{2}{N^2}{k_1}^2{k_2}^2{k_3}^2\right].
\end{aligned}$$

Next, before evaluating the details of G_{ijklm}, we need to perform the integrals over the $\boldsymbol{k}$ modes. Note that

$$\begin{aligned}
&\delta(\boldsymbol{x}) = \int \frac{d^N k}{(2\pi)^N} e^{i\boldsymbol{k}\cdot\boldsymbol{x}}, \qquad \tilde{f}_1 = \tilde{f}(\boldsymbol{k}_1) = \int d^N x e^{-i\boldsymbol{k}_1\boldsymbol{x}} f(\boldsymbol{x}), \quad {k_1}^2\tilde{f}_1 = \int d^N x e^{-i\boldsymbol{k}_1\boldsymbol{x}}(-\Delta f(\boldsymbol{x}))\\
&(\boldsymbol{k}_1\cdot\boldsymbol{k}_2)\tilde{f}_1\tilde{f}_2 = -\int d^N x_1 \int d^N x_2 e^{-i\boldsymbol{k}_1\boldsymbol{x}_1} e^{-i\boldsymbol{k}_2\cdot\boldsymbol{x}_2}(\nabla f(\boldsymbol{x}_1)\cdot\nabla f(\boldsymbol{x}_2)).
\end{aligned}$$

These properties are used in the derivation. There are nine terms in total in the correction to $\bar{n}^\star(\nu)$; here we show two for reference:

$$\begin{aligned}
G_{10100}: \qquad & \int_{\boldsymbol{k}_1,\boldsymbol{k}_2,\boldsymbol{k}_3} \langle\tilde{f}_1\tilde{f}_2\tilde{f}_3\rangle \left[-\frac{2}{\sigma_0\sigma_1^2}[(\boldsymbol{k}_1\cdot\boldsymbol{k}_2) + (\boldsymbol{k}_2\cdot\boldsymbol{k}_3) + (\boldsymbol{k}_1\cdot\boldsymbol{k}_3)]\right] = -\frac{2}{\sigma_0\sigma_1^2}\frac{3}{2}\langle f^2\Delta f\rangle = \sigma_0\cdot 4S^{(1)}\\
G_{01010}: \qquad & \int_{\boldsymbol{k}_1,\boldsymbol{k}_2,\boldsymbol{k}_3} \langle\tilde{f}_1\tilde{f}_2\tilde{f}_3\rangle \frac{2N}{(N-1)\sigma_2^3}\left[(\boldsymbol{k}_1\cdot\boldsymbol{k}_2)^2{k_3}^2 + \text{cyc.} - \frac{3}{N}{k_1}^2{k_2}^2{k_3}^2\right]\\
&= \frac{2N}{(N-1)\sigma_2^3}\left(3(-\langle f_{ij}f_{ij}\Delta f\rangle) - \frac{3}{N}(-\langle(\Delta f)^3\rangle)\right) = \sigma_0\cdot\frac{6}{N-1}\gamma^3(NS_2^{(3)} - S_1^{(3)}).
\end{aligned}$$

Here we have defined the skewness parameters that will be used below:

$$\begin{aligned}
&S^{(0)} \equiv \frac{\langle f^3\rangle_{\rm c}}{\sigma_0^4}, \qquad S^{(1)} \equiv -\frac{3}{4}\frac{\langle f^2\Delta f\rangle_{\rm c}}{\sigma_0^2\sigma_1^2}, \qquad S^{(2)} \equiv -\frac{3N}{2(N-1)}\frac{\langle(\nabla f\cdot\nabla f)\Delta f\rangle_{\rm c}}{\sigma_1^4},\\
&S_2^{(2)} \equiv \frac{\langle f(\Delta f)^2\rangle_{\rm c}}{\sigma_1^4}, \qquad S_1^{(3)} \equiv -\frac{\sigma_0^2}{\sigma_1^6}\langle(\Delta f)^3\rangle_{\rm c}, \qquad S_2^{(3)} \equiv -\frac{\sigma_0^2}{\sigma_1^6}\langle f_{ij}f_{ij}\Delta f\rangle_{\rm c}.
\end{aligned} \tag{3.24}$$

We also use

$$\langle f\nabla f\cdot\nabla f\rangle_{\rm c}=\frac{2}{3}\sigma_0^2\sigma_1^2S^{(1)},\quad \langle ff_{ij}f_{ij}\rangle_{\rm c}=\sigma_1^4\left(S_2^{(2)}-\frac{N-1}{N}S^{(2)}\right)$$
$$\langle f_if_jf_{ij}\rangle_{\rm c}=\frac{N-1}{3N}\sigma_1^4S^{(2)},\qquad \langle f_{ij}f_{jk}f_{ki}\rangle_{\rm c}=\frac{\sigma_1^6}{2\sigma_0^2}\left(S_1^{(3)}-3S_2^{(3)}\right).$$

Substituting these relations into the critical-point counts gives

$$\begin{aligned}\bar n^\star(\nu)=&G_{00000}(\nu)+\frac{\sigma_0}{6}\Bigg[G_{30000}S^{(0)}+4\gamma G_{21000}S^{(1)}+3\gamma^2G_{12000}S_2^{(2)}+\gamma^3G_{03000}S_1^{(3)}\\&+4G_{10100}S^{(1)}+\frac{4(N-1)}{N}\gamma G_{01100}S^{(2)}+6\gamma^2G_{10010}\left(S_2^{(2)}-S^{(2)}\right)\\&+\frac{6}{N-1}\gamma^3G_{01010}\left(NS_2^{(3)}-S_1^{(3)}\right)+\frac{3(N-2)(N+2)^2}{2(N+4)}\gamma^3G_{00001}\left(\frac{N+2}{3}S_1^{(3)}-NS_2^{(3)}\right)\Bigg].\end{aligned}$$

3.3.3 Leading-order corrections for CPs in 2D and 3D

For $N=2$, the leading-order ensemble average of the critical-point number density takes the form

$$\begin{aligned}\bar n^\star(\nu)=G_{0000}+\frac{\sigma_0}{6}\Big[&G_{3000}S^{(0)}+4\gamma G_{2100}S^{(1)}+3\gamma^2G_{1200}S_2^{(2)}+\gamma^3G_{0300}S_1^{(3)}+4G_{1010}S^{(1)}\\&+2\gamma G_{0110}S^{(2)}+6\gamma^2G_{1001}(S_2^{(2)}-S^{(2)})+6\gamma^3G_{0101}(2S_2^{(3)}-S_1^{(3)})\Big],\end{aligned}\tag{3.25}$$

where $\gamma\equiv\sigma_1^2/(\sigma_0\sigma_2)$, and the coefficients are

$$G_{ijkl}\equiv G_{ijkl0}(\nu)=\frac{1}{4\pi}\left(\frac{\sigma_2}{\sigma_1}\right)^2X_k\int_{a^\star}^{b^\star}dx\,\mathcal{N}(\nu,x)H_{i-1,j}(\nu,x)f_{l0}^\star(x),\tag{3.26}$$

where $X_0=1$, $X_1=-1$. Here $x\equiv\lambda_1+\lambda_2$, with integration limits $[a^\star,b^\star]=[0,\infty)$ for peaks, $(-\infty,\infty)$ for saddles and χ, and $(-\infty,0]$ for voids.

For $N=3$, we write the differential number density rather than the cumulative form used in the 2D case:

$$\begin{aligned}-\frac{d\bar n^\star(\nu)}{d\nu}=G'_{00000}+\frac{\sigma_0}{6}\Bigg[&G'_{30000}S^{(0)}+4\gamma G'_{21000}S^{(1)}+3\gamma^2G'_{12000}S_2^{(2)}+\gamma^3G'_{03000}S_1^{(3)}+4G'_{10100}S^{(1)}\\+\frac{8}{3}\gamma G'_{01100}S^{(2)}+6\gamma^2G'_{10010}(S_2^{(2)}-S^{(2)})&+3\gamma^3G'_{01010}(3S_2^{(3)}-S_1^{(3)})+\frac{75}{14}\gamma^3G'_{00001}\left(\frac{5}{3}S_1^{(3)}-3S_2^{(3)}\right)\Bigg].\end{aligned}\tag{3.27}$$

The corresponding 2D differential expression is obtained by replacing $H_{i-1,j}\to H_{ij}$ in Eq. (3.26). We then define a new set of coefficients G'_{ijklm} for 3D:

$$G'_{ijklm}(\nu)=\frac{1}{(2\pi)^{3/2}}\left(\frac{\sigma_2}{\sqrt{3}\sigma_1}\right)^3X_k\int_{a^\star}^{b^\star}dx\,\mathcal{N}(\nu,x)H_{i,j}(\nu,x)f_{lm}^\star(x),\tag{3.28}$$

where $X_0=1$, $X_1=-3/2$, $x=\lambda_1+\lambda_2+\lambda_3$, and the integration limits are $[0,\infty)$ for peaks, $(-\infty,\infty)$ for filaments, walls and χ, and $(-\infty,0]$ for voids.

3.4 Numerical implementation

This section describes the numerical pipeline used to test the theory summarized in the previous section. We first construct the smoothed 2D and 3D fields. We then define the threshold-binning scheme and the validation metrics used to compare both cumulative and differential critical-set number densities. Finally, we present the results and discussion. The direct critical-point counting algorithm is introduced in Appendix A.2.

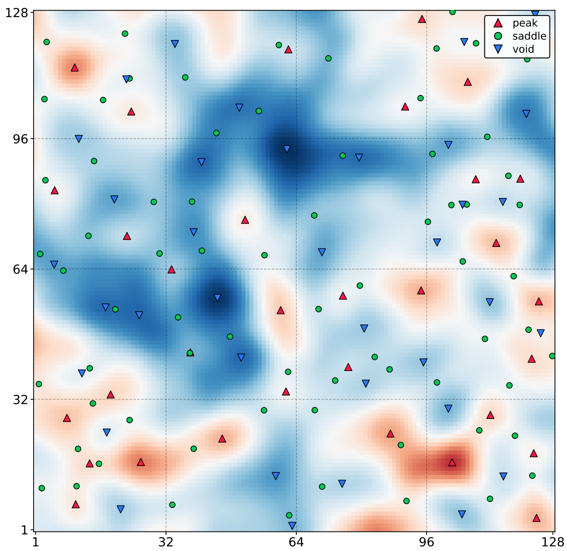

Figure 3.2: Example of a two-dimensional toy-model non-Gaussian field with $\mathcal{A} = 0.14$. The identified critical points are marked, and the figure shows a cropped region from the full 2048×2048 grid.

3.4.1 Field construction

2D toy NG fields

As a representative weak-lensing application, we consider a toy model for a two-dimensional NG field defined on angular coordinates. We first use `CAMB` [44] to compute the matter power spectrum $P_{\mathrm{m}}(k; z)$ at a given redshift. We then construct the weak-lensing convergence power spectrum

$$P_\kappa(\ell) = \int d\chi\, \chi^2 q^2(\chi) P_{\mathrm{m}}\left(\frac{\ell}{\chi}; \chi\right),$$

where the lensing kernel is

$$q(\chi) \equiv \frac{3H_0^2 \Omega_{\mathrm{m},0}}{2a(\chi)c^2} \frac{\chi_{\mathrm{s}} - \chi}{\chi \chi_{\mathrm{s}}}.$$

Here χ is the comoving line-of-sight distance, $\Omega_{\mathrm{m},0}$ is the present-day matter density parameter, $a(\chi)$ is the scale factor, H_0 is the Hubble constant, c is the speed of light, and χ_{s} is the source-galaxy distance. We then define the smoothed field power spectrum

$$P_{\mathrm{2D}}(\ell) = W^2(\ell \vartheta_R) P_\kappa(\ell), \tag{3.29}$$

where $W(\ell\vartheta_R) \equiv \exp\left(-\ell^2\vartheta_R^2/2\right)$ is a Gaussian smoothing kernel and ϑ_R is the smoothing scale. To generate different realizations, we multiply a factor $s = (a + bi)/\sqrt{2}$, which is a normalized complex Gaussian random variable with $a, b \sim \mathrm{Normal}(0, 1)$ using `np.random.default_rng.normal`. This factor assigns random phases and amplitudes to the Fourier modes, enabling a stochastic realization with the target power spectrum. We then generate the Gaussian field $\kappa_g(\boldsymbol{\vartheta})$ from $P_\kappa(\ell)$ by inverse fast Fourier transform. The local-type NG potential field is then constructed as

$$\kappa(\boldsymbol{\vartheta}) = \kappa_g(\boldsymbol{\vartheta}) + \frac{\mathcal{A}}{\sigma_g} {\kappa_g}^2(\boldsymbol{\vartheta}),$$

where $\mathcal{A}$ controls the NG amplitude and $\sigma_g^2 \equiv \langle {\kappa_g}^2 \rangle$. A realization with $\mathcal{A} = 0.14$ gives an example of the field and its identified critical points. At first order in $\mathcal{A}$, the analytic power spectrum is unchanged, while the bispectrum becomes

$$B_{\mathrm{2D}}(\boldsymbol{\ell}_1, \boldsymbol{\ell}_2, \boldsymbol{\ell}_3) = \frac{2\mathcal{A}}{\sigma_g} W(\ell_1\vartheta_R) W(\ell_2\vartheta_R) W(\ell_3\vartheta_R) \times \left[P_\kappa(\ell_1) P_\kappa(\ell_2) + \mathrm{cyc.}\right], \tag{3.30}$$

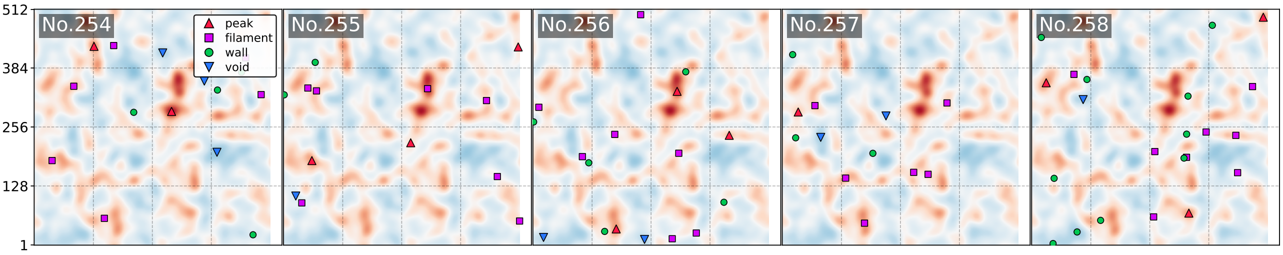

Figure 3.3: Snapshot of the three-dimensional simulated density field after smoothing with $R = 20\,h^{-1}$Mpc, together with the identified critical points in selected slices.

where "cyc." denotes cyclic permutations. The spectra $P_{\rm 2D}$ and $B_{\rm 2D}$ are then used to evaluate the spectral moments in Eq. (3.5) and the generalized skewness parameters in Eq. (3.24). This toy 2D field has nonzero skewness and zero kurtosis, and is applied here in the context of weak-lensing fields.

3D cosmological density fields

The 3D matter field is taken from the `QUIJOTE` N-body simulation suite [45] at $z = 0$ in the fiducial cosmology, and is therefore intrinsically NG. On the theory side, we evaluate the required inputs from modeled polyspectra. We use the Halofit nonlinear power spectrum in `CAMB` for $P_{\rm 3D}(k)$, together with the tree-level bispectrum in standard Eulerian perturbation theory [6]:

$$B_{\rm 3D}(\boldsymbol{k}_1, \boldsymbol{k}_2, \boldsymbol{k}_3) = 2W(k_1R)W(k_2R)W(k_3R)[F_2(\boldsymbol{k}_1, \boldsymbol{k}_2)P_{\rm L}(k_1)P_{\rm L}(k_2) + \text{cyc.}], \tag{3.31}$$

where $P_{\rm L}$ is the linear power spectrum computed with `CAMB`, and the window function is $W(kR) \equiv \exp\left(-k^2R^2/2\right)$. The kernel is

$$F_2(\boldsymbol{k}_1, \boldsymbol{k}_2) = \frac{5}{7} + \frac{1}{2}\frac{\boldsymbol{k}_1 \cdot \boldsymbol{k}_2}{k_1k_2}\left(\frac{k_1}{k_2} + \frac{k_2}{k_1}\right) + \frac{2}{7}\left(\frac{\boldsymbol{k}_1 \cdot \boldsymbol{k}_2}{k_1k_2}\right)^2.$$

Figure 3.3 shows several slices of the simulation after smoothing with $R = 20\,h^{-1}$Mpc, together with the identified critical points. The remaining critical points in the snapshot are located in other slices.

3.4.2 Threshold binning and testing metrics

We compare both cumulative and differential number densities, but only the differential case requires threshold binning. After the counting algorithm identifies the critical sets in a smoothed field, we build a catalog and assign each critical set a normalized height:

$$\nu_a^\star = \frac{f_a^\star - \bar{f}}{\sigma_0}, \quad \text{with} \quad a = 1, 2, \cdots, N^\star,$$

where $\bar{f}$ is the field mean, which is typically zero, and the field is normalized by σ_0. Here $N^\star$ is the length of the catalog, i.e. the total number of critical sets. Thus, for a given realization, the cumulative density above a threshold is

$$n^\star(\nu_i) = \frac{1}{A}\sum_{j=i}^{N_{\rm bin}} N_j^\star, \quad \text{with} \quad i = 1, 2, \cdots, N_{\rm bin}, \tag{3.32}$$

where ν_i is the center of the ith threshold bin, $N_{\rm bin}$ is the number of bins, A denotes the total area in 2D or the total volume in 3D, and $N_j^\star$ is the number of critical points of a given type in that bin, so $N^\star = \sum_i N_i^\star$. The differential density is then estimated by threshold binning:

$$-\frac{dn^\star}{d\nu}(\nu_i) \approx \frac{N_i^\star}{A\Delta\nu}, \tag{3.33}$$

where $\Delta\nu$ is the bin width. In both 2D and 3D, we adopt the threshold range $[-5, 5]$ with $N_{\rm bin} = 100$, corresponding to $\Delta\nu = 0.05$.

For each species, threshold bin, and realization, we denote the estimators in Eqs. (3.32) and (3.33) by $q_s^\star(\nu_i)$. The ensemble-mean differential density is then

$$\bar{q}^\star(\nu_i) = \frac{1}{N_{\rm seed}} \sum_{s=1}^{N_{\rm seed}} q_s^\star(\nu_i), \tag{3.34}$$

and its standard error of the mean is

$$\sigma_{\bar{q}}^\star(\nu_i) = \sqrt{\frac{1}{N_{\rm seed}(N_{\rm seed}-1)} \sum_{s=1}^{N_{\rm seed}} \Big[q_s^\star(\nu_i) - \bar{q}^\star(\nu_i)\Big]^2},$$

where $N_{\rm seed}$ is the number of realizations.

We quantify the agreement between theory and measurement using

$$\Delta^\star_{\rm NG/G}(\nu_i) \equiv \bar{n}^\star_{\rm NG/G} - \bar{q}^\star, \tag{3.35}$$

which means the difference in cumulative critical point density between the non-Gaussian leading-order or Gaus- sian prediction and the ensemble mean identified from simulation fields at threshold ν_i. If this is close to zero, it indicates that the theory prediction is consistent with the simulation result; if not, the higher-order non-Gaussian corrections may be needed. We also use this metric for the differential critical-point density.

3.4.3 Results and discussion

Both analyses assume the same fiducial ΛCDM cosmology for the generation of the 2D and 3D fields, with $h = 0.6711$, $\Omega_{\rm m} = 0.3175$, $\Omega_{\rm b} = 0.049$, $n_s = 0.9624$, and $\sigma_8 = 0.834$. The analytic formulas for the number densities of critical sets are given in Eqs. (3.25) and (3.27). The analytic polyspectra required to evaluate the skewness parameters in Eq. (3.24) are described in Eqs. (3.29), (3.30), and (3.31). The numerical finding method, including the treatment of periodic boundaries, is described in Appendix A.2. We compare both cumulative and differential number densities; the latter is evaluated using the threshold-binning procedure described in Sec. 3.4.2. The comparison between the Gaussian and NG predictions is quantified by the metric in Eq. (3.35). For visual comparison, all curves are normalized by the maximum of the corresponding simulated curve for each critical set.

2D

For the 2D analysis, we use $N_{\rm seed} = 100$, grid size per direction $N_{\rm grid} = 2048$, box size $\vartheta = 1200\,$arcmin, and smoothing scale $\vartheta_R = 20\,$arcmin. We vary only the NG amplitude, taking $\mathcal{A} = 0.06$, 0.10, and 0.14 to compare cases from weakly to mildly NG; see Figures 3.4, 3.5, and 3.6, respectively. We note that the error bars are so small that they are barely visible in the figures. As expected, most values of Δ for the NG prediction fluctuate more closely around zero than the Gaussian prediction, indicating that the NG term provides a useful correction. For lower values, $\mathcal{A} \lesssim 0.06$, the leading-order NG prediction is sufficiently accurate. For higher $\mathcal{A}$, however, the panels for $-d\bar{n}^{\rm peak}/d\nu$ and $-d\bar{n}^{\rm void}/d\nu$ show noticeable oscillatory behavior, because the higher Hermite polynomials receive larger amplitudes. For voids, we find that for $\mathcal{A} = 0.14$ the numerical curves, shown by white points, are flat before $\nu \approx -2$, followed by a steep decrease in $\bar{n}^{\rm void}$ and an even steeper increase in $-d\bar{n}^{\rm void}/d\nu$. This behavior comes from the lower bound of the field, $\kappa \geqslant -\sigma_g/(4\mathcal{A})$, which corresponds to $\nu \geqslant -1/\big(4\mathcal{A}\sqrt{1+3\mathcal{A}^2}\big)$. Thus, the analytic prediction cannot reproduce this behavior because it does not capture the hard lower bound. For the same reason, the other critical-point densities $-d\bar{n}^\star/d\nu$ also exhibit non-continuous behavior around $\nu \approx -2$. By contrast, the saddles appear to be only weakly affected by A in the weakly non-Gaussian field. This may be because saddles predominantly re- side in $\nu \approx -2$ to 2 and are therefore less affected by non-Gaussianity, unlike peaks and voids, which reside in high- and low-density tails of the distribution. The Euler characteristic shows the corresponding behavior for different values of $\mathcal{A}$. In addition, no critical points are found for normalized field values $\nu \gtrsim 5$. The next-to-leading-order correction is required for better predictions in mildly NG fields. For a nonzero kurtosis field, the trispectrum, or four-point correlation functions are needed.

3D

For the 3D analysis, we use $N_{\rm seed} = 400$, $N_{\rm grid} = 512$, and box size $L = 1000\,h^{-1}$Mpc. Figures 3.7, 3.8, and 3.9 show the corresponding results for the 3D matter field with smoothing radii $R = 40$, 30, and $20\,h^{-1}$Mpc, respectively. The results are similar to those in 2D. Most values of Δ fluctuate more closely around zero for the NG prediction, indicating that the leading-order expansion improves the accuracy of the analytic prediction. There are no critical points for normalized field values $\nu \gtrsim 5$. The leading-order analytic prediction for peaks and voids shows stronger oscillatory behavior and deviates more from the numerical result. Both the filaments and walls are not affected by changes in the smoothing radius as strongly as the peaks and voids, possibly because they predomi- nantly reside in the bulk of the distribution and are less affected by non-Gaussianity. Together with the behavior of the Euler characteristic, these discrepancies again indicate that higher-order corrections to the statistics and polyspectra are required, as has already been considered and tested for Minkowski functionals in Ref. [46].

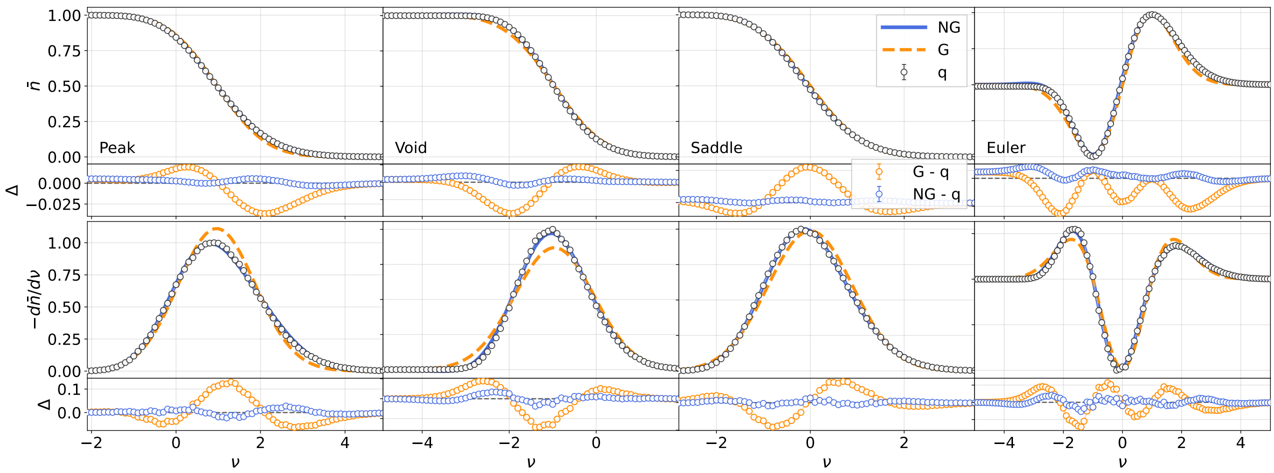

Figure 3.4: Cumulative and differential number densities of four species (peak, void, saddle, and Euler characteristic) in the two-dimensional non-Gaussian field for $\mathcal{A} = 0.06$. The orange dashed curve shows the Gaussian analytic prediction, the blue solid curve shows the leading-order NG prediction, and the white points show the numerical results. The curves are normalized by the maximum numerical value. The Δ panels show the difference between theory and data. Because the error bars are very small, they are barely visible in the figures.

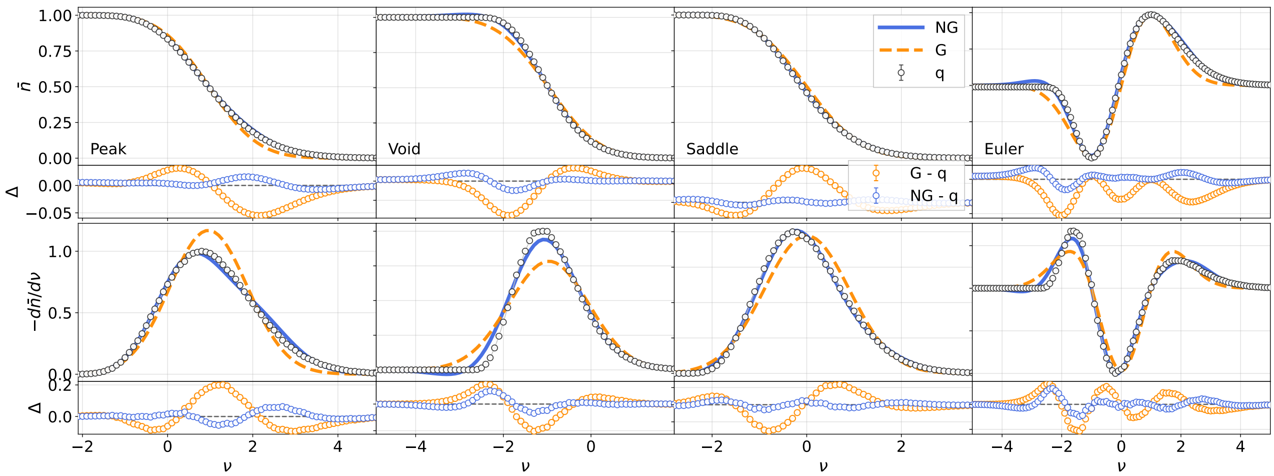

Figure 3.5: Same as Fig. 3.4, but for $\mathcal{A} = 0.10$.

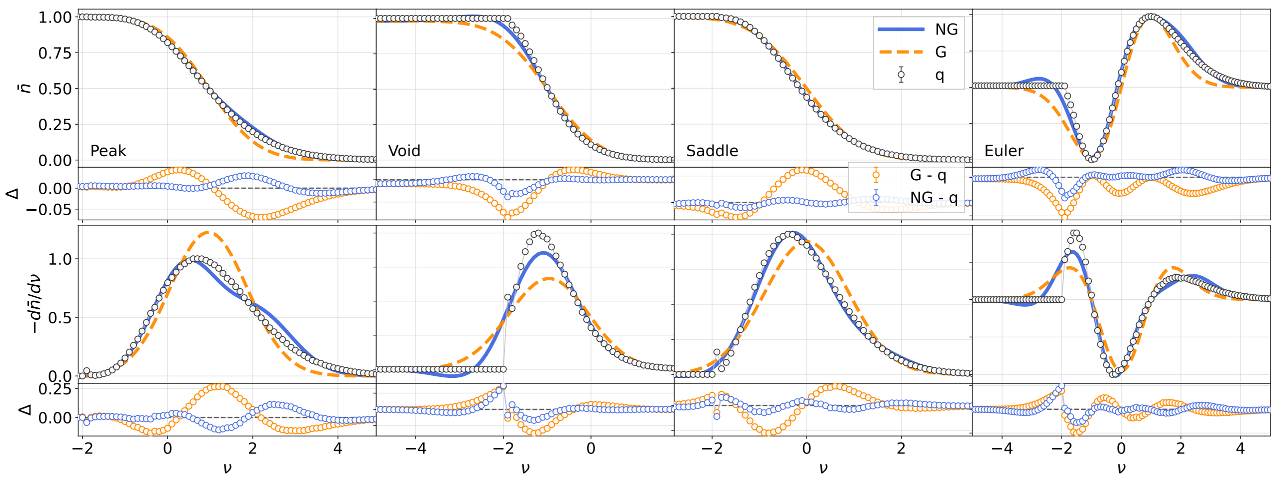

Figure 3.6: Same as Fig. 3.4, but for $\mathcal{A} = 0.14$.

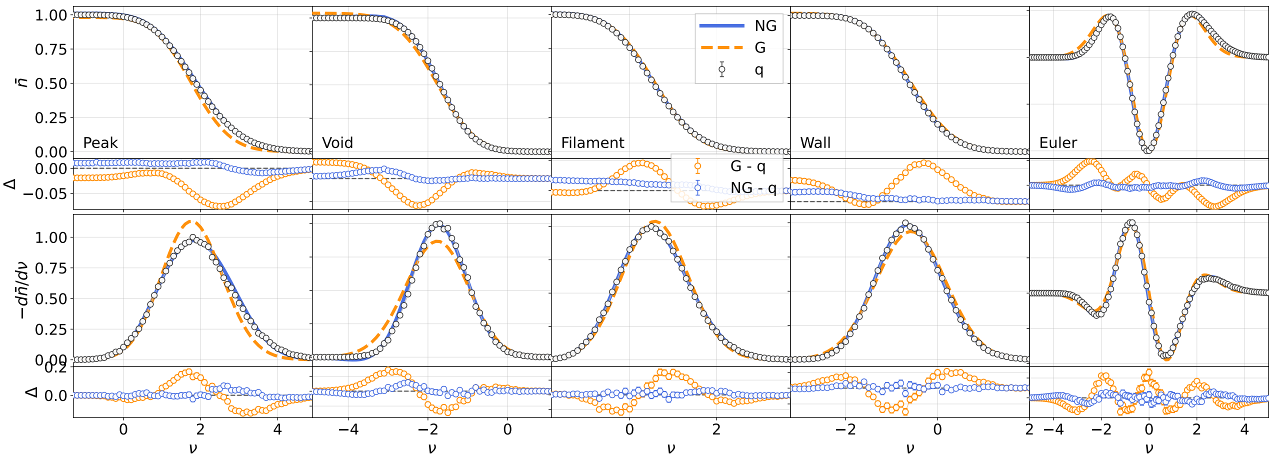

Figure 3.7: Normalized cumulative and differential number densities of five species (peak, void, filament, wall, and Euler characteristic) in the three-dimensional fiducial N-body simulated field for smoothing radius $R = 40\,h^{-1}$Mpc. The plotting conventions are the same as in the 2D case; see the caption of Fig. 3.4.

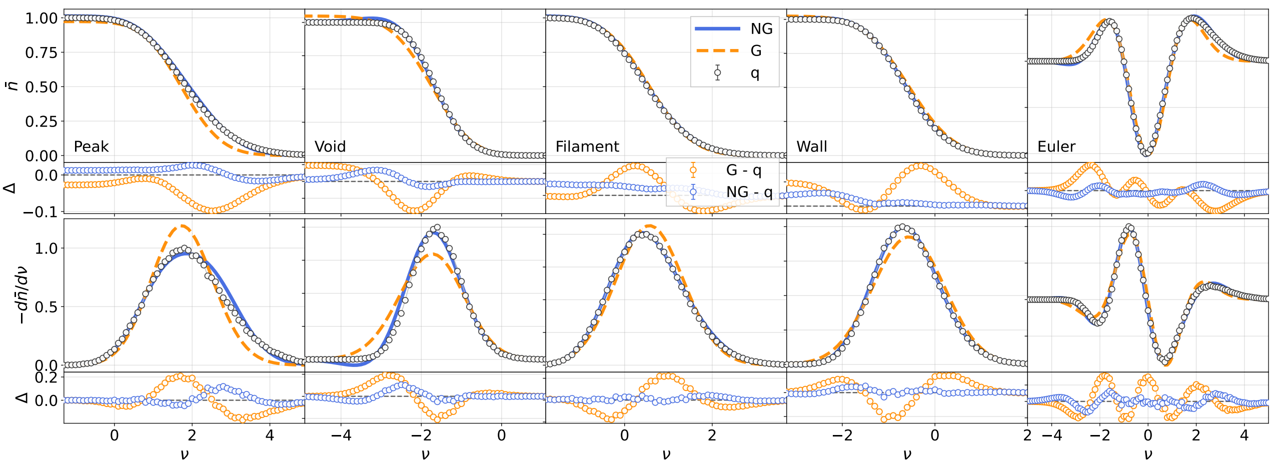

Figure 3.8: Same as Fig. 3.7, but for $R = 30\,h^{-1}$Mpc.

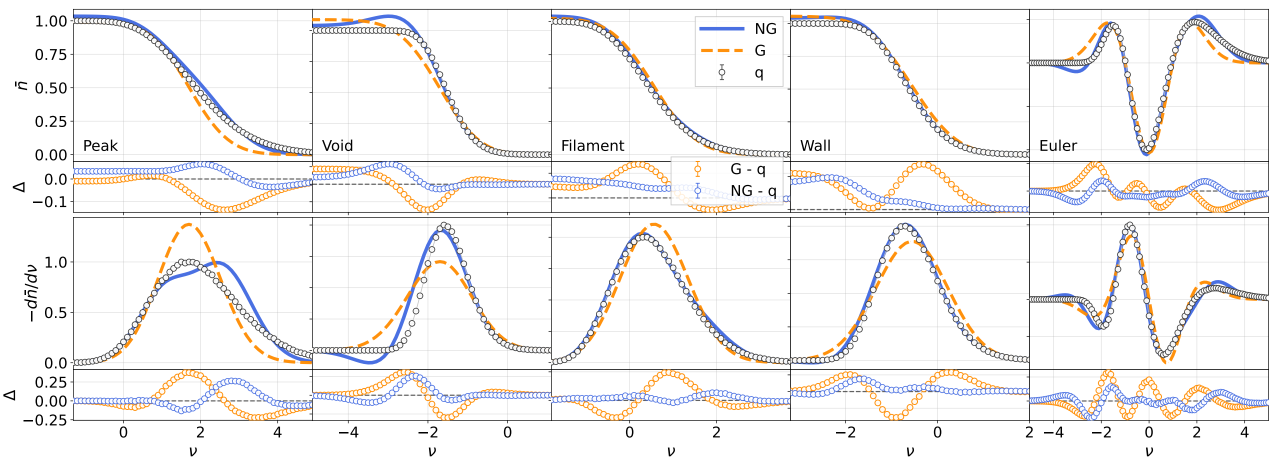

Figure 3.9: Same as Fig. 3.7, but for $R = 20\,h^{-1}$Mpc.

Chapter 4

Application of Cosmological Inference

This chapter connects the theoretical predictions to cosmological parameter inference using matter fields from N-body simulations. We begin by examining the cosmological sensitivity of the threshold-dependent differential critical-point number density through parameter-derivative analyses, which provide a first indication of the information content without accounting for parameter degeneracies. We then establish a Bayesian inference framework and assess the convergence of the inferred constraints through a series of stability tests with varying numbers of simulation realizations. Finally, using the covariance matrix estimated from $N_{\rm sim} = 1000$ realizations, we perform a posterior analysis for $(\Omega_{\rm m}, \sigma_8)$ and $(\Omega_{\rm m}, \Omega_{\rm b}, h, n_s, \sigma_8)$. We find that combining all critical points can provide tighter constraints than using any single critical-point class alone. We also discuss why σ_8 can be degenerate with other parameters in non-Gaussian statistics.

4.1 Sensitivity of CPs to cosmological parameters

To quantify the dependence of CPs on cosmology, we analyze variations around the fiducial cosmology, varying $\Omega_{\rm m}$, $\Omega_{\rm b}$, h, n_s, and σ_8 individually. For simplicity, in this chapter we define

$$\mu^\star(\nu, \theta) \equiv -\frac{d\bar{n}^\star}{d\nu}(\nu, \theta)$$

and compare it with the fiducial prediction through

$$\Delta\mu^\star = \mu^\star(\theta) - \mu^\star(\theta_{\rm fid})$$

where $\Delta\mu^\star$ is the deviation of the CPs curve for a given cosmology θ from the fiducial cosmology $\theta_{\rm fid}$. To further demonstrate this quantifies how well CPs can distinguish different cosmologies, we also compute the numerical derivatives of CPs with respect to θ at $\theta_{\rm fid}$,

$$\frac{\partial\mu^\star}{\partial\theta}(\theta_{\rm fid}) \approx \frac{\mu^\star(\theta_{\rm fid} + \Delta\theta) - \mu^\star(\theta_{\rm fid} - \Delta\theta)}{2\Delta\theta}$$

These derivatives characterize the local response around the fiducial cosmology and form the basic ingredients of Fisher-information analyses.

Figure 4.1 shows the sensitivity of the Gaussian and NG predictions for a smoothed matter density field with $R = 40\,h^{-1}$Mpc, where the perturbative prediction is most accurate; the qualitative parameter dependence should remain similar at smaller smoothing scales. The response is qualitatively consistent with the Betti-curve analysis of Ref. [31]: both CPs and Betti numbers encode the related morphology of the cosmic web, although they summarize it in different ways as explained in Sec. 3.1.3. In the present calculation, $\Omega_{\rm m}$ and $\Omega_{\rm b}$ produce the largest changes in the critical-point number density, while the responses to h and n_s are weaker.

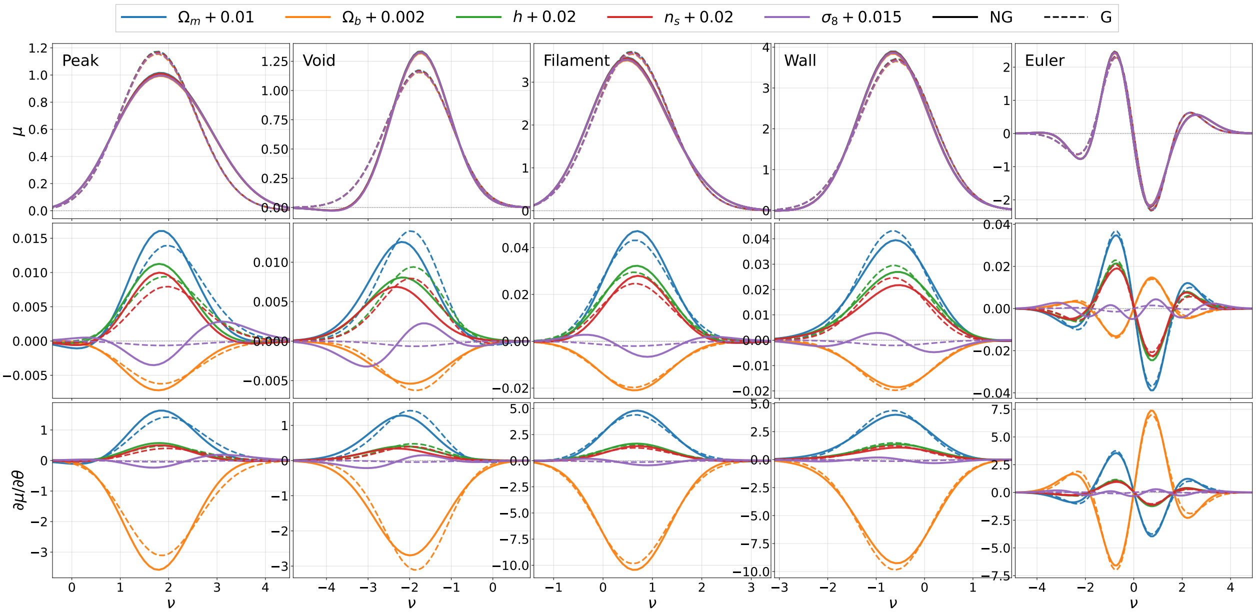

Figure 4.1: Differential number density sensitivity of five parameters with Gaussian and non-Gaussian predictions in a smoothed matter density field with $R = 40\,h^{-1}$Mpc.

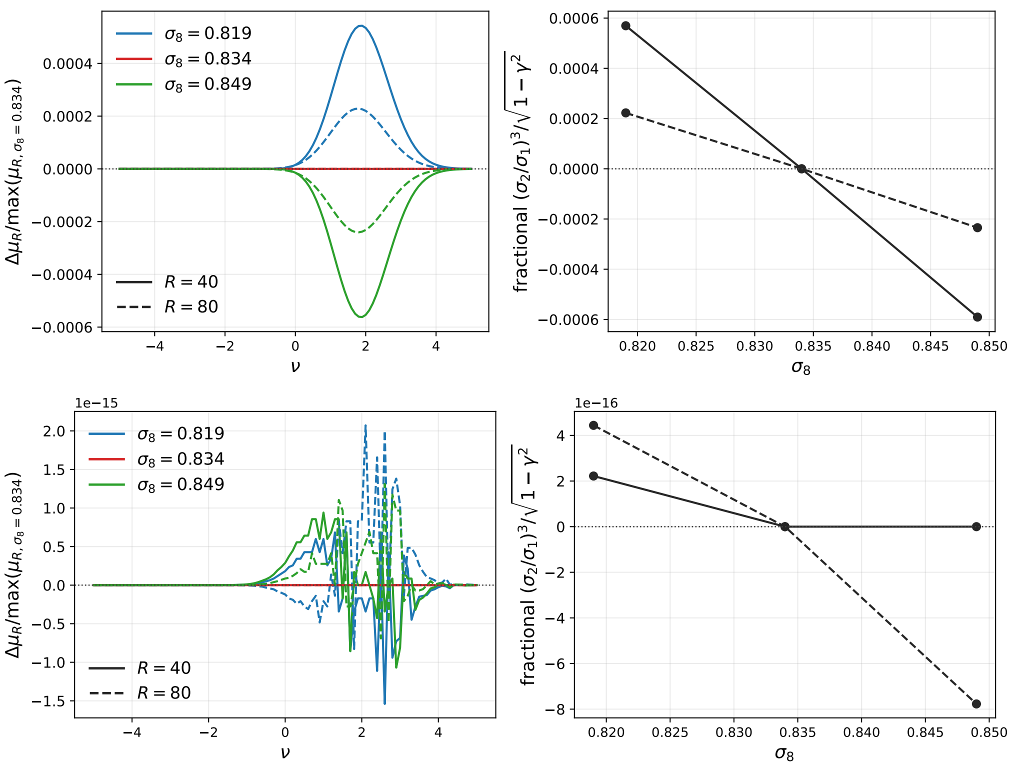

Figure 4.2: Gaussian σ_8 sensitivity test in linear and nonlinear fields. Left: difference in the Gaussian differential peak number density relative to the fiducial σ_8 model for $R = 40,\ 80\,h^{-1}$Mpc. Right: the spectral-moment factor $(\sigma_2/\sigma_1)^3/\sqrt{1-\gamma^2}$ for different values of σ_8 and R.

The Gaussian prediction is almost insensitive to σ_8. This is expected because, for a field expressed in terms of the normalized threshold ν, Gaussian CPs depend primarily on the shape of the power spectrum through the spectral moments and their ratios, whereas σ_8, defined as the standard deviation of the present-day linear matter field averaged in spheres of radius $R_8 = 8\,h^{-1}$Mpc with a spherical top-hat windown function,

$$\sigma_8^2 = \int_0^\infty \frac{dk}{2\pi^2}\, k^2 P_{\rm L}(k, z=0) \left[\frac{3\big(\sin(kR_8) - kR_8\cos(kR_8)\big)}{(kR_8)^3}\right]^2 .$$

mainly changes the overall fluctuation amplitude. Consequently, the Gaussian contribution alone has limited constraining power on σ_8. The residual fluctuations visible in Figure 4.1 arise from the nonlinear power spectrum $P_{\rm NL}(k)$, computed using the halo model in Eq. (2.17), which does not scale strictly proportionally with σ_8. The left two panels of Figure 4.2 show the difference in the Gaussian differential peak number density relative to the fiducial σ_8 model and the factor $(\sigma_2/\sigma_1)^3/\sqrt{1-\gamma^2}$ for different values of σ_8 and R using $P_{\rm NL}(k)$, while the right two panels show the corresponding results using $P_{\rm L}(k)$. The leading NG correction is also proportional to σ_0 and higher-order connected moments, and therefore retains information about the fluctuation amplitude. As a result, NG critical-point statistics remain sensitive to σ_8 even when the Gaussian prediction gives only weak constraints.

The signs of the parameter responses admit a natural physical interpretation. Increasing $\Omega_{\rm m}$ enhances gravitational clustering, boosting small-scale power and generally increasing the abundance of CPs. By contrast, increasing $\Omega_{\rm b}$ suppresses the transfer function and enhances baryon acoustic oscillation features, leading to a lower overall matter power spectrum. Since the suppression is stronger at high k, σ_2 decreases faster than σ_1, reducing the characteristic abundance scale $\bar{n}^\star \propto (\sigma_2/\sigma_1)^3$ and consequently the abundance of CPs. The sensitivity to h arises because changing h modifies the physical matter and baryon densities, $\Omega_{\rm m}h^2$ and $\Omega_{\rm b}h^2$, shifts the turnover scale of the matter power spectrum, and changes the physical scale corresponding to a fixed smoothing radius. Similarly, increasing n_s tilts the primordial power spectrum toward smaller scales, enhancing small-scale fluctuations and increasing the abundance of small-scale structures.

The response to σ_8 is qualitatively different. Unlike the other parameters, σ_8 primarily affects the non-Gaussian correction rather than the Gaussian contribution. The non-Gaussian terms consist of Hermite-polynomial contributions with different threshold dependence, so increasing σ_8 does not simply rescale the critical-point abundance. Instead, it redistributes CPs across threshold, producing a characteristic sign-changing response as a function of ν. For example, increasing σ_8 shifts peaks toward higher thresholds, enhancing the high-density tail while reducing their abundance at intermediate ν. This behavior is consistent with the oscillatory structure of the Hermite-polynomial expansion and can be seen directly by comparing the non-Gaussian correction with the Gaussian prediction. An interesting feature is that, for nonlinear structure probes, $\Omega_{\rm m}$ and σ_8 are often strongly degenerate. For example, in the case of the power spectrum, the parameter responses $\partial\mu^\star/\partial\theta$ typically exhibit similar trends, leading to the well-known S_8-like degeneracy. In contrast, for critical-point statistics we find qualitatively different behavior. Around $\nu \sim 1.5$, the peak abundance responds to $\Omega_{\rm m}$ and σ_8 with opposite trends, and similar behavior is observed for the other critical-point classes in different threshold ranges.

For visual comparison, $\mu^\star$ is normalized by the maximum value of the NG peak prediction over the threshold range. With this normalization, the filament and wall amplitudes are roughly three times larger than those of peaks and voids, and their derivatives $\partial\mu^\star/\partial\theta$ are correspondingly larger. This indicates stronger raw parameter sensitivity, although the final constraining power depends on both the response and the covariance matrix. It is worth emphasizing that the interpretation presented here is derived from an analytic theory of geometric statistics, rather than inferred empirically from numerical simulations as in Ref. [31]. In the next sections, we construct the inference pipeline, test the stability of CPs constraints for different values of $N_{\rm sim}$, and compare their additional constraining power beyond the power spectrum.

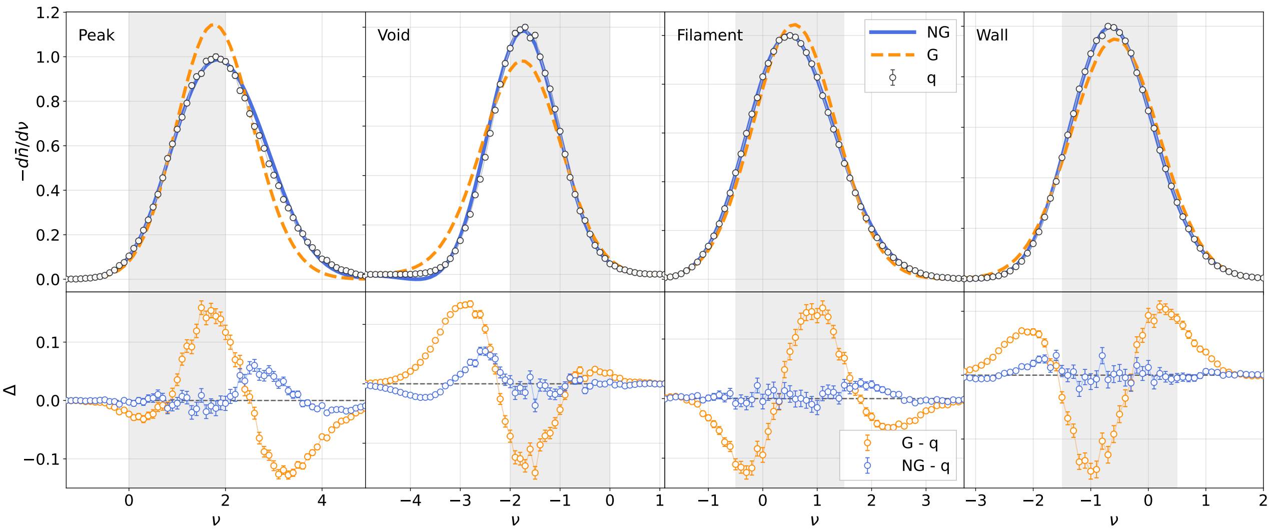

Figure 4.3: The differential number density of peaks, voids, filaments, and walls for 1000 realizations. The shaded regions are used in the analysis. Blue solid curves are leading-order non-Gaussian prediction, while orange dashed curves are only Gaussian prediction. q is calculated from numerical results.

4.2 Bayesian inference framework

The cosmological parameter inference pipeline consists of five steps: (i) constructing the summary-statistic data vector, (ii) estimating the sample covariance from simulation realizations, (iii) building a forward model $\boldsymbol{\mu}(\boldsymbol{\theta})$ from a cosmological parameter grid, (iv) defining the likelihood function, and (v) obtaining posterior constraints through sampling or direct evaluation on the parameter grid.

4.2.1 Likelihood Analysis

Data vector construction

We construct a summary-statistic data vector from the differential critical-point abundance measured as a function of the normalized threshold ν. To ensure a stable covariance estimate and suppress noise-dominated modes, the threshold range is restricted for each critical-point class and the measurements are rebinned into a reduced set of bins. The final data vector is defined as

$$\boldsymbol{d} \equiv \left[\bar{q}^{\rm peak}, \bar{q}^{\rm filament}, \bar{q}^{\rm wall}, \bar{q}^{\rm void}\right],$$

where each component denotes the differential number density measured over a selected threshold range. To balance cosmological information against covariance stability, the threshold ranges for each CPs class are peak $\in [0, 2]$, void $\in [-2, 0]$, filament $\in [-0.5, 1.5]$, and wall $\in [-1.5, 0.5]$, as shown by the shaded regions in Fig. 4.3. The bin spacing is 0.2, so the data vector for each critical-point class has dimension 11, giving $p = 44$ in total.

Mean and covariance estimation

The forward model predicts the mean critical-point statistics as a function of cosmological parameters

$$\boldsymbol{\theta} = (\Omega_{\rm m}, \Omega_{\rm b}, h, n_s, \sigma_8).$$

The sample mean $\bar{\boldsymbol{d}}$ is estimated from $N_{\rm sim}$ realizations by Eq. (3.34), and the sample covariance matrix is

$$\hat{C} = \frac{1}{N_{\rm sim} - 1} \sum_{s=1}^{N_{\rm sim}} (\boldsymbol{d}_s - \bar{\boldsymbol{d}})(\boldsymbol{d}_s - \bar{\boldsymbol{d}})^T. \tag{4.1}$$

In practice, the true covariance matrix is unknown and must be estimated from the finite number of realizations used here. Consequently, the inverse covariance estimator is biased. For Gaussian likelihood analyses, we therefore apply the Hartlap correction [47],

$$\hat{\Psi}_{\rm H} = \alpha\, \hat{C}^{-1}, \qquad \alpha = \frac{N_{\rm sim} - p - 2}{N_{\rm sim} - 1}, \tag{4.2}$$

where p is the dimension of the data vector.

Likelihood function

Given an observed data mean vector $\boldsymbol{d}$ and a model prediction $\boldsymbol{\mu}(\boldsymbol{\theta})$, we define

$$\Delta(\boldsymbol{\theta}) = \boldsymbol{d} - \boldsymbol{\mu}(\boldsymbol{\theta}).$$

For a Gaussian likelihood, the chi-square statistic is

$$\chi^2_{\rm H}(\boldsymbol{\theta}) = \Delta^T \hat{\Psi}_{\rm H} \Delta.$$

The corresponding likelihood is

$$\ln \mathcal{L}_{\rm G} = -\frac{1}{2}\chi^2_{\rm H} + \text{const}.$$

Equivalently, one may use the relative likelihood

$$-2\ln \mathcal{L}_{\rm G} \propto \Delta\chi^2_{\rm H} = \chi^2_{\rm H}(\boldsymbol{\theta}) - \chi^2_{\rm H,min}.$$

Because the covariance matrix is estimated from a finite number of realizations, a more rigorous treatment is provided by the multivariate Student-t likelihood of Sellentin & Heavens [48],

$$\ln \mathcal{L}_{\rm t} = -\frac{N_{\rm sim}}{2} \ln\left(1 + \frac{\chi^2}{N_{\rm sim} - 1}\right) + \text{const}, \tag{4.3}$$

where $\chi^2 = \Delta^T \hat{C}^{-1}\Delta$. In the limit $N_{\rm sim} \to \infty$, the Student-t likelihood asymptotically approaches the Gaussian likelihood. Here the covariance is estimated using Eq. (4.1).

In Appendix. A.3, we explain further about the correction of Hartlap covariance and Student-t likelihood.

Prior and posterior distribution

Assuming uniform priors within the sampled parameter range, Bayes' theorem gives

$$\mathcal{P}(\boldsymbol{\theta}|\boldsymbol{d}) = \frac{\mathcal{L}(\boldsymbol{\theta})\mathcal{P}_{\rm prior}(\boldsymbol{\theta})}{Z}, \qquad Z = \int \mathcal{L}(\boldsymbol{\theta})\mathcal{P}_{\rm prior}(\boldsymbol{\theta})\, d\boldsymbol{\theta},$$

where Z is the Bayesian evidence. The marginalized posterior distribution for parameter θ_i is obtained by integrating over the remaining parameters,

$$\mathcal{P}(\theta_i|\boldsymbol{d}) = \int \mathcal{P}(\boldsymbol{\theta}|\boldsymbol{d}) \prod_{j\neq i} d\theta_j.$$

To avoid confusion with the simulation-realization index s, we use r to label posterior samples in parameter space. Let $\boldsymbol{\theta}_r$ be the r-th posterior sample, with $r = 1, \ldots, N_{\rm post}$, and let $\theta_{i,r}$ be its i-th parameter component. The normalized posterior weight of this sample is

$$w_r = \frac{\tilde{w}_r}{\sum_{q=1}^{N_{\rm post}} \tilde{w}_q}, \qquad \tilde{w}_r = \mathcal{L}(\boldsymbol{\theta}_r)\mathcal{P}_{\rm prior}(\boldsymbol{\theta}_r).$$

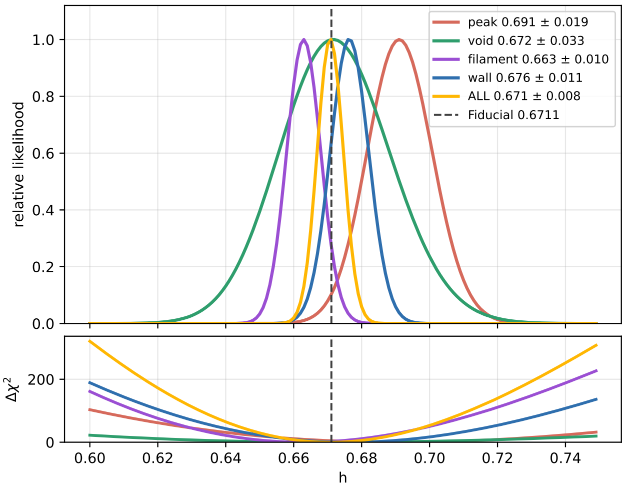

Figure 4.4: Relative likelihood and $\Delta\chi^2$ of h using 1000 realizations and different data vectors.

For uniform priors and equally spaced grid points, this reduces to

$$w_r = \frac{\mathcal{L}(\boldsymbol{\theta}_r)}{\sum_{q=1}^{N_{\rm post}} \mathcal{L}(\boldsymbol{\theta}_q)}.$$

The posterior mean of parameter θ_i is then estimated as

$$\bar{\theta}_i = \sum_{r=1}^{N_{\rm post}} w_r \theta_{i,r}.$$

The parameter covariance is

$$C^{\theta}_{ij} = \sum_{r=1}^{N_{\rm post}} w_r(\theta_{i,r} - \bar{\theta}_i)(\theta_{j,r} - \bar{\theta}_j). \tag{4.4}$$

The marginalized uncertainty of each parameter is therefore

$$\sigma_{\theta_i} = \sqrt{C^{\theta}_{ii}}. \tag{4.5}$$

The final results are shown as corner plots, with the diagonal panels showing the one-dimensional marginalized posteriors and the off-diagonal panels showing the two-dimensional credible contours. The five-parameter cosmological estimation is presented in the next section.

Results in One-Dimensional Parameter Space

For visualization, we present the marginalized one-dimensional posterior distributions for h, together with the corresponding posterior means for all CPs; see Fig. 4.4. The prior range is $h \in [0.600, 0.749]$ with spacing 0.001, giving 150 grid points. We use $N_{\rm sim} = 1000$. The posterior constraints are 0.691 ± 0.019 for peaks, 0.672 ± 0.033 for voids, 0.663 ± 0.010 for filaments, 0.676 ± 0.011 for walls, and 0.671 ± 0.008 for all CPs combined. The fiducial parameter value, $h = 0.6711$, is overplotted as a reference line. The credible intervals for all critical-point predictions include the fiducial value. Combining all critical points yields a narrower constraint than any individual critical-point class.

4.2.2 Stability test

To assess whether the number of realizations is sufficient for cosmological parameter inference for $(\Omega_{\rm m}, \Omega_{\rm b}, h, n_s, \sigma_8)$, we perform a subsampling stability test. The dimension of the data vector is

$p = 44$. We consider the subsampling sizes $N_{\rm sub} \in [100, 1000]$ in steps of 50, with the corresponding number of repetitions $N_{\rm repeat}$ decreasing from 200 to 100. This allows us to estimate error bars for the stability-test quantities. For each value of $N_{\rm sub}$, we randomly draw a subset of realizations without replacement from the full sample, and repeat this procedure $N_{\rm repeat}$ times. We perform three complementary stability tests based on the Fisher matrix, the marginalized parameter uncertainties, and the posterior central values.

Fisher matrix

The Fisher matrix is defined as

$$F_{\theta_i\theta_j} = -\left\langle \frac{\partial^2 \ln \mathcal{L}}{\partial\theta_i \partial\theta_j} \right\rangle.$$

Assuming that the covariance matrix is independent of cosmological parameters, the Fisher matrix for a Gaussian likelihood is

$$F_{\theta_i\theta_j} = \left(\frac{\partial \boldsymbol{\mu}}{\partial\theta_i}\right)^T \hat{\Psi}_{\rm H} \left(\frac{\partial \boldsymbol{\mu}}{\partial\theta_j}\right),$$

where $\boldsymbol{\mu}$ denotes the theoretical prediction and $\hat{\Psi}_{\rm H}$ is the Hartlap-corrected precision matrix, given in Eq. (4.2). The parameter covariance matrix is estimated as $\hat{C}_\theta = \hat{F}^{-1}$, similar to Eq. (4.4), but here it is estimated around the fiducial cosmology. To characterize the overall constraining power, we define the Figure of Merit (FoM) as

$$\mathrm{FoM} = \frac{1}{\sqrt{\det \hat{C}_\theta}} = \sqrt{\det \hat{F}}.$$

Since the volume of the five-dimensional parameter error ellipsoid scales as $V \propto \sqrt{\det C_\theta}$, a larger FoM corresponds to a smaller posterior volume and therefore tighter cosmological constraints. To quantify convergence, we plot $\mathrm{FoM}(N_{\rm sub})/\mathrm{FoM}(N_{\rm ref})$, using $N_{\rm sim} = 1000$ as the reference sample.

We use the parameter prior ranges

$$\Omega_{\rm m} \in [0.2, 0.4], \quad \Omega_{\rm b} \in [0.04, 0.06], \quad h \in [0.6, 0.8], \quad n_s \in [0.8, 1.1], \quad \sigma_8 \in [0.6, 1.0] \tag{4.6}$$

with five grid points for each parameter, giving 5^5 theoretical prediction curves in total. The first panel of Fig. 4.5 shows that the FoM increases as $N_{\rm sub}$ increases, indicating that the overall constraining power also increases.

Uncertainties

The marginalized uncertainties of parameter estimation are defined in Eq. (4.5). In the second panel of Fig. 4.5, each parameter uncertainty σ_θ decreases as the number of samples increases, especially for σ_8. This suggests that larger simulation samples substantially improve the σ_8 constraint. $\Omega_{\rm b}$ is stable when $N_{\rm sub} \geqslant 400$, while $\Omega_{\rm m}$ and n_s become stable only near $N_{\rm sub} \geqslant 850$. For n_s and σ_8, the constraints depend more strongly on degeneracy breaking in the non-Gaussian regime, as discussed in Sec. 4.1, so more samples are needed for stability.

Central values

In addition to the parameter uncertainties, we examine the stability of the posterior central values. For each subsample, the posterior mean of parameter θ_i is denoted by $\bar{\theta}_i$. We quantify its deviation from the full-sample result using

$$\frac{\bar{\theta}_i(N_{\rm sub}) - \bar{\theta}_i(N_{\rm sim})}{\sigma_{\theta_i}(N_{\rm sim})}.$$

A stable parameter inference requires both the marginalized uncertainties and the posterior central values to converge as $N_{\rm sub}$ approaches $N_{\rm sim}$.

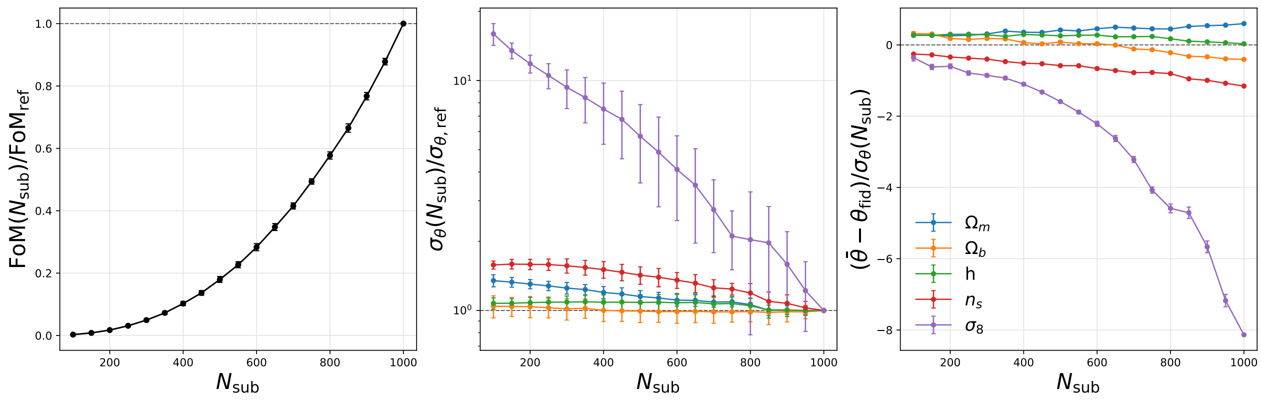

Figure 4.5: Stability test for the Figure of Merit, marginalized parameter uncertainties, and posterior central values as functions of sample number.

However, the third panel shows that only h is clearly converging. We estimate that this may be caused by the limited accuracy of the grid-based estimate of $\bar{\theta}$, together with the smaller σ_θ values shown in the second panel. In the next section, we use the nested-sampling method for parameter estimation.

4.3 Parameter estimation

4.3.1 Nested sampling method

After validating the stability of the covariance matrix, we perform the final cosmological parameter inference using nested sampling. The posterior distribution is sampled using a nested-sampling technique, see Ref.[49] for a tutorial. In the implementation, the parameter names, prior ranges, likelihood type, number of realizations, covariance mode, and selected critical-point components, all discussed previously are passed to the nested-sampling routine.

Interpolation of the grid

The theoretical predictions are computed on a discrete cosmological grid. However, nested sampling requires evaluating the likelihood at arbitrary parameter values within the prior volume defined in Eq. (4.6). Therefore, during the likelihood evaluation, the theoretical data vector $\boldsymbol{\mu}(\boldsymbol{\theta})$ is obtained by interpolating the precomputed cosmological grid.

For each proposed parameter point $\boldsymbol{\theta}$, the code first evaluates the corresponding model prediction $\boldsymbol{\mu}(\boldsymbol{\theta})$ from the grid interpolation. This interpolated prediction is then used to compute $\Delta(\boldsymbol{\theta})$ and hence the likelihood. The interpolation is applied only in the parameter-estimation stage, while the covariance matrix is estimated independently from the simulation realizations.

Sampling pipeline

We use the Python package `ultranest` [1]. The pipeline is as follows: first, draw an initial set of live points randomly from the prior and compute the likelihood of each live point; second, identify the live point with the lowest likelihood and mark it as a dead point, assigning it a weight toward the posterior and evidence; third, sample a new point with likelihood greater than that of the dead point and repeat the process, which is the nested procedure. We use `ultranest.ReactiveNestedSampler` as the main sampler; it is more advanced than the conventional nested sampler because it determines where additional sampling is needed, when to stop, and how to estimate uncertainties. `ultranest.MLFriends`

[1] https://johannesbuchner.github.io/UltraNest/

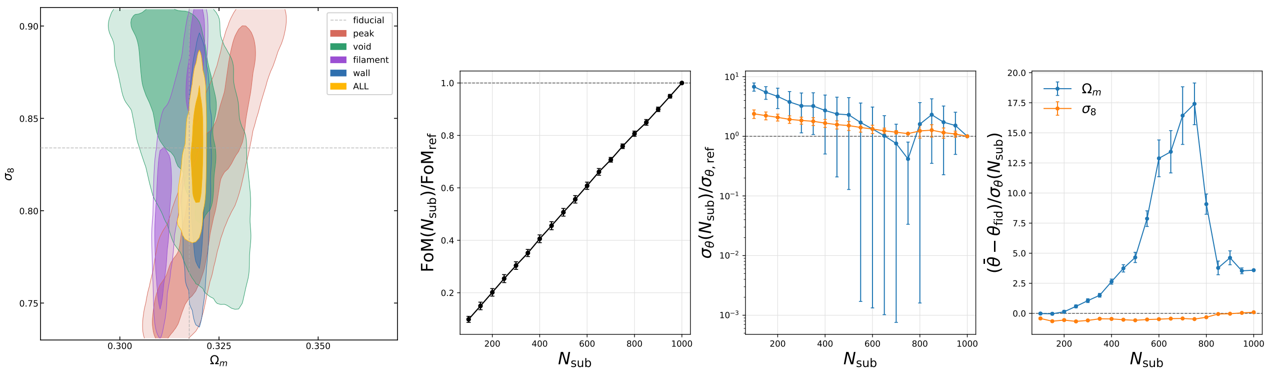

Figure 4.6: Posterior and stability tests for $(\Omega_{\mathrm{m}}, \sigma_8)$.

is used to propose new live points by estimating a high-likelihood ellipsoidal region from the current live-point distribution. Finally, we store the posterior samples and produce corner plots.

There are several important parameters to adjust the final posterior:

- `min_num_live_points`: the minimum number of stored live points, the more, the posterior resolution the higher, the region of possible likelihood is not missed, the posterior weight is more stable;
- `min_ess`: effective sample size can weight the posterior samples, this gives the target number of effective posterior samples, if setting too small, the contour is noisy;
- `dlogz`: evidence uncertainty, the target of standard varaince between booststrapped logZ integrators, the smaller, the more accurate of evidence;
- `max_ncalls`: the upper limitation of likelihood evaluation times, it usually not the fianl control of the stop;
- `frac_remian`: one of the stop criterian, it will integrate untill the remain the live points can contribute how much the present integration.

4.3.2 Posterior constraints

The nested-sampling output provides posterior samples and their associated weights. From these weighted samples, we construct the one-dimensional marginalized posterior distributions and two-dimensional credible contours.

Posterior for $(\Omega_{\mathrm{m}}, \sigma_8)$

In Fig. 4.6, we show the posterior and stability tests for $(\Omega_{\mathrm{m}}, \sigma_8)$. We use the prior here for $\Omega_{\mathrm{m}} \in [0.25, 0.37]$ and $\sigma_8 \in [0.63, 0.91]$ The combined ALL constraint lies inside those from the individual probes. Ω_{m} is constrained to a narrow region because the two-parameter setup excludes the influence of the remaining cosmological parameters. The CPs statistics also help break the degeneracy with σ_8. For peaks and voids, the contours are slightly tilted because of the opposite parameter sensitivities in the chosen threshold regions.

Posterior for $(\Omega_{\mathrm{m}}, \Omega_{\mathrm{b}}, h, n_s, \sigma_8)$

Fig. 4.7 shows the posterior for $(\Omega_{\mathrm{m}}, \Omega_{\mathrm{b}}, h, n_s, \sigma_8)$. As expected, σ_8 is degenerate with the other parameters. Unlike in the $(\Omega_{\mathrm{m}}, \sigma_8)$ case, Ω_{m} is no longer isolated from the remaining parameters and becomes degenerate with them.

These results demonstrate the potential of critical-point statistics for cosmological parameter estimation in the next era of precision observations.

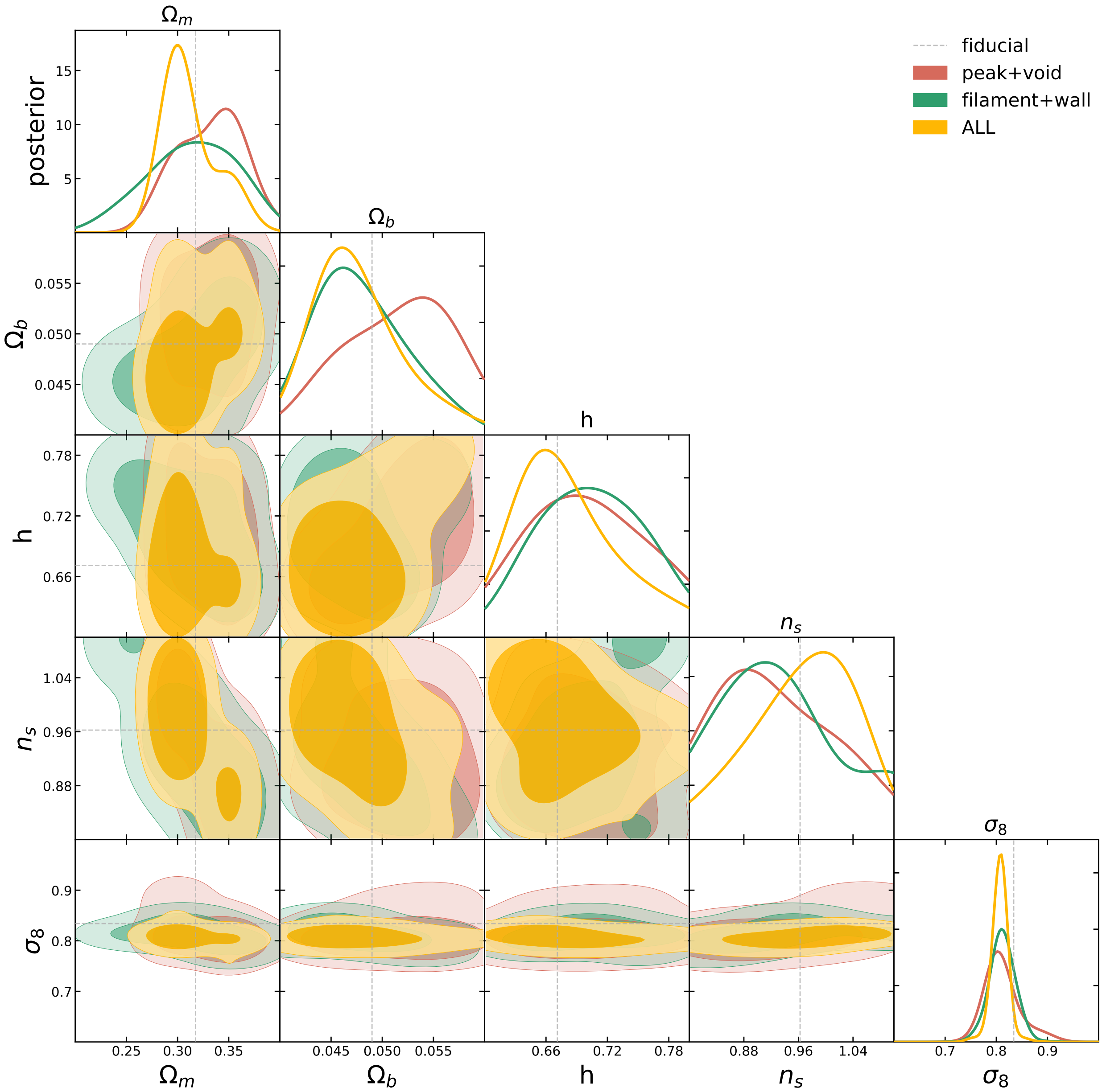

Figure 4.7: Posterior for $(\Omega_{\rm m}, \Omega_{\rm b}, h, n_s, \sigma_8)$.

Chapter 5

Summary

This thesis studied critical-point statistics as geometric probes of weakly non-Gaussian cosmological fields. The main motivation is that conventional two-point statistics, although central to modern cosmology, do not exhaust the information contained in the morphology of the late-time matter distribution. Critical points (CPs) provide a local and physically interpretable description of this morphology. In a 3D matter density field, they correspond to peaks, voids, filaments, and walls, which form the basic components of the cosmic web. We found that Leading-order non-Gaussian corrections improve the prediction in weakly non-Gaussian fields and CPs are sensitive to σ_8 in a way that differs from the Gaussian prediction, giving them potential power for breaking parameter degeneracies.

Chapter 2 reviewed the cosmological background motivated for this thesis. Inflation provides nearly Gaussian primordial perturbations, while interactions during inflation and nonlinear gravitational evolution generate non-Gaussian corrections (note that this non-Gaussian property is not considering in the later main contents). The chapter then connected these initial conditions to late-time large-scale structure through Eulerian and Lagrangian perturbation theory, the nonlinear power spectrum, higher-order moments, and the halo model. Finally, weak gravitational lensing and the cosmic web were discussed as two random cosmic matter fields in which critical-point statistics can be applied in the later numerical and inference analyses.

Chapter 3 developed the theoretical and numerical framework for leading order non-Gaussian perturbative critical-point statistics. Starting from the Kac–Rice formula and the Morse index, the number density of critical points was written in terms of the joint statistics of the field and its derivatives. The Gaussian one-point theory was reviewed in both two and three dimensions, following the BBKS poineered work and Matsubara's framework. The leading-order non-Gaussian correction was then organized through the generalized Wiener–Hermite expansion and the corresponding skewness parameters. In 2D toy weak-lensing-like field $\kappa = \kappa_g + \mathcal{A}\kappa_g^2$, where κ_g are normal fields, the leading-order non-Gaussian prediction improved over the Gaussian prediction for weak non-Gaussianity, but failed at roughly $\mathcal{A} \gtrsim 0.10$. In the 3D N-body simulated fields, peaks and voids were more sensitive to nonlinear effects and showed stronger deviations for smoothing radius $R \gtrsim 30\,h^{-1}$Mpc, while filaments and walls remained more stable even around $R \gtrsim 20\,h^{-1}$Mpc. This is expected since both peaks and voids live in the higher tails of the distribution where affected the high-threshold and low threshold a lot, while the saddles (filaments and walls) reside in the middle of the field value distribution. These results indicate that leading-order non-Gaussian theory captures part of the relevant morphology but that higher-order corrections are needed for accurate predictions in more nonlinear regimes.

Chapter 4 connected critical-point statistics to cosmological parameter inference. The threshold-dependent differential number density was used as the summary statistic, and its response to Ω_m, Ω_b, h, n_s, and σ_8 was examined. The response of CPs was found to be qualitatively consistent with other morphology-based probes, such as Betti curves [31], but CPs also provide a distinct interpretation through local field geometry. In particular, the Gaussian prediction is almost insensitive to σ_8 when expressed in terms of the normalized threshold ν, because it mainly depends on the shape of the power spectrum through spectral-moment ratios. The non-Gaussian correction, however, retains amplitude information and therefore restores sensitivity to σ_8. We used simulation-estimated covariances, Hartlap-corrected Gaussian likelihoods, and the multivariate Student-t likelihood as a finite-covariance alternative. Stability tests showed how the inferred constraints depend on the number of simulation realizations. Using $N_{\rm sim} = 1000$ realizations for the covariance estimate, the posterior analyses demonstrated that combining all critical-point classes gives tighter constraints than using any single class alone. In the one-dimensional example for h, the combined CPs constraint was narrower than those from peaks, voids, filaments, or walls separately. In the two-parameter $(\Omega_{\rm m}, \sigma_8)$ analysis, the combined constraint also lay inside the individual constraints, showing the complementarity of the different CPs classes. In the five-parameter case, degeneracies with σ_8 remained important, but the results still demonstrated that critical-point statistics can be incorporated into a standard forward-modeling and posterior-sampling pipeline.

All in all, critical-point statistics using an analytical approach provide a complementary way to analyze the information contained in cosmic fields. Compared to the numerical emulator method, it may not be as accurate or precise, but it is transparent and physically explainable. We hope that we can extend this technique to many other fields.

Acknowledgements

I would like to express my sincere gratitude to my supervisor, MATSUBARA Takahiko, for his guidance, support, and constructive feedback throughout my master's studies. I am truly grateful for the opportunity to work with him, and I deeply respect his attitude toward scientific research. Our first conversation was on Zoom, where we discussed my bachelor's thesis project on the BAO power spectrum and the halo model. I have often doubted my own ability, but his encouragement has always motivated me to pursue my studies further. The ChatGPT subscription he provided contributed greatly to my studies in coding and writing. I would also like to thank Kuriki-san and Iso-san for their helpful discussions, especially when I first presented my results seriously to others. I also thank the cosmophysics group in Theory Center, where I learnt a lot beyond my study.

I do not consider myself a particularly brave person, even though people think that studying abroad is a brave decision. At first, I simply wanted to leave the stressful environment in my home country. I am grateful to have met so many kind people here. KEK has also been a very good place for me. For many reasons, I was able to concentrate on my studies. I would also like to thank myself for never regretting the choices I made before and thank my family and close friends for their support.

Appendix A

Technical Details

A.1 Fast Fourier transform normalization

This section relates the fast Fourier transform (FFT) normalization used in Python (NumPy/SciPy) to the normalization of the continuous Fourier transform (CFT).

For the continuous one-dimensional Fourier transform, we use

$$\tilde{f}(\omega) = \int e^{-i\omega t} f(t) dt, \qquad f(t) = \frac{1}{2\pi} \int e^{i\omega t} \tilde{f}(\omega) d\omega$$

In NumPy/SciPy, the forward and inverse discrete Fourier transforms are defined by

$$X_k = \sum_{n=0}^{N-1} x_n e^{-2\pi i k n/N}, \qquad x_n = \frac{1}{N} \sum_{k=0}^{N-1} X_k e^{2\pi i k n/N}$$

With this convention, the forward transform is an unnormalized discrete sum. For N samples with spacing $\Delta t = L/N$, let $x_n = f(n\Delta t)$ on the interval $t \in [0, L]$, where L is the box size. The relation between the CFT and the discrete FFT is then

$$\tilde{f}(\omega_k) = \sum_{n=0}^{N-1} f(n\Delta t) e^{-i\omega_k n \Delta t} \Delta t = \Delta t \cdot X_k, \qquad \omega_k = \frac{2\pi k}{N \Delta t} = \frac{2\pi k}{L}$$

This is the basic normalization relation for the forward transform in the discrete case.

In two dimensions, the finite-box Fourier series of the κ field can be written as

$$\kappa(\theta) = \frac{1}{L^2} \sum_{\ell} \kappa_{\boldsymbol{\ell}} e^{i\boldsymbol{\ell}\cdot\boldsymbol{\theta}}$$

where $\boldsymbol{\ell} = (\ell_x, \ell_y) = \frac{2\pi}{L}(n_x, n_y)$ is the finite-box Fourier-series wavevector for a flat-sky field. The physical Fourier modes occur in $\pm\ell$ pairs, but only one complex degree of freedom in each pair is independent. Therefore,

$$\langle \kappa(\boldsymbol{\theta})^2 \rangle = \frac{1}{L^4} \sum_{\boldsymbol{\ell}} \langle \kappa_{\boldsymbol{\ell}} \kappa^*_{\boldsymbol{\ell}} \rangle e^{i(\boldsymbol{\ell}-\boldsymbol{\ell})\cdot\boldsymbol{\theta}} = \frac{1}{L^4} \sum_{\boldsymbol{\ell}} \langle |\kappa_{\boldsymbol{\ell}}|^2 \rangle$$

where the κ field is generated from the smoothed convergence power spectrum $P(\ell)$. The corresponding variance is given by

$$\langle \kappa(\boldsymbol{\theta})^2 \rangle = \int \frac{d^2\ell}{(2\pi)^2} P(\ell) = \sum_{\ell} \frac{(d\ell)^2}{(2\pi)^2} P(\ell) = \frac{1}{L^2} \sum_{\ell} P(\ell)$$

Here the sum includes both positive and negative Fourier modes. Thus this sum determines the variance σ_0. At first sight, one might infer that $\langle|\kappa_{\boldsymbol{\ell}}|^2\rangle = L^2 P(\ell)$. However, because the conjugate modes satisfy $\kappa_{-\boldsymbol{\ell}} = \kappa^*_{\boldsymbol{\ell}}$, the power assigned to each stored mode must be split between the conjugate pair. In other words, $\kappa_{\boldsymbol{\ell}}$ and $\kappa_{-\boldsymbol{\ell}}$ are not independent degrees of freedom, so each mode cannot be assigned the full variance $L^2 P(\ell)$. Therefore,

$$\langle|\kappa_{\boldsymbol{\ell}}|^2\rangle + \langle|\kappa_{-\boldsymbol{\ell}}|^2\rangle = L^2 P(\ell) \quad \Rightarrow \quad \langle|\kappa_{\boldsymbol{\ell}}|^2\rangle = \frac{L^2}{2} P(\ell)$$

With the Python FFT conventions, `rfft2` and `irfft2` already account for the full set of frequencies, so no extra factor of 2 is needed. In the two-dimensional implementation, we have

$$\kappa(\mathbf{n}) = \texttt{ifft2}(F) = \frac{1}{N^2} \sum_{\boldsymbol{\ell}} F_{\boldsymbol{\ell}} e^{i2\pi \boldsymbol{\ell}\cdot\boldsymbol{n}/N} \quad \Rightarrow \quad \langle\kappa(\mathbf{n})^2\rangle = \frac{1}{N^4} \sum_{\boldsymbol{\ell}} \langle|F_{\boldsymbol{\ell}}|^2\rangle$$

To obtain $\langle\kappa^2\rangle \approx (1/L^2)\sum_\ell P(\ell)$, we require $\langle|F_{\boldsymbol{\ell}}|^2\rangle = \frac{N^4}{2L^2} P(\ell)$, where the factor of 2 comes from the conjugate-mode pair. Thus, for a real two-dimensional Gaussian κ field generated with `irfft2`,

$$F_\ell = \frac{N^2}{\sqrt{2}L} \sqrt{P(\ell)}(a_\ell + ib_\ell), \qquad a_\ell, b_\ell \sim \mathcal{N}(0, 1)$$

This normalization gives the desired variance, $\langle\kappa^2\rangle = \int_\ell P(\ell)$. The factor $1/\sqrt{2}$ appears when both members of a conjugate pair ($\pm k$) are stored explicitly, in order to avoid double counting the power.

In the three-dimensional case, the $1/2$ factor is absent if one uses an `rfft` along a single axis. In this case,

$$\langle\delta(\boldsymbol{x})^2\rangle = \int \frac{d^3k}{(2\pi)^3} P(k) = \sum_k \frac{(\Delta k)^3}{(2\pi)^3} P(k) = \frac{1}{L^3} \sum_k P(k)$$

To match the FFT definition,

$$\langle\delta(\mathbf{n})^2\rangle = \frac{1}{N^6} \sum_k \langle|F_{\mathbf{k}}|^2\rangle$$

so in the three-dimensional case we use the normalization

$$F_{\mathbf{k}} = \frac{N^3}{L^{3/2}} P(k)$$

In the code, we use `fftfreq` and `rfftfreq` for different axes because `rfft2` stores the full set of modes along one axis and only the non-negative modes along the other. Therefore, the frequency grids must be generated with `fftfreq` for the full axis and `rfftfreq` for the reduced axis, such that

$$\sum_{\text{full plane}} P(k) \;=\; \sum_{\text{rfft plane}} w(k_y)\, P(k).$$

This resolves the factor-of-two issue. Another issue arises from the accuracy of the power-spectrum integration. Because the map is defined on a finite box with a discrete grid and a smoothing or pixel window, there is an unavoidable mismatch in the mode domain. If one integrates $P_\kappa(\ell)$ beyond the FFT Nyquist domain, the theoretical variance becomes too large, leading to ratios such as 0.8–0.9. The Nyquist frequency, which is the maximum resolvable frequency, is $\ell_{\mathrm{Nyq}} = \frac{\pi}{\Delta\theta} = \frac{\pi N}{L}$. The FFT domain should therefore be $|\ell_x| \leqslant \ell_{\mathrm{Nyq}}, \quad 0 \leqslant \ell_y \leqslant \ell_{\mathrm{Nyq}}$. Using $2\pi N/L$ would double-count aliased power.

The remaining deviation at the $\sim$ 1–2% level comes from the discrete grid, the continuum approximation, the discretization of the Limber integral, and the discrete sampling of the smoothing window in Fourier space. For example, Figure 4.2 shows that the Gaussian prediction for changing σ_8 should theoretically be zero, but numerical discretization at nonlinear scales causes it to deviate from zero.

A.2 Critical-point finding algorithm

Critical points are identified from local gradients and Hessians using Newton's method [30]. A candidate is first identified on the grid, refined once by a local Taylor expansion, and then classified by the Hessian eigenvalues at the refined position. In one dimension, the method iterates toward the root of $f(x) = 0$. For a point $(x_0, f(x_0))$, the next estimate is $x_1 = x_0 - f(x_0)/f'(x_0)$. For a system of two equations, such as $\partial_x f = 0$ and $\partial_y f = 0$, we use a discrete version of this procedure. We define $\delta_x = x_i - x_c$ and $\delta y = y_i - y_c$ as offsets from a grid center (x_c, y_c) at which the gradient approximately vanishes and which is identified as a critical point. The Taylor expansion of the field around the central grid point is

$$f(x, y) = f_c + f_{,x}\delta x + f_{,y}\delta y + \frac{1}{2}f_{,xx}\delta x^2 + f_{,xy}\delta_x\delta_y + \frac{1}{2}f_{,yy}\delta y^2 + \cdots , \tag{A.1}$$

where $f_c = f(x_c, y_c)$, the grid positions are $x_i = x_0 + i\Delta x$ and $y_j = y_0 + j\Delta y$, where Δx and Δy are grid spacings. The central finite-difference derivatives in 2D are

$$\begin{aligned} f_{,x} &= \frac{f_{i+1,j} - f_{i-1,j}}{2\Delta x}, \qquad & f_{,y} &= \frac{f_{i,j+1} - f_{i,j-1}}{2\Delta y}, \\ f_{,xx} &= \frac{f_{i+1,j} - 2f_{i,j} + f_{i-1,j}}{(\Delta x)^2}, \qquad & f_{,xy} &= \frac{f_{i+1,j+1} - f_{i+1,j-1} - f_{i-1,j+1} + f_{i-1,j-1}}{4\Delta x\Delta y}. \end{aligned}$$

We also define a block in a local 3×3 patch of the field, such as `(i+[-1,0,1])%nx`, to enforce periodic wrapping in the x direction, where `nx` is the number of columns. The critical-point condition is

$$\frac{\partial f}{\partial x} \approx f_{,x} + f_{,xx}\delta x + f_{,xy}\delta y = 0, \qquad \frac{\partial f}{\partial y} \approx f_{,y} + f_{,yy}\delta y + f_{,xy}\delta x = 0,$$

or written as

$$\begin{pmatrix} f_{,xx} & f_{,xy} \\ f_{,xy} & f_{,yy} \end{pmatrix}\begin{pmatrix} \delta x \\ \delta y \end{pmatrix} = -\begin{pmatrix} f_{,x} \\ f_{,y} \end{pmatrix},$$

where the leftmost matrix is the 2D Hessian matrix H. We then solve this equation numerically.

A critical point (x_i, y_i) is accepted in a cell when the Newton offset satisfies $\delta x \lesssim \Delta x/2$ and $\delta y \lesssim \Delta y/2$. Defining $D = \det H = f_{,xx}f_{,yy} - f_{,xy}^2$ and $T = \operatorname{tr} H = f_{,xx} + f_{,yy}$, for peak $D > 0, T < 0$; for void $D > 0, T > 0$; for saddle $D < 0$. After identifying the grid cell containing a critical point, we use the Newton offset to refine its position and Eq. (A.1) to refine its field amplitude, thereby reducing incorrect bin assignments. In the tests, we adopt an offset equal to 0.6 to avoid missing points, since the field may be relatively flat around some positions. However, after refinement, some grid points may gather at very similar positions and should therefore be identified as the same point. Accordingly, when appending a new position, we also compute the closest distance to existing points to avoid duplicates.

The pipeline is the same in 3D, except that the numbers of negative eigenvalues of the 3D Hessian from 3 to 0 are classified as peak, filament, wall, and void, respectively. Reference [24] presents an algorithm for counting the Euler characteristic directly, but we do not adopt that approach here. Instead, we use Eq. (3.7). All the code are given by ChatGPT models.

A.3 Hartlap covariance and Student-t likelihood

This appendix explains how we treat the covariance matrix used in the likelihood analysis of Chap. 4. The main issue is that the true covariance matrix is not known exactly. Instead, it is estimated from a finite number of simulations. As a result, both the sample covariance and its inverse contain statistical noise.

Let the data vector have dimension p, and let $\boldsymbol{d}_s$ be the measurement from the s-th simulation realization. The sample mean and covariance are defined in Eqs. (3.34) and (4.1). If the true covariance matrix were known, the Gaussian likelihood would involve the true precision matrix $\Psi \equiv C^{-1}$. In practice, however, we only have the sample covariance $\hat{C}$, and the naive choice would be to use $\hat{C}^{-1}$ in the likelihood. This is not quite correct, because the inverse of an estimated covariance matrix is a biased estimator of the true precision matrix. To see this, assume that the simulation measurements are independent Gaussian draws, $\boldsymbol{d}_s \sim \mathcal{N}_p(\boldsymbol{\mu}, C)$, where C is the true covariance matrix. If the true mean $\boldsymbol{\mu}$ were known, the scatter matrix formed from the simulations would be a sum of Gaussian outer products. Such a matrix follows a Wishart distribution. In reality the true mean is not known and is replaced by the sample mean $\bar{\boldsymbol{d}}$. This uses up one degree of freedom, so we define

$$A \equiv \sum_{s=1}^{N_{\rm sim}} (\boldsymbol{d}_s - \bar{\boldsymbol{d}})(\boldsymbol{d}_s - \bar{\boldsymbol{d}})^T = n\hat{C}, \qquad n \equiv N_{\rm sim} - 1.$$

For Gaussian data vectors, this scatter matrix follows a Wishart distribution, $A \sim \mathcal{W}_p(C, n)$, which is the multivariate analogue of the χ^2 distribution. It describes how covariance matrices fluctuate when they are estimated from a finite number of Gaussian samples. The first moment of a Wishart matrix is $\langle A \rangle = nC$. Therefore $\langle \hat{C} \rangle = \langle \frac{A}{n} \rangle = C$ and the sample covariance itself is an unbiased estimator of the true covariance.

The inverse covariance behaves differently. Matrix inversion is nonlinear, so in general $\langle \hat{C}^{-1} \rangle \neq \langle \hat{C} \rangle^{-1}$. For a Wishart matrix, the inverse moment is

$$\langle A^{-1} \rangle = \frac{1}{n-p-1} C^{-1}, \qquad n > p+1.$$

Thus,

$$\langle \hat{C}^{-1} \rangle = n \langle A^{-1} \rangle = \frac{n}{n-p-1} C^{-1} = \frac{N_{\rm sim} - 1}{N_{\rm sim} - p - 2} C^{-1}.$$

The factor multiplying C^{-1} is larger than unity. Hence the naive inverse covariance systematically overestimates the true precision matrix. In a Gaussian likelihood, this would make the χ^2 artificially too large and would lead to overly tight constraints. Finally, we have the Hartlap factor in Eq. (4.2) that corrects for this bias. For the analysis in Chap. 4, the full combined critical-point data vector has dimension $p = 44$, and we use $N_{\rm sim} = 1000$ simulations. The Hartlap factor is therefore

$$\alpha = \frac{1000 - 44 - 2}{1000 - 1} = \frac{954}{999} \simeq 0.955.$$

Thus the Hartlap correction reduces the naive precision matrix by about 4.5%. It fixes the mean bias of the inverse covariance, but it does not account for the remaining uncertainty in the covariance estimate itself.

The Student-t likelihood is obtained by integrating over the possible true covariance matrices compatible with $\hat{C}$:

$$P(\boldsymbol{d}|\boldsymbol{\mu}, \hat{C}) = \int dC\, P(\boldsymbol{d}|\boldsymbol{\mu}, C) P(C|\hat{C}).$$

This marginalization propagates the uncertainty of the covariance estimate into the likelihood. The sampling distribution of $\hat{C}$ is Wishart:

$$P(\hat{C}|C) \propto |C|^{-n/2} \exp\left[-\frac{n}{2} \mathrm{tr}(C^{-1}\hat{C})\right], \qquad n = N_{\rm sim} - 1.$$

Combining the Gaussian likelihood with the covariance posterior, $P(C|\hat{C}) \propto P(\hat{C}|C)P(C)$, we obtain an integrand proportional to $P(\boldsymbol{d}|\boldsymbol{\mu}, C)\, P(\hat{C}|C)\, P(C)$. The dependence of the exponential part on (C) is

$$-\frac{1}{2}\Delta^T C^{-1}\Delta - \frac{n}{2}\mathrm{tr}(C^{-1}\hat{C}) = -\frac{1}{2}\mathrm{tr}\Big[C^{-1}\Big(n\hat{C} + \Delta\Delta^T\Big)\Big].$$

The prior $P(C)$ is proportional to $|C|^{-(p+1)/2}$, then

$$\int_{C>0} dC\, |C|^{-(n+p+2)/2} \exp\left[-\frac{1}{2}\mathrm{tr}(C^{-1}(n\hat{C} + \Delta\Delta^T))\right] \propto |n\hat{C} + \Delta\Delta^T|^{-(n+1)/2}.$$

which gives a determinant factor involving this matrix (we omitted details here). Using the matrix determinant lemma,

$$\det\Big(n\hat{C} + \Delta\Delta^T\Big) = \det\Big(n\hat{C}\Big)\left[1 + \frac{1}{n}\Delta^T\hat{C}^{-1}\Delta\right] \propto 1 + \frac{\chi^2}{n},$$

where $\chi^2 = \Delta^T\hat{C}^{-1}\Delta$, then we obtains the marginalized likelihood

$$\mathcal{L}_\mathrm{t} = \frac{\Gamma(N_\mathrm{sim}/2)}{\Gamma[(N_\mathrm{sim} - p)/2]\, [\pi(N_\mathrm{sim} - 1)]^{p/2} |\hat{C}|^{1/2}} \left[1 + \frac{\chi^2}{N_\mathrm{sim} - 1}\right]^{-N_\mathrm{sim}/2}.$$

When $\hat{C}$ is fixed during parameter inference, the prefactor does not depend on the model parameters and can be absorbed into the normalization. The log-likelihood then becomes Eq. (4.3). The connection to the ordinary Gaussian likelihood is clear in the limit of many simulations. For large N_sim,

$$\ln\left[1 + \frac{\chi^2}{N_\mathrm{sim} - 1}\right] \simeq \frac{\chi^2}{N_\mathrm{sim} - 1}.$$

Therefore

$$\ln\mathcal{L}_\mathrm{t} \to -\frac{1}{2}\chi^2 + \mathrm{const}, \qquad N_\mathrm{sim} \to \infty.$$

Thus the Student-t likelihood reduces to the usual Gaussian likelihood when the covariance is known accurately.

At finite N_sim, the Student-t likelihood has heavier tails than a Gaussian likelihood. This means that large residuals are not interpreted solely as evidence for a poor model fit, since part of the discrepancy may instead originate from uncertainty in the estimated covariance matrix. In this sense, the Student-t likelihood gives a more conservative treatment of covariance uncertainty.

The Hartlap correction and the Student-t likelihood therefore address different aspects of the same finite-simulation problem. The Hartlap factor corrects the mean bias of the inverse covariance used in a Gaussian likelihood. The Student-t likelihood instead marginalizes over the unknown true covariance matrix and modifies the shape of the likelihood. In the limit $N_\mathrm{sim} \to \infty$, both approaches converge to the ordinary Gaussian likelihood with the true covariance.

A.4 Specific coefficients

This appendix collects the explicit coefficient functions entering Eqs. (3.26) and (3.28).

First, we introduce the 2D coefficients obtained from Eq. (3.26) by replacing $H_{i-1,j} \to H_{i,j}$ for $-d\bar{n}^\star/d\nu$:

$$G'_{ijkl0}(\nu) = \frac{1}{4\pi}\left(\frac{\sigma_2}{\sigma_1}\right)^2 X_k \int_{a^\star}^{b^\star} dx\, H_{i,j}(\nu, x)\mathcal{N}(\nu, x) f^\star_{l0}(x),$$

where $X_0 = 1$ and $X_1 = -1$. Using Eqs. (3.22) and (3.13), and suppressing the common prefactor $(\sigma_2/\sigma_1)^2/(4\pi)$, the required terms are given in Eq. (A.2), with species-dependent kernel functions

$$f_{00}^{\text{peak/void}}(x) = e^{-x^2} + x^2 - 1, \qquad f_{10}^{\text{peak/void}}(x) = (1 + x^2)e^{-x^2} - 1,$$
$$f_{00}^{\text{saddle}}(x) = -e^{-x^2}, \qquad f_{10}^{\text{saddle}}(x) = -e^{-x^2}(1 + x^2),$$
$$f_{00}^{\chi}(x) = x^2 - 1, \qquad f_{10}^{\chi}(x) = -1.$$

For the 3D coefficients defined in Eq. (3.28), we similarly suppress the common prefactor $\left(\sigma_2/(\sqrt{3}\sigma_1)\right)^3/(2\pi)^{3/2}$ The explicit terms needed in Eq. (3.27) are given in Eq. (A.2). For peaks and voids, $f^\star_{lm}(x)$ take the common form

$$f_{00}^{\text{peak/void}} = f_{00a} + f_{00b} + f_{00c} + f_{00d}$$
$$\equiv \frac{x}{2}(x^2 - 3)\,\text{erf}\left(\frac{1}{2}\sqrt{\frac{5x}{2}}\right) + \frac{x}{2}(x^2 - 3)\,\text{erf}\left(\sqrt{\frac{5x}{2}}\right) + \sqrt{\frac{2}{5\pi}}\left(\frac{x^2}{2} - \frac{8}{5}\right)e^{-5x^2/2} + \sqrt{\frac{2}{5\pi}}\left(\frac{31x^2}{4} + \frac{8}{5}\right)e^{-5x^2/8}$$
$$f_{10}^{\text{peak/void}} = f_{10a} + f_{10b} + f_{10c} + f_{10d}$$
$$\equiv -\frac{3x}{2}\,\text{erf}\left(\frac{1}{2}\sqrt{\frac{5}{2}}x\right) - \frac{3x}{2}\,\text{erf}\left(\sqrt{\frac{5}{2}}x\right) - \frac{12}{5}\sqrt{\frac{2}{5\pi}}e^{-5x^2/2} + \frac{12}{5}\sqrt{\frac{2}{5\pi}}\left(1 + \frac{15x^2}{8}\right)\left(1 + \frac{15x^2}{16}\right)e^{-5x^2/8}$$
$$f_{01}^{\text{peak/void}} = f_{01a} + f_{01b} + f_{01c} + f_{01d}$$
$$\equiv -\frac{21}{25}\,\text{erf}\left(\frac{1}{2}\sqrt{\frac{5}{2}}x\right) - \frac{21}{25}\,\text{erf}\left(\sqrt{\frac{5}{2}}x\right) + \frac{27x}{10}\sqrt{\frac{2}{5\pi}}\left(\frac{2}{15}e^{-5x^2/2}\right) + \frac{27x}{10}\sqrt{\frac{2}{5\pi}}\left(\frac{11}{5} + \frac{x^2}{4} + \frac{5x^4}{16}\right)e^{-5x^2/8}.$$

Here $f_{00a}, \ldots, f_{01d}$ denote the successive additive terms in the corresponding peak/void expressions above. Thus, for filaments and walls it is convenient to split the functions into positive- and negative-x branches, since the integration range is $(-\infty, \infty)$:

$$f_{00+}^{\text{fila}} = -\frac{x}{2}(x^2 - 3) + f_{00a} + f_{00d}, \qquad f_{00-}^{\text{fila}} = -\frac{x}{2}(x^2 - 3) - f_{00b} - f_{00c},$$
$$f_{10+}^{\text{fila}} = \frac{3x}{2} + f_{10a} + f_{10d}, \qquad f_{10-}^{\text{fila}} = \frac{3x}{2} - f_{10b} - f_{10c},$$
$$f_{01+}^{\text{fila}} = \frac{21}{25} + f_{01a} + f_{01d}, \qquad f_{01-}^{\text{fila}} = \frac{21}{25} - f_{01b} - f_{01c},$$

$$f_{00+}^{\text{wall}} = \frac{x}{2}(x^2 - 3) - f_{00b} - f_{00c}, \qquad f_{00-}^{\text{wall}} = \frac{x}{2}(x^2 - 3) + f_{00a} + f_{00d},$$
$$f_{10+}^{\text{wall}} = -\frac{3x}{2} - f_{10b} - f_{10c}, \qquad f_{10-}^{\text{wall}} = -\frac{3x}{2} + f_{10a} + f_{10d},$$
$$f_{01+}^{\text{wall}} = -\frac{21}{25} - f_{01b} - f_{01c}, \qquad f_{01-}^{\text{wall}} = -\frac{21}{25} + f_{01a} + f_{01d}.$$

The $+$ branch is integrated over $x \in [0, \infty)$, whereas the $-$ branch is integrated over $x \in (-\infty, 0]$. For the Euler characteristic, the corresponding kernel functions are

$$f_{00}^{\chi} = x(x^2 - 3), \qquad f_{10}^{\chi} = -3x, \qquad f_{01}^{\chi} = -\frac{42}{25}.$$

$$
\begin{aligned}
&G'_{00000} \propto X_0 \int_{a^\star}^{b^\star} dx\, \mathcal{N}(\nu,x) f^\star_{00}(x), \qquad G'_{00001} \propto X_0 \int_{a^\star}^{b^\star} dx\, \mathcal{N}(\nu,x) f^\star_{01}(x),\\
&G'_{30000} \propto X_0 \int_{a^\star}^{b^\star} dx\, (-1)\mathcal{N}(\nu,x) f^\star_{00}(x) \frac{-\gamma^3 x^3 + 3\gamma^2 \nu x^2 - 3\gamma(\nu^2+\gamma^2-1)x + \nu^3 + 3(\gamma^2-1)\nu}{(\gamma^2-1)^3},\\
&G'_{21000} \propto X_0 \int_{a^\star}^{b^\star} dx\, (-1)\mathcal{N}(\nu,x) f^\star_{00}(x) \frac{\gamma^2 x^3 - \gamma(2+\gamma^2)\nu x^2 + (1+2\gamma^2)(\nu^2+\gamma^2-1)x - \gamma\nu(\nu^2+3\gamma^2-3)}{(\gamma^2-1)^3},\\
&G'_{12000} \propto X_0 \int_{a^\star}^{b^\star} dx\, (-1)\mathcal{N}(\nu,x) f^\star_{00}(x)\\
&\qquad\qquad \times \frac{-\gamma x^3 + (2\gamma^2\nu+\nu)x^2 + (-\gamma^3\nu^2 - 2\gamma\nu^2 - 3\gamma^3 + 3\gamma)x + (2\gamma^4\nu + \gamma^2\nu^3 - \gamma^2\nu - \nu)}{(\gamma^2-1)^3},\\
&G'_{03000} \propto X_0 \int_{a^\star}^{b^\star} dx\, (-1)\mathcal{N}(\nu,x) f^\star_{00}(x) \frac{x^3 - 3\gamma\nu x^2 + 3\gamma^2\nu^2 x + 3(\gamma^2-1)x - \gamma^3\nu^3 - 3\gamma(\gamma^2-1)\nu}{(\gamma^2-1)^3},\\
&G'_{10100} \propto X_1 \int_{a^\star}^{b^\star} dx\, (-1)\mathcal{N}(\nu,x) f^\star_{00}(x) \frac{-\gamma x + \nu}{\gamma^2-1}, \qquad G'_{01100} \propto X_1 \int_{a^\star}^{b^\star} dx\, (-1)\mathcal{N}(\nu,x) f^\star_{00}(x) \frac{x - \gamma\nu}{\gamma^2-1},\\
&G'_{10010} \propto X_0 \int_{a^\star}^{b^\star} dx\, (-1)\mathcal{N}(\nu,x) f^\star_{10}(x) \frac{-\gamma x + \nu}{\gamma^2-1}, \qquad G'_{01010} \propto X_0 \int_{a^\star}^{b^\star} dx\, (-1)\mathcal{N}(\nu,x) f^\star_{10}(x) \frac{x - \gamma\nu}{\gamma^2-1}.
\end{aligned}
\tag{A.2}
$$

These expressions provide all coefficient functions needed for the analytic formula in the main text.

Appendix B

Attempts to derive kurtosis-order corrections

This chapter records an attempted derivation of the kurtosis-order correction. The calculation is included as a reference for the intermediate steps rather than as a final result. We follow the pipeline of Ref. [9]. First, we calculate the second-order response term involving $\mathcal{G}_4$, no $\mathcal{G}_6$,

$$\frac{1}{24}\int \frac{d^N k_1}{(2\pi)^N}\frac{d^N k_2}{(2\pi)^N}\frac{d^N k_3}{(2\pi)^N}\frac{d^N k_4}{(2\pi)^N}\left\langle \tilde{f}(\boldsymbol{k}_1)\tilde{f}(\boldsymbol{k}_2)\tilde{f}(\boldsymbol{k}_3)\tilde{f}(\boldsymbol{k}_4)\right\rangle_{\mathrm{c}}\mathcal{G}_4(\boldsymbol{k}_1,\boldsymbol{k}_2,\boldsymbol{k}_3,\boldsymbol{k}_4) \tag{B.1}$$

which requires the fourth derivatives of $\mathcal{P}_{\mathrm{G}}$ with respect to η and $\tilde{\zeta}$, followed by the full functional derivative and the rotational average. The main difficulty is the construction of the rank-8 symmetric tensor. We suspect that this step may be the source of the error. The steps are analogous to the leading-order correction, but the algebra is difficult to carry out by hand, so we developed Mathematica code[1] for the calculation.

B.1 Intermediate steps

B.1.1 The $\mathcal{G}_4$ calculation

To calculate $\mathcal{G}_4$, we need the first- through fourth-order partial derivatives of $\mathcal{P}_{\mathrm{G}}$ with respect to η and $\tilde{\zeta}$. The first- to third-order results are given in the appendix of Ref. [9]; here we give the fourth-order results.

For η, the result is

$$\begin{aligned}\frac{\partial^4}{\partial\eta_i\partial\eta_j\partial\eta_k\partial\eta_l}\mathcal{P}_{\mathrm{G}} = \Big[&4(\delta_{ij}\delta_{kl}+\delta_{ik}\delta_{jl}+\delta_{il}\delta_{jk})\frac{\partial^2}{\partial(\eta^2)^2}\\ &+8(\delta_{ij}\eta_k\eta_l+\delta_{ik}\eta_j\eta_l+\delta_{il}\eta_j\eta_k+\delta_{jk}\eta_i\eta_l+\delta_{jl}\eta_i\eta_k+\delta_{kl}\eta_i\eta_j)\frac{\partial^3}{\partial(\eta^2)^3}+16\eta_i\eta_j\eta_k\eta_l\frac{\partial^4}{\partial(\eta^2)^4}\Big]\mathcal{P}_{\mathrm{G}}\end{aligned}$$

The corresponding angular average, treated as a fully symmetric rank-4 tensor, is

$$\langle\eta_i\eta_j\eta_k\eta_l\rangle_\Omega = \frac{\eta^4}{N(N+2)}(\delta_{ij}\delta_{kl}+\delta_{ik}\delta_{jl}+\delta_{il}\delta_{jk})$$

Next, for $\tilde{\zeta}$, we need rank-8 symmetric tensors that are $SO(N)$ rotationally invariant, symmetric within each index pair, and symmetric under exchange of index pairs. We define the following tensor

[1] The Mathematica code is available at `github.com/banligo/CPsNLO`.

basis:

$$
\begin{aligned}
\mathcal{D}^{(4)}_{ij,kl,mn,pq} &\equiv \delta_{ij}\delta_{kl}\delta_{mn}\delta_{pq}, \\
\mathcal{S}^{(2,2)}_{ij,kl,mn,pq} &\equiv \delta_{iq}\delta_{jp}\delta_{kn}\delta_{lm} + \delta_{in}\delta_{jm}\delta_{kq}\delta_{lp} + \delta_{il}\delta_{jk}\delta_{mq}\delta_{np}, \\
\mathcal{S}^{(3,1)}_{ij,kl,mn,pq} &\equiv \delta_{iq}\delta_{jp}\delta_{kl}\delta_{mn} + \delta_{ij}\delta_{kq}\delta_{lp}\delta_{mn} + \delta_{ij}\delta_{kl}\delta_{mq}\delta_{np} + \delta_{in}\delta_{jm}\delta_{kl}\delta_{pq} + \delta_{ij}\delta_{kn}\delta_{lm}\delta_{pq} + \delta_{il}\delta_{jk}\delta_{mn}\delta_{pq}, \\
\mathcal{T}^{(1)}_{ij,kl,mn,pq} &\equiv \delta_{kl}\delta_{mn}\delta_{pq}\,\tilde{\zeta}_{ij} + \delta_{ij}\delta_{mn}\delta_{pq}\,\tilde{\zeta}_{kl} + \delta_{ij}\delta_{kl}\delta_{pq}\,\tilde{\zeta}_{mn} + \delta_{ij}\delta_{kl}\delta_{mn}\,\tilde{\zeta}_{pq}, \\
\mathcal{T}^{(2)}_{ij,kl,mn,pq} &\equiv \delta_{kq}\delta_{lp}\delta_{mn}\,\tilde{\zeta}_{ij} + \delta_{kl}\delta_{mq}\delta_{np}\,\tilde{\zeta}_{ij} + \delta_{kn}\delta_{lm}\delta_{pq}\,\tilde{\zeta}_{ij} + \delta_{iq}\delta_{jp}\delta_{mn}\,\tilde{\zeta}_{kl} + \delta_{ij}\delta_{mq}\delta_{np}\,\tilde{\zeta}_{kl} + \delta_{in}\delta_{jm}\delta_{pq}\,\tilde{\zeta}_{kl} \\
&\quad + \delta_{iq}\delta_{jp}\delta_{kl}\,\tilde{\zeta}_{mn} + \delta_{ij}\delta_{kq}\delta_{lp}\,\tilde{\zeta}_{mn} + \delta_{il}\delta_{jk}\delta_{pq}\,\tilde{\zeta}_{mn} + \delta_{in}\delta_{jm}\delta_{kl}\,\tilde{\zeta}_{pq} + \delta_{ij}\delta_{kn}\delta_{lm}\,\tilde{\zeta}_{pq} + \delta_{il}\delta_{jk}\delta_{mn}\,\tilde{\zeta}_{pq}, \\
\mathcal{U}^{(2)}_{ij,kl,mn,pq} &\equiv \delta_{mn}\delta_{pq}\,\tilde{\zeta}_{ij}\tilde{\zeta}_{kl} + \delta_{kl}\delta_{pq}\,\tilde{\zeta}_{ij}\tilde{\zeta}_{mn} + \delta_{ij}\delta_{pq}\,\tilde{\zeta}_{kl}\tilde{\zeta}_{mn} + \delta_{kl}\delta_{mn}\,\tilde{\zeta}_{ij}\tilde{\zeta}_{pq} + \delta_{ij}\delta_{mn}\,\tilde{\zeta}_{kl}\tilde{\zeta}_{pq} + \delta_{ij}\delta_{kl}\,\tilde{\zeta}_{mn}\tilde{\zeta}_{pq}, \\
\mathcal{U}^{(2,\mathrm{ex})}_{ij,kl,mn,pq} &\equiv \delta_{mq}\delta_{np}\,\tilde{\zeta}_{ij}\tilde{\zeta}_{kl} + \delta_{kq}\delta_{lp}\,\tilde{\zeta}_{ij}\tilde{\zeta}_{mn} + \delta_{iq}\delta_{jp}\,\tilde{\zeta}_{kl}\tilde{\zeta}_{mn} + \delta_{kn}\delta_{lm}\,\tilde{\zeta}_{ij}\tilde{\zeta}_{pq} + \delta_{in}\delta_{jm}\,\tilde{\zeta}_{kl}\tilde{\zeta}_{pq} + \delta_{il}\delta_{jk}\,\tilde{\zeta}_{mn}\tilde{\zeta}_{pq}, \\
\mathcal{V}^{(3)}_{ij,kl,mn,pq} &\equiv \delta_{pq}\,\tilde{\zeta}_{ij}\tilde{\zeta}_{kl}\tilde{\zeta}_{mn} + \delta_{mn}\,\tilde{\zeta}_{ij}\tilde{\zeta}_{kl}\tilde{\zeta}_{pq} + \delta_{kl}\,\tilde{\zeta}_{ij}\tilde{\zeta}_{mn}\tilde{\zeta}_{pq} + \delta_{ij}\,\tilde{\zeta}_{kl}\tilde{\zeta}_{mn}\tilde{\zeta}_{pq}.
\end{aligned}
$$

Here, $\mathcal{D}^{(4)}$ is completely disconnected, with each pair contracted only with itself; $\mathcal{S}^{(3,1)}$ contains one crossed pair contraction plus two trivial trace contractions, summed over all choices; and $\mathcal{S}^{(2,2)}$ contains two crossed pair contractions. The tensor $\mathcal{T}^{(1)}$ contains one $\tilde{\zeta}$ with disconnected δ factors, while $\mathcal{T}^{(2)}$ contains one $\tilde{\zeta}$ multiplied by one crossed contraction and one trivial contraction, summed over all choices. For $\mathcal{U}^{(2,\mathrm{ex})}$, "ex" denotes exchange or crossed contraction of the remaining two index pairs. Then, the fourth-order results for $\tilde{\zeta}$ is

$$
\begin{aligned}
\frac{\partial^4 \mathcal{P}_G}{\partial\zeta_{ij}\partial\zeta_{kl}\partial\zeta_{mn}\partial\zeta_{pq}} = \Bigg\{ & \mathcal{D}^{(4)}_{ij,kl,mn,pq}\frac{\partial^4}{\partial J_1^4} + \frac{4}{(N-1)^2}\left[N^2\mathcal{S}^{(2,2)}_{ij,kl,mn,pq} - NS^{(3,1)}_{ij,kl,mn,pq} + 3D^{(4)}_{ij,kl,mn,pq}\right]\frac{\partial^2}{\partial J_2^2} \\
& + \frac{2}{(N-1)}\left[N\mathcal{S}^{(3,1)}_{ij,kl,mn,pq} - 6\mathcal{D}^{(4)}_{ij,kl,mn,pq}\right]\frac{\partial^3}{\partial J_1^2\partial J_2} + \frac{16N^4}{(N-1)^4}\tilde{\zeta}_{ji}\tilde{\zeta}_{lk}\tilde{\zeta}_{nm}\tilde{\zeta}_{qp}\frac{\partial^4}{\partial J_2^4} \\
& + \frac{4N}{(N-1)^2}\left[-N\mathcal{T}^{(2)}_{ij,kl,mn,pq} + 3\mathcal{T}^{(1)}_{ij,kl,mn,pq}\right]\frac{\partial^3}{\partial J_1\partial J_2^2} \\
& - \frac{2N}{N-1}\mathcal{T}^{(1)}_{ij,kl,mn,pq}\frac{\partial^4}{\partial J_1^3\partial J_2} + \frac{8N^2}{(N-1)^3}\left[N\mathcal{U}^{(2,\mathrm{ex})}_{ij,kl,mn,pq} - \mathcal{U}^{(2)}_{ij,kl,mn,pq}\right]\frac{\partial^3}{\partial J_2^3} \\
& + \frac{4N^2}{(N-1)^2}\mathcal{U}^{(2)}_{ij,kl,mn,pq}\frac{\partial^4}{\partial J_1^2\partial J_2^2} - \frac{8N^3}{(N-1)^3}\mathcal{V}^{(3)}_{ij,kl,mn,pq}\frac{\partial^4}{\partial J_1\partial J_2^3}\Bigg\}\mathcal{P}_G.
\end{aligned}
$$

The fourth moment angular average for $\tilde{\zeta}$ is

$$
\langle\tilde{\zeta}_{ij}\tilde{\zeta}_{kl}\tilde{\zeta}_{mn}\tilde{\zeta}_{pq}\rangle_\Omega = \frac{12(N-1)J_2^2}{N^2(N+2)(N^2+N+2)}\left(\frac{1}{N^2}\mathcal{A}^{(4)} - \frac{1}{6N}\mathcal{B}^{(4)} + \frac{1}{12}\mathcal{D}^{(4)}\right) \tag{B.2}
$$

where

$$
\begin{aligned}
\mathcal{A}^{(4)}_{ij,kl,mn,pq} &= \delta_{ij}\delta_{kl}\delta_{mn}\delta_{pq} \\
\mathcal{B}^{(4)}_{ij,kl,mn,pq} &= \delta_{ij}\delta_{kl}(\delta_{mp}\delta_{nq} + \delta_{mq}\delta_{np}) + \delta_{ij}\delta_{mn}(\delta_{kp}\delta_{lq} + \delta_{kq}\delta_{lp}) \\
&\quad + \delta_{ij}\delta_{pq}(\delta_{km}\delta_{ln} + \delta_{kn}\delta_{lm}) + \delta_{kl}\delta_{mn}(\delta_{ip}\delta_{jq} + \delta_{iq}\delta_{jp}) \\
&\quad + \delta_{kl}\delta_{pq}(\delta_{im}\delta_{jn} + \delta_{in}\delta_{jm}) + \delta_{mn}\delta_{pq}(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}) \\
\mathcal{D}^{(4)}_{ij,kl,mn,pq} &= (\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk})(\delta_{mp}\delta_{nq} + \delta_{mq}\delta_{np}) \\
&\quad + (\delta_{im}\delta_{jn} + \delta_{in}\delta_{jm})(\delta_{kp}\delta_{lq} + \delta_{kq}\delta_{lp}) \\
&\quad + (\delta_{ip}\delta_{jq} + \delta_{iq}\delta_{jp})(\delta_{km}\delta_{ln} + \delta_{kn}\delta_{lm})
\end{aligned} \tag{B.3}
$$

The construction method for rank-$2r$ symmetric traceless tensor is described in Sec. B.1.3.

Applying the functional derivative with the code gives $\mathcal{G}_4$:

$$
\begin{aligned}
\mathcal{G}_4(\boldsymbol{k}_1,\boldsymbol{k}_2,\boldsymbol{k}_3,\boldsymbol{k}_4) &= \frac{G_{40000}}{\sigma_0^4} + \frac{G_{31000}}{\sigma_0^3\sigma_2}\sum_a {k_a}^2 + \frac{G_{22000}}{\sigma_0^2\sigma_2^2}\sum_{a<b} {k_a}^2{k_b}^2 + \frac{G_{13000}}{\sigma_0\sigma_2^3}\sum_{a<b<c} {k_a}^2{k_b}^2{k_c}^2 + \frac{G_{04000}}{\sigma_2^4}{k_1}^2{k_2}^2{k_3}^2{k_4}^2 \\
&- \frac{2G_{20100}}{\sigma_0^2\sigma_1^2}\sum_{a<b}(\boldsymbol{k}_a\cdot\boldsymbol{k}_b) - \frac{2G_{11100}}{\sigma_0\sigma_1^2\sigma_2}\sum_a {k_a}^2\sum_{\substack{b<c\\ b,c\neq a}}(\boldsymbol{k}_b\cdot\boldsymbol{k}_c) - \frac{2G_{02100}}{\sigma_1^2\sigma_2^2}\sum_{a<b}(\boldsymbol{k}_a\cdot\boldsymbol{k}_b)\prod_{c\neq a,b}{k_c}^2 \\
&+ \frac{8NG_{00200}}{(N+2)\sigma_1^4}\sum_{(ab)(cd)}(\boldsymbol{k}_a\cdot\boldsymbol{k}_b)(\boldsymbol{k}_c\cdot\boldsymbol{k}_d) + \frac{2G_{20010}}{(N-1)\sigma_0^2\sigma_2^2}\left[N\sum_{a<b}(\boldsymbol{k}_a\cdot\boldsymbol{k}_b)^2 - \sum_{a<b}{k_a}^2{k_b}^2\right] \\
&+ \frac{2G_{11010}}{(N-1)\sigma_0\sigma_2^3}\Big[N\sum_a {k_a}^2\sum_{\substack{b<c\\ b,c\neq a}}(\boldsymbol{k}_b\cdot\boldsymbol{k}_c)^2 - 3\sum_{a<b<c}{k_a}^2{k_b}^2{k_c}^2\Big] + \frac{2G_{02010}}{(N-1)\sigma_2^4}\left[N\sum_{a<b}(\boldsymbol{k}_a\cdot\boldsymbol{k}_b)^2\prod_{c\neq a,b}{k_c}^2 - 6{k_1}^2{k_2}^2{k_3}^2{k_4}^2\right] \\
&+ \frac{4G_{00110}}{(N-1)\sigma_1^2\sigma_2^2}\left[-N\sum_{(ab)(cd)}(\boldsymbol{k}_a\cdot\boldsymbol{k}_b)(\boldsymbol{k}_c\cdot\boldsymbol{k}_d)[(\boldsymbol{k}_a\cdot\boldsymbol{k}_b)+(\boldsymbol{k}_c\cdot\boldsymbol{k}_d)] + \sum_{a<b}(\boldsymbol{k}_a\cdot\boldsymbol{k}_b)\prod_{c\neq a,b}{k_c}^2\right] \\
&+ \frac{N+2}{(N-1)\sigma_2^4}\left(\frac{(N-1)(N+2)}{4}G_{00000} - 2G_{00010}\right)\Big[N^2\sum_{(ab)(cd)}(\boldsymbol{k}_a\cdot\boldsymbol{k}_b)^2(\boldsymbol{k}_c\cdot\boldsymbol{k}_d)^2 \\
&\qquad - N\sum_{a<b}(\boldsymbol{k}_a\cdot\boldsymbol{k}_b)^2\prod_{c\neq a,b}{k_c}^2 + 3{k_1}^2{k_2}^2{k_3}^2{k_4}^2\Big] \\
&+ \frac{(N+2)^2G_{10001}}{(N+4)\sigma_0\sigma_2^3}\Big[N^2\sum_{\triangle(abc)}(\boldsymbol{k}_a\cdot\boldsymbol{k}_b)(\boldsymbol{k}_b\cdot\boldsymbol{k}_c)(\boldsymbol{k}_c\cdot\boldsymbol{k}_a) - N\sum_a {k_a}^2\sum_{\substack{b<c\\ b,c\neq a}}(\boldsymbol{k}_b\cdot\boldsymbol{k}_c)^2 + 2\sum_{a<b<c}{k_a}^2{k_b}^2{k_c}^2\Big] \\
&+ \frac{(N+2)^2G_{01001}}{(N+4)\sigma_2^4}\Big[N^2\sum_a {k_a}^2\prod_{\substack{b<c\\ b,c\neq a}}(\mathbf{k}_b\cdot\mathbf{k}_c) - 2N\sum_{a<b}(\mathbf{k}_a\cdot\mathbf{k}_b)^2\prod_{c\neq a,b}{k_c}^2 + 8{k_1}^2{k_2}^2{k_3}^2{k_4}^2\Big] \\
&+ \frac{(N+2)}{4(N^3+N-2)\sigma_2^4}\big((N-1)(N+2)(N^2+N+2)G_{00000} + 8(N^2+N+2)G_{00010} + 32G_{00020}\big) \\
&\quad \times\left[N^2\sum_{(ab)(cd)}(\boldsymbol{k}_a\cdot\boldsymbol{k}_b)^2(\boldsymbol{k}_c\cdot\boldsymbol{k}_d)^2 - N\sum_{a<b}(\boldsymbol{k}_a\cdot\boldsymbol{k}_b)^2\prod_{c\neq a,b}{k_c}^2 + 3{k_1}^2{k_2}^2{k_3}^2{k_4}^2\right],
\end{aligned}
\tag{B.4}
$$

where the four external indices are $a,b,c,d \in \{1,2,3,4\}$. The notation $(ab)(cd)$ denotes pairings, i.e. $(12)(34)+(13)(24)+(14)(23)$; $\triangle(abc)$ denotes triangles, i.e. $(123)+(124)+(134)+(234)$; and $\square(abcd)$ denotes squares.

B.1.2 N-dimensional number density

Going back to Eq. (B.1), putting the result from Eq. (B.4) into it, we need to perform the integral over k modes. For example, $f_{,i}f_{,i} = \int_{\boldsymbol{k}_1\boldsymbol{k}_2}(ik_{1i})(ik_{2i})\tilde{f}_1\tilde{f}_2e^{i(\boldsymbol{k}_1+\boldsymbol{k}_2)\cdot\mathbf{x}}$ gives

$$\int_{\boldsymbol{k}_1,\boldsymbol{k}_2,\boldsymbol{k}_3,\boldsymbol{k}_4}\langle(\boldsymbol{k}_1\cdot\boldsymbol{k}_2)(\boldsymbol{k}_3\cdot\boldsymbol{k}_4)\tilde{f}_1\tilde{f}_2\tilde{f}_3\tilde{f}_4\rangle_{\mathrm{c}} = \langle(\nabla f\cdot\nabla f)^2\rangle_c = \langle|\nabla f|^4\rangle_c\,.$$

Similarly, $f_{,ij}f_{,ij} = \int_{\boldsymbol{k}_1\boldsymbol{k}_2}(ik_{1i})(ik_{2i})(ik_{1j})(ik_{2j})\tilde{f}_1\tilde{f}_2e^{i(\boldsymbol{k}_1+\boldsymbol{k}_2)\cdot\mathbf{x}}$ gives

$$\int_{\boldsymbol{k}_1,\boldsymbol{k}_2,\boldsymbol{k}_3,\boldsymbol{k}_4}\langle(\boldsymbol{k}_1\cdot\boldsymbol{k}_2)^2\tilde{f}_1\tilde{f}_2\tilde{f}_3\tilde{f}_4\rangle_{\mathrm{c}} = \langle f^2f_{,ij}f_{,ij}\rangle_c = \langle f^2(\Delta f)^2\rangle + 3\langle f|\nabla f|^2\Delta f\rangle + \langle|\nabla f|^4\rangle.$$

Therefore, the final result in N-dimensional space is

$$
\begin{aligned}
\bar{n}^\star(\nu) = \text{Gaussian} + \text{1st LO} + \frac{\sigma_0^2}{24}\Bigg\{ & G_{40000}K^{(0)} + 6\gamma G_{31000}K^{(1)} + 6\gamma^2 G_{22000}K_3^{(2)} \\
& + \frac{-4}{N^2}\gamma^3 G_{13000}\left[(N^2+4)K_1^{(3)} + \frac{3N+2}{3}K_2^{(3)}\right] + \gamma^4 G_{04000}K_1^{(4)} \\
& + 6G_{20100}K^{(1)} + 12\gamma G_{11100}\left(K_1^{(2)} - \frac{1}{N}K_2^{(2)}\right) + 12\gamma^2 G_{02100}K_4^{(3)} + 12G_{00200}\left(K_1^{(2)} - K_2^{(2)}\right) \\
& + 6\gamma^2 G_{20010}\left(2K_3^{(2)} - 2K_1^{(2)} - K_2^{(2)}\right) + \frac{-4(N+2)}{N^2}\gamma^3 G_{11010}\left[12K_1^{(3)} + (N+2)K_2^{(3)}\right] \\
& + \frac{12}{N-1}\gamma^4 G_{02010}\left(NK_2^{(4)} - K_1^{(4)}\right) + \frac{24}{N-1}\gamma^2 G_{00110}\left(NK_3^{(3)} - K_4^{(3)}\right) \\
& + \frac{3(N+2)}{(N-1)}\gamma^4\left(\frac{(N-1)(N+2)}{4}G_{00000} - 2G_{00010}\right)\left(N^2K_4^{(4)} - 2NK_2^{(4)} + K_1^{(4)}\right) \\
& + \frac{-2(N-2)(N-1)(N+2)^3}{N^2}\gamma^3 G_{10001}\left(K_1^{(3)} + \frac{1}{6}K_2^{(3)}\right) \\
& + \frac{4(N+2)^2}{(N+4)}\gamma^4 G_{01001}\left(N^2K_3^{(4)} - 3NK_2^{(4)} + 2K_1^{(4)}\right) \\
& + \frac{3(N+2)}{4(N-1)(N^2+N+2)}\gamma^4\big((N-1)(N+2)(N^2+N+2)G_{00000} + 8(N^2+N+2)G_{00010} + 32G_{00020}\big) \\
& \quad \times \left[N^2K_4^{(4)} - 2NK_2^{(4)} + K_1^{(4)}\right]\Bigg\} + \frac{\sigma_0^2}{72}(\cdots)
\end{aligned}
\tag{B.5}
$$

where the kurtosis parameters are defined as follows; see Sec. B.1.4 for the detailed construction:

$$
\begin{aligned}
K^{(0)} &\equiv \frac{\langle f^4\rangle_c}{\sigma_0^6}, \qquad K^{(1)} \equiv -\frac{2}{3}\frac{\langle f^3\Delta f\rangle_c}{\sigma_0^4\sigma_1^2}, \qquad K_3^{(2)} \equiv \frac{\langle f^2(\Delta f)^2\rangle_c}{\sigma_0^2\sigma_1^4} \\
K_1^{(2)} &\equiv \frac{-2N}{(N+2)(N-1)}\frac{(N+2)\langle f|\nabla f|^2\Delta f\rangle_c + \langle|\nabla f|^4\rangle_c}{\sigma_0^2\sigma_1^4} \\
K_2^{(2)} &\equiv \frac{-2N}{(N+2)(N-1)}\frac{(N+2)\langle f|\nabla f|^2\Delta f\rangle_c + N\langle|\nabla f|^4\rangle_c}{\sigma_0^2\sigma_1^4} \\
K_1^{(3)} &\equiv \frac{N^2}{(N-2)(N-1)(N+2)(N+4)\sigma_1^6}\left[(N^2+3N+6)\langle f(\Delta f)^3\rangle_c - (6N+4)\langle f f_{,ij}f_{,ij}\Delta f\rangle_c\right] \\
K_2^{(3)} &\equiv \frac{6N^2}{(N-2)(N-1)(N+2)(N+4)\sigma_1^6}\left[-(3N+2)\langle f(\Delta f)^3\rangle_c + (N^2+4)\langle f f_{,ij}f_{,ij}\Delta f\rangle_c\right] \\
K_3^{(3)} &\equiv \frac{\langle|\nabla f|^2 f_{,ij}f_{,ij}\rangle_c}{\sigma_1^6}, \qquad K_4^{(3)} \equiv \frac{\langle|\nabla f|^2(\Delta f)^2\rangle_c}{\sigma_1^6} \\
K_1^{(4)} &\equiv \frac{\sigma_0^2}{\sigma_1^8}\langle(\Delta f)^4\rangle_c, \qquad K_2^{(4)} \equiv \frac{\sigma_0^2}{\sigma_1^8}\langle f_{,ij}f_{,ij}(\Delta f)^2\rangle_c, \qquad K_3^{(4)} \equiv \frac{\sigma_0^2}{\sigma_1^8}\langle f_{,ij}f_{,jk}f_{ki}(\Delta f)\rangle_c \\
K_4^{(4)} &\equiv \frac{\sigma_0^2}{\sigma_1^8}\langle f_{,ij}f_{,ij}f_{,kl}f_{,kl}\rangle_c .
\end{aligned}
\tag{B.6}
$$

For $N=2$, note that

$$
K_1^{(3)} = K_2^{(3)} \equiv \frac{3\langle f(\Delta f)^3\rangle_c}{8\sigma_1^6}.
$$

The field cumulants are

$$\langle f^4\rangle_c = \langle f^4\rangle - 3\sigma_0^4, \qquad \langle f^3\Delta f\rangle_c = \langle f^3\Delta f\rangle + 3\sigma_0^2\sigma_1^2, \qquad \langle f^2|\nabla f|^2\rangle_c = \langle f^2|\nabla f|^2\rangle - \sigma_0^2\sigma_1^2,$$
$$\langle f|\nabla f|^2\Delta f\rangle_c = \langle f|\nabla f|^2\Delta f\rangle + \sigma_1^4, \quad \langle|\nabla f|^4\rangle_c = \langle|\nabla f|^4\rangle - \frac{N+2}{N}\sigma_1^4,$$
$$\langle f^2(\Delta f)^2\rangle_c = \langle f^2(\Delta f)^2\rangle - \sigma_0^2\sigma_2^2 - 2\sigma_1^4$$
$$\langle f(\Delta f)^3\rangle_c = \langle f(\Delta f)^3\rangle + 3\sigma_1^2\sigma_2^2, \qquad \langle f f_{,ij} f_{,ij}\Delta f\rangle_c = \langle f f_{,ij} f_{,ij}\Delta f\rangle + \frac{N+2}{N}\sigma_1^2\sigma_2^2,$$
$$\langle|\nabla f|^2 f_{,ij} f_{,ij}\rangle_c = \langle|\nabla f|^2 f_{,ij} f_{,ij}\rangle - \sigma_1^2\sigma_2^2, \qquad \langle|\nabla f|^2(\Delta f)^2\rangle_c = \langle|\nabla f|^2(\Delta f)^2\rangle - \sigma_1^2\sigma_2^2,$$
$$\langle(\Delta f)^4\rangle_c = \langle(\Delta f)^4\rangle - 3\sigma_2^4, \quad \langle f_{,ij} f_{,ij}(\Delta f)^2\rangle_c = \langle f_{,ij} f_{,ij}(\Delta f)^2\rangle - \frac{N+2}{N}\sigma_2^4,$$
$$\langle f_{,ij} f_{,jk} f_{,ki}(\Delta f)\rangle_c = \langle f_{,ij} f_{,jk} f_{,ki}(\Delta f)\rangle - \frac{3}{N}\sigma_2^4$$
$$\langle f_{,ij} f_{,ij} f_{,kl} f_{,kl}\rangle_c = \langle f_{,ij} f_{,ij} f_{,kl} f_{,kl}\rangle - \left(1 + \frac{6}{N(N+2)}\right)\sigma_2^4,$$

The general proof follows from

$$\langle X_1X_2X_3X_4\rangle_c = \langle X_1X_2X_3X_4\rangle - \sum_{\text{pairing}} \langle X_iX_j\rangle\langle X_kX_l\rangle$$

where $X_i \in \{f, f_{,i}, f_{,ij}\}$, together with isotropy relations such as $\langle ff\rangle = \sigma_0^2$, $\langle f\Delta f\rangle = -\langle|\nabla f|^2\rangle = -\sigma_1^2$, $\langle f_{,i}f_{,j}\rangle = \sigma_1^2\delta_{ij}/N$, and $\langle f_{,ij}f_{kl}\rangle = \sigma_2^2(\delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk})/(N(N+2))$.

B.1.3 Rank-$2r$ isotropic traceless tensor projector

First, for any matrix X, note that the rotational invariants can be written as $p_1^2 \equiv X_{ii} = \delta_{ij}\delta_{kl}X_{ij}X_{kl}$ and $p_2 \equiv X_{ij}X_{ji} = \frac{1}{2}(\delta_{ik}\delta_{jl}+\delta_{il}\delta_{jk})X_{ij}X_{kl}$. We can then construct the polynomial $Q[X] = ap_2+bp_1^2 = P^{(2)}_{ij,kl}X_{ij}X_{kl}$, so that

$$P^{(2)}_{ij,kl} = a(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}) + b\delta_{ij}\delta_{kl} \qquad \text{(traceful)}$$

forms a basis for any rank-4 symmetric tensor, but it is not traceless. Next, we want a projector that is both symmetric and traceless, so we construct $\tilde{Q}[X] = \tilde{P}^{(2)}_{ij,kl}X_{ij}X_{kl}$. Define $\tilde{X} = X - \frac{p_1}{N}\mathbf{1}$, and note that $Q[\tilde{X}] = \alpha\tilde{p}_2 = P^{(2)}_{ij,kl}\tilde{X}_{ij}\tilde{X}_{kl} = \tilde{P}^{(2)}_{ij,kl}X_{ij}X_{kl}$. Since $\tilde{p}_2 = p_2 - \frac{1}{N}p_1^2$, this gives

$$\tilde{P}^{(2)}_{ij,kl} = \frac{\alpha}{2}\left(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk} - \frac{2}{N}\delta_{ij}\delta_{kl}\right) \qquad \text{(traceless)}$$

This constructs a traceless projector, which provides a basis for traceless symmetric rank-4 tensors. The traceless property of $\tilde{X}$ leads directly to $\tilde{P}^{(2)}$, the corresponding traceless tensor representative.

Next, for rank-6, we need the fully connected (no-trace), single-trace, pairwise-contraction, and triple-trace structures:

$$\mathcal{A}_{ij,kl,mn} = \delta_{ij}\delta_{kl}\delta_{mn}$$
$$\mathcal{B}_{ij,kl,mn} = \delta_{ij}(\delta_{km}\delta_{ln} + \delta_{kn}\delta_{lm}) + \delta_{kl}(\delta_{im}\delta_{jn} + \delta_{in}\delta_{jm}) + \delta_{mn}(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk})$$
$$\mathcal{C}_{ij,kl,mn} = \delta_{ik}\delta_{jm}\delta_{ln} + \delta_{ik}\delta_{jn}\delta_{lm} + \delta_{il}\delta_{jm}\delta_{kn} + \delta_{il}\delta_{jn}\delta_{km} + \delta_{jk}\delta_{im}\delta_{ln} + \delta_{jk}\delta_{in}\delta_{lm} + \delta_{jl}\delta_{im}\delta_{kn} + \delta_{jl}\delta_{in}\delta_{km}$$

so that $p_3 \equiv X_{ij}X_{jk}X_{ki} = \frac{1}{8}\mathcal{C}_{ij,kl,mn}X_{ij}X_{kl}X_{mn}$, $p_1p_2 \equiv X_{ii}X_{kl}X_{lk} = \frac{1}{6}\mathcal{B}_{ij,kl,mn}X_{ij}X_{kl}X_{mn}$, and $p_1^3 \equiv \mathcal{A}_{ij,kl,mn}X_{ij}X_{kl}X_{mn}$. Therefore, $Q[X] = ap_3 + bp_1p_2 + cp_1^3 = P^{(3)}_{ij,kl,mn}X_{ij}X_{kl}X_{mn}$, so that $P^{(3)}_{ij,kl,mn} = a\mathcal{A} + \frac{b}{6}\mathcal{B}\frac{c}{8}\mathcal{C}$. Since $\tilde{p}_3 = p_3 - \frac{3}{N}p_1p_2 + \frac{2}{N^2}p_1^3$ and $Q[\tilde{X}] = \alpha\tilde{p}_3$, we obtain the traceless symmetric rank-6 tensor representative $\tilde{P}^{(3)}$ as

$$\tilde{P}^{(3)}_{ij,kl,mn} = \frac{\alpha}{8}\left(\mathcal{C}_{ij,kl,mn} - \frac{4}{N}\mathcal{B}_{ij,kl,mn} + \frac{16}{N^2}\mathcal{A}_{ij,kl,mn}\right)$$

Similarly, for a rank-8 isotropic tensor, we introduce four blocks $A = (ij)$, $B = (kl)$, $C = (mn)$, and $D = (pq)$. We write $Q[X] = ap_1^4 + bp_1^2p_2 + cp_1p_3 + dp_2^2 + ep_4 = P^{(4)}_{ij,kl,mn,pq}X_{ij}X_{kl}X_{mn}X_{pq}$. For $N \leqslant 3$, we know that $Q[\tilde{X}] = \alpha\tilde{p}_4 + \beta(\tilde{p}_2)^2$, with $\tilde{p}_4 = p_4 - \frac{4}{N}p_1p_3 + \frac{6}{N^2}p_1^2p_2 - \frac{3}{N^3}p_1^4$ and $\tilde{p}_2 = p_2 - \frac{1}{N}p_1^2$. Thus, $p_1^4 = \mathcal{A}^{(4)}_{ij,kl,mn,pq}X_{ij}X_{kl}X_{mn}X_{pq}$, $p_1^2p_2 = \frac{1}{12}\mathcal{B}^{(4)}XXXX$, and $p_2^2 = \frac{1}{12}\mathcal{D}^{(4)}XXXX$. The tensors $\mathcal{A}^{(4)}$, $\mathcal{B}^{(4)}$, and $\mathcal{D}^{(4)}$ are defined in Eq. (B.3).

We impose the constraint $2\hat{\alpha} = \hat{\beta}$. For $N = 1$, this gives $\hat{\alpha} = 0$ and $\hat{\beta} = 0$; for $N = 2$, it gives $\hat{\alpha} = \frac{2}{3}(\alpha + 2\beta)$ and $\hat{\beta} = \frac{4}{3}(\alpha + 2\beta)$; and for $N = 3$, it gives $\hat{\alpha} = \frac{35}{12}(\alpha + 2\beta)$ and $\hat{\beta} = \frac{35}{6}(\alpha + 2\beta)$. Therefore, we define the new parameter

$$\omega \equiv \frac{1}{2}\alpha + \beta$$

or equivalently,

$$\hat{\alpha} = \frac{(N-1)(N+2)(2N^2+3N-6)}{12N}\omega, \qquad \hat{\beta} = \frac{1}{12}(N-1)(N+2)(N^2+N+2)\omega$$

which gives

$$2\hat{\alpha} - \hat{\beta} = -\frac{(N-1)(N-2)(N-3)(2+N)^2}{12N}\omega$$

Moreover, $Q[\tilde{X}] = \omega(\tilde{p}_2)^2$ gives

$$\tilde{P}^{(4)} = \omega\left(\frac{1}{N^2}\mathcal{A}^{(4)} - \frac{1}{6N}\mathcal{B}^{(4)} + \frac{1}{12}\mathcal{D}^{(4)}\right)$$

Assume $\langle\tilde{\zeta}\tilde{\zeta}\tilde{\zeta}\tilde{\zeta}\rangle_\Omega = \mathcal{N}_4\tilde{P}^{(4)}$. Using the constraint

$$\langle\tilde{\zeta}_{ij}\tilde{\zeta}_{ji}\rangle_\Omega\,\langle\tilde{\zeta}_{kl}\tilde{\zeta}_{lk}\rangle_\Omega = \langle\tilde{\zeta}_{ij}\tilde{\zeta}_{ji}\tilde{\zeta}_{kl}\tilde{\zeta}_{lk}\rangle_\Omega$$

we finally obtain Eq. (B.2).

B.1.4 Definition of the kurtosis parameters

For $K^{(0)}$, $K^{(1)}$, and $K^{(2)}$

For a field expanded as $f = f^{(1)} + f^{(2)} + \cdots$, we have $f^{(1)} \sim \sigma_0$ and $f^{(2)} \sim \sigma_0^2$. A connected n-point tree diagram has $n-1$ edges, and each edge contributes a power of σ_0^2, so $\langle f^n\rangle_c \sim \sigma_0^{2n-2}$. We therefore parametrize

$$\langle\alpha^4\rangle_c = \tilde{K}^{(0)}\sigma_0^2 \quad\Rightarrow\quad K^{(0)} \equiv \tilde{K}^{(0)} = \frac{\langle f^4\rangle_c}{\sigma_0^6}$$

Also, $\langle f^2 f_{,i}f_{,j}\rangle_c \sim \sigma_0^4\sigma_1^2$, so

$$\langle\alpha^2\eta_i\eta_j\rangle_c = \frac{1}{N}\tilde{K}^{(1)}\delta_{ij}\sigma_0^2 \quad\Rightarrow\quad K^{(1)} \equiv 2\tilde{K}^{(1)} = 2\frac{\langle f^2|\nabla f|^2\rangle_c}{\sigma_0^4\sigma_1^2} = -\frac{2}{3}\frac{\langle f^3\Delta f\rangle_c}{\sigma_0^4\sigma_1^2}$$

Since $\langle ff_{,i}f_{,j}f_{,kl}\rangle \sim \sigma_0^3\sigma_1^2\sigma_2$, we next parametrize

$$\langle\alpha\eta_i\eta_j\zeta_{kl}\rangle_c = \frac{\gamma}{N^2}\left[\tilde{K}_1^{(2)}\delta_{ij}\delta_{kl} + \tilde{K}_2^{(2)}\frac{1}{2}(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk})\right]\sigma_0^2$$

Contracting the indices gives $N\langle f|\nabla f|^2\Delta f\rangle/(\sigma_1^4\sigma_0^2) = \tilde{K}_1^{(2)}N + \tilde{K}_2^{(2)}$. The identity $\langle f_{,i}f_{,i}f_{,j}f_{,j}\rangle + \langle ff_{,ii}f_{,j}f_{,j}\rangle + 2\langle ff_{,i}f_{,j}f_{,ij}\rangle = 0$ gives $-N\Big(\langle|\nabla f|^4\rangle + \langle f|\nabla f|^2\Delta f\rangle\Big)/(2\sigma_1^4\sigma_0^2) = \tilde{K}_1^{(2)} + \tilde{K}_2^{(2)}(N+1)/2$. Solving the resulting linear equations, we redefine

$$K_1^{(2)} \equiv -2\tilde{K}_1^{(2)} = \frac{-2N}{(N+2)(N-1)}\frac{(N+2)\langle f|\nabla f|^2\Delta f\rangle_c + \langle|\nabla f|^4\rangle_c}{\sigma_0^2\sigma_1^4}$$

$$K_2^{(2)} \equiv 2\tilde{K}_2^{(2)} = \frac{-2N}{(N+2)(N-1)}\frac{(N+2)\langle f|\nabla f|^2\Delta f\rangle_c + N\langle|\nabla f|^4\rangle_c}{\sigma_0^2\sigma_1^4}$$

Equivalently, $\langle f|\nabla f|^2\Delta f\rangle_c = \frac{\sigma_1^4\sigma_0^2}{2N}\Big(-K_1^{(2)}N + K_2^{(2)}\Big)$, and $\langle|\nabla f|^4\rangle_c = \frac{\sigma_1^4\sigma_0^2(N+2)}{2N}\Big(K_1^{(2)} - K_2^{(2)}\Big)$.

We can express the other second-order quadratic terms using these two relations. The identity $\partial_i\langle f^2 f_{,i}f_{,jj}\rangle = 0$ gives $2\langle ff_{,i}f_{,i}f_{,jj}\rangle + \langle f^2 f_{,ii}f_{,jj}\rangle + \langle f^2 f_{,i}f_{,ijj}\rangle$, $\partial_j\langle f^2 f_{,i}f_{,ij}\rangle = 0$ gives $2\langle ff_{,i}f_{,j}f_{,ij}\rangle + \langle f^2 f_{,ij}f_{,ij}\rangle + \langle f^2 f_{,i}f_{,ijj}\rangle = 0$, and $\partial_j\langle ff_{,i}f_{,i}f_{,j}\rangle = 0$ gives $2\langle ff_{,i}f_{,j}f_{,ij}\rangle = -\langle f_{,i}f_{,i}f_{,j}f_{,j}\rangle - \langle ff_{,i}f_{,i}f_{,jj}\rangle$, so that

$$\begin{aligned}\langle ff_{,i}f_{,j}f_{,ij}\rangle_c &= \frac{-1}{2}\Big(\langle|\nabla f|^4\rangle_c + \langle f|\nabla f|^2\Delta f\rangle_c\Big)\\ \langle f^2 f_{,ij}f_{,ij}\rangle_c &= \langle f^2(\Delta f)^2\rangle + 3\langle f|\nabla f|^2\Delta f\rangle_c + \langle|\nabla f|^4\rangle_c\end{aligned}$$

We need one additional cumulant from $\langle f^2 f_{,ij}f_{,kl}\rangle \sim \sigma_0^4\sigma_2^2 \sim \sigma_0^2\sigma_1^4$, so

$$\langle\alpha^2\zeta_{ii}\zeta_{kk}\rangle = \sigma_0^2\gamma^2\tilde{K}_3^{(2)} \quad\Rightarrow\quad K_3^{(2)} \equiv \tilde{K}_3^{(2)} = \frac{\langle f^2(\Delta f)^2\rangle}{\sigma_0^2\sigma_1^4}$$

For $K^{(3)}$

Next, we consider higher-order terms. Since $\langle f_{,i}f_{,j}f_{,kl}f_{,pq}\rangle_c \sim \sigma_0^2\sigma_1^2\sigma_2^2 \sim \sigma_1^6$, we can parametrize

$$\langle\eta_i\eta_j\zeta_{kl}\zeta_{pq}\rangle_c = \frac{\gamma^2}{N^3}\Big[\tilde{K}_1^{(3)}\delta_{ij}\delta_{kl}\delta_{pq} + \frac{1}{2}\tilde{K}_2^{(3)}(\Delta_{ij,kl}\delta_{pq} + \Delta_{ij,pq}\delta_{kl}) + \tilde{K}_3^{(3)}\Delta_{kl,pq}\delta_{ij} + \tilde{K}_4^{(3)}\Delta_{ij,kl,pq}\Big]\sigma_0^2$$

where $\Delta_{ij,kl} \equiv 1/2(\delta_{ik}\delta_{jl}+\delta_{il}\delta_{jk})$ and $\Delta_{ij,kl,mn} \equiv 1/8[(\delta_{jk}\delta_{lp}\delta_{qi} + \delta_{jk}\delta_{lp}\delta_{pi} + \delta_{jl}\delta_{kp}\delta_{qi} + \delta_{jl}\delta_{kq}\delta_{pi}) + (i \leftrightarrow j)]$. We perform the contractions

$$\begin{aligned}\frac{\langle|\nabla f|^2(\Delta f)^2\rangle_c}{\sigma_1^2\sigma_2^2} &= \langle\eta_i\eta_i\zeta_{kk}\zeta_{pp}\rangle_c = \frac{\gamma^2}{N^2}\Big[N^2\tilde{K}_1^{(3)} + N\tilde{K}_2^{(3)} + N\tilde{K}_3^{(3)} + \tilde{K}_4^{(3)}\Big]\\ \frac{\langle f_{,i}f_{,k}f_{,ik}(\Delta f)\rangle_c}{\sigma_1^2\sigma_2^2} &= \langle\eta_i\eta_k\zeta_{ik}\zeta_{pp}\rangle_c = \frac{\gamma^2}{N^2}\Big[N\tilde{K}_1^{(3)} + \frac{(N^2+N+2)}{2}\tilde{K}_2^{(3)} + \tilde{K}_3^{(3)} + \frac{N+1}{2}\tilde{K}_4^{(3)}\Big]\\ \frac{\langle|\nabla f|^2 f_{,kp}f_{,kp}\rangle_c}{\sigma_1^2\sigma_2^2} &= \langle\eta_i\eta_i\zeta_{kp}\zeta_{kp}\rangle_c = \frac{\gamma^2}{N^2}\Big[N\tilde{K}_1^{(3)} + \tilde{K}_2^{(3)} + \frac{N^2+N}{2}\tilde{K}_3^{(3)} + \frac{N+1}{2}\tilde{K}_4^{(3)}\Big]\\ \frac{\langle f_{,i}f_{,k}f_{,kp}f_{,pi}\rangle_c}{\sigma_1^2\sigma_2^2} &= \langle\eta_i\eta_k\zeta_{kp}\zeta_{kp}\rangle_c = \frac{\gamma^2}{N^2}\Big[N\tilde{K}_1^{(3)} + \frac{N+1}{2}\tilde{K}_2^{(3)} + \frac{N+1}{2}\tilde{K}_3^{(3)} + \frac{N^2+3N+4}{8}\tilde{K}_4^{(3)}\Big]\end{aligned}$$

and obtain the relation

$$0 = (\langle\eta_i\eta_j\zeta_{kl}\zeta_{pq}\rangle + \langle\eta_i\eta_k\zeta_{jl}\zeta_{pq}\rangle + \langle\eta_j\eta_k\zeta_{il}\zeta_{pq}\rangle) - (l \leftrightarrow q) = 4\tilde{K}_1^{(3)} + 2\tilde{K}_2^{(3)} - 2\tilde{K}_3^{(3)} - \tilde{K}_4^{(3)}$$

for $N \geqslant 2$.

Similarly to $\tilde{K}_3^{(2)}$, we introduce a parameter from $\langle ff_{,ij}f_{,kl}f_{,mn}\rangle \sim \sigma_0^3\sigma_2^3 \sim \sigma_1^6$:

$$\langle\alpha\zeta_{ii}\zeta_{jj}\zeta_{kk}\rangle = \sigma_0^2\gamma^3\tilde{K}_5^{(3)} \quad\Rightarrow\quad K_5^{(3)} \equiv \tilde{K}_5^{(3)} = \frac{\langle f(\Delta f)^3\rangle}{\sigma_1^6}$$

There does not appear to be a simple way to define the remaining cumulants from this relation alone.

We decompose and parametrize the tensor

$$\langle\alpha\zeta_{ij}\zeta_{kl}\zeta_{pq}\rangle_c = \frac{\gamma^3}{N^3}\Big[\tilde{K}_a^{(3)}\delta_{ij}\delta_{kl}\delta_{pq} + \frac{1}{3}\tilde{K}_b^{(3)}(\Delta_{ij,kl}\delta_{pq} + \Delta_{ij,pq}\delta_{kl} + \Delta_{kl,pq}\delta_{ij}) + \tilde{K}_c^{(3)}\Delta_{ij,kl,pq}\Big]\sigma_0^2$$

For $N \geqslant 2$, we obtain the relation

$$0 = (\langle\alpha\zeta_{ij}\zeta_{kl}\zeta_{pq}\rangle + \langle\alpha\zeta_{ik}\zeta_{jl}\zeta_{pq}\rangle + \langle\alpha\zeta_{jk}\zeta_{il}\zeta_{pq}\rangle) - (l \leftrightarrow q) = 12\tilde{K}_a^{(3)} + 2\tilde{K}_b^{(3)} - 3\tilde{K}_c^{(3)}$$

and

$$\frac{\langle f(\Delta f)^3\rangle_c}{\sigma_0\sigma_2^3} = \langle \alpha\zeta_{ii}\zeta_{jj}\zeta_{kk}\rangle_c = \frac{\gamma^3}{N^2}\Big[N^2\tilde{K}_a^{(3)} + N\tilde{K}_b^{(3)} + \tilde{K}_c^{(3)}\Big]\sigma_0^2$$
$$\frac{\langle ff_{,ij}f_{,ij}(\Delta f)\rangle_c}{\sigma_0\sigma_2^3} = \langle \alpha\zeta_{ij}\zeta_{ij}\zeta_{kk}\rangle_c = \frac{\gamma^3}{N^2}\left[N\tilde{K}_a^{(3)} + \frac{N^2+N+4}{6}\tilde{K}_b^{(3)} + \frac{N+1}{2}\tilde{K}_c^{(3)}\right]\sigma_0^2$$
$$\frac{\langle ff_{,ij}f_{,jk}f_{ki}\rangle_c}{\sigma_0\sigma_2^3} = \langle \alpha\zeta_{ij}\zeta_{jk}\zeta_{ki}\rangle_c = \frac{\gamma^3}{N^2}\left[\tilde{K}_a^{(3)} + \frac{N+1}{2}\tilde{K}_b^{(3)} + \frac{N^2+3N+4}{8}\tilde{K}_c^{(3)}\right]\sigma_0^2$$

Therefore, we define

$$K_1^{(3)} \equiv \tilde{K}_a^{(3)} = \frac{N^2}{(N-2)(N-1)(N+2)(N+4)\sigma_1^6}\left[(N^2+3N+6)\langle f(\Delta f)^3\rangle_c - (6N+4)\langle ff_{,ij}f_{,ij}(\Delta f)\rangle\right]$$
$$K_2^{(3)} \equiv \tilde{K}_b^{(3)} = \frac{6N^2}{(N-2)(N-1)(N+2)(N+4)\sigma_1^6}\left[-(3N+2)\langle f(\Delta f)^3\rangle_c + (N^2+4)\langle ff_{,ij}f_{,ij}(\Delta f)\rangle\right]$$

Thus,

$$\langle f(\Delta f)^3\rangle_c = \frac{\sigma_1^6}{N^2}\left[(N^2+4)K_1^{(3)} + \frac{3N+2}{3}K_2^{(3)}\right]$$
$$\langle ff_{,ij}f_{,ij}(\Delta f)\rangle_c = \frac{\sigma_1^6}{N^2}\left[(3N+2)K_1^{(3)} + \frac{N^2+3N+6}{6}K_2^{(3)}\right]$$
$$\langle ff_{,ij}f_{jk}f_{ki}\rangle = \frac{\sigma_1^6}{N^2}\left[\frac{N^2+3N+6}{2}K_1^{(3)} + \frac{N^2+9N+10}{12}K_2^{(3)}\right]$$

When $N=2$, the symmetry constraint gives $\langle f(\Delta f)^3\rangle_c = \langle ff_{,ij}f_{,ij}(\Delta f)\rangle_c = \langle ff_{,ij}f_{,jk}f_{,ki}\rangle_c$, so the three invariants collapse into one, i.e.

$$\tilde{K}_a^{(3)} = \frac{\tilde{K}_c^{(3)} - \langle f(\Delta f)^3\rangle_c/\sigma_1^6}{2}, \quad \tilde{K}_b^{(3)} = \frac{3\langle f(\Delta f)^3\rangle_c}{\sigma_1^6} - \frac{3}{2}\tilde{K}_c^{(3)}$$

We can choose $\tilde{K}_c^{(3)}$ freely and define

$$K_1^{(3)} = K_2^{(3)} \equiv \tilde{K}_a^{(3)} = \tilde{K}_b^{(3)} = \frac{3\langle f(\Delta f)^3\rangle_c}{8\sigma_1^6}$$

We can define two additional independent cumulants:

$$\langle \eta_k\eta_k\zeta_{ij}\zeta_{ij}\rangle = \sigma_0^2\gamma^2\tilde{K}_3^{(3)} \quad \Rightarrow \quad K_3^{(3)} \equiv \tilde{K}_3^{(3)} = \frac{\langle |\nabla f|^2 f_{,ij}f_{,ij}\rangle}{\sigma_1^6}$$
$$\langle \eta_i\eta_i\zeta_{jj}\zeta_{kk}\rangle = \sigma_0^2\gamma^2\tilde{K}_4^{(3)} \quad \Rightarrow \quad K_4^{(3)} \equiv \tilde{K}_4^{(3)} = \frac{\langle |\nabla f|^2(\Delta f)^2\rangle}{\sigma_1^6}$$

For $K^{(4)}$

Next, we consider $K^{(4)}$. We define

$$\langle \zeta_{ii}\zeta_{jj}\zeta_{kk}\zeta_{ll}\rangle = \sigma_0^2\gamma^4\tilde{K}_1^{(4)} \quad \Rightarrow \quad K_1^{(4)} \equiv \tilde{K}_1^{(4)} = \frac{\sigma_0^2}{\sigma_1^8}\langle(\Delta f)^4\rangle_c$$
$$\langle \zeta_{ij}\zeta_{ij}\zeta_{kk}\zeta_{ll}\rangle = \sigma_0^2\gamma^4\tilde{K}_2^{(4)} \quad \Rightarrow \quad K_2^{(4)} \equiv \tilde{K}_2^{(4)} = \frac{\sigma_0^2}{\sigma_1^8}\langle f_{,ij}f_{,ij}(\Delta f)^2\rangle_c$$
$$\langle \zeta_{ij}\zeta_{jk}\zeta_{ki}\zeta_{ll}\rangle = \sigma_0^2\gamma^4\tilde{K}_3^{(4)} \quad \Rightarrow \quad K_3^{(4)} \equiv \tilde{K}_3^{(4)} = \frac{\sigma_0^2}{\sigma_1^8}\langle f_{,ij}f_{,jk}f_{,ki}(\Delta f)\rangle_c$$
$$\langle \zeta_{ij}\zeta_{ij}\zeta_{kl}\zeta_{kl}\rangle = \sigma_0^2\gamma^4\tilde{K}_4^{(4)} \quad \Rightarrow \quad K_4^{(4)} \equiv \tilde{K}_4^{(4)} = \frac{\sigma_0^2}{\sigma_1^8}\langle f_{,ij}f_{,ij}f_{,kl}f_{,kl}\rangle_c$$
$$\langle \zeta_{ij}\zeta_{jk}\zeta_{kl}\zeta_{li}\rangle = \sigma_0^2\gamma^4\tilde{K}_5^{(4)} \quad \Rightarrow \quad K_5^{(4)} \equiv \tilde{K}_5^{(4)} = \frac{\sigma_0^2}{\sigma_1^8}\langle f_{,ij}f_{,jk}f_{,kl}f_{,li}\rangle_c$$

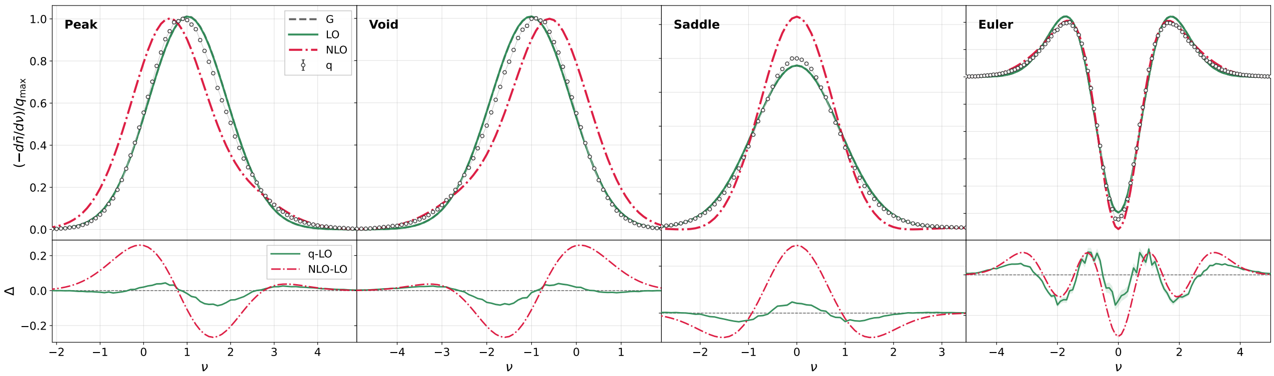

Figure B.1: Kurtosis-order correction results for $\mathcal{A}=0$ and $\mathcal{B}=0.02$.

We can reduce the parameters further under dimension-dependent conditions. Let $p_1=\operatorname{tr}H$, $p_2=\operatorname{tr}H^2$, $p_3=\operatorname{tr}H^3$, and $p_4=\operatorname{tr}H^4$. For generic $N\geqslant 4$, five pure-Hessian kurtosis parameters are needed. The Cayley-Hamilton theorem gives the identity

$$p_4=\frac{1}{6}p_1^4-p_1^2p_2+\frac{4}{3}p_1p_3+\frac{1}{2}p_2^2 \quad\Rightarrow\quad K_3^{(4)}=\frac{3}{4}\left[K_5^{(4)}-\frac{1}{6}K_1^{(4)}+K_2^{(4)}-\frac{1}{2}K_4^{(4)}\right]$$

for both $N=2$ and $N=3$. For $N=2$, we also have the identity

$$p_3=-\frac{1}{2}p_1^3+\frac{3}{2}p_1p_2 \quad\Rightarrow\quad p_4=-\frac{1}{2}p_1^4+p_1^2p_2+\frac{1}{2}p_2^2$$

Thus, for a two-dimensional field,

$$K_3^{(4)}=-\frac{1}{2}K_1^{(4)}+\frac{3}{2}K_2^{(4)},\qquad K_5^{(4)}=-\frac{1}{2}K_1^{(4)}+K_2^{(4)}+\frac{1}{2}K_4^{(4)}$$

B.2 False results in two-dimensional fields

2D Formula

For $N=2$, we can reduce Eq. (B.5) to

$$\begin{aligned}\bar{n}^{\star}_{\mathrm{2D}}(\nu)=&\text{Gaussian}+\text{1st LO}+\frac{\sigma_0^2}{24}\Big[G_{40000}K^{(0)}+6\gamma G_{31000}K^{(1)}+6\gamma^2G_{22000}K_3^{(2)}\\&-\frac{32}{3}\gamma^3(G_{13000}+6G_{11010})K_1^{(3)}+\gamma^4G_{04000}K_1^{(4)}+6G_{20100}K^{(1)}+12\gamma G_{11100}\Big(K_1^{(2)}-\frac{1}{2}K_2^{(2)}\Big)\\&+12\gamma^2G_{02100}K_4^{(3)}+12G_{00200}\Big(K_1^{(2)}-K_2^{(2)}\Big)+6\gamma^2G_{20010}\Big(2K_3^{(2)}-2K_1^{(2)}-K_2^{(2)}\Big)\\&+12\gamma^4G_{02010}\Big(2K_2^{(4)}-K_1^{(4)}\Big)+24\gamma^2G_{00110}\Big(2K_3^{(3)}-K_4^{(3)}\Big)\\&+12\gamma^4(2G_{00000}+G_{00020})\Big(4K_4^{(4)}-4K_2^{(4)}+K_1^{(4)}\Big)\Big]+\frac{\sigma_0^2}{72}(\cdots)\end{aligned}$$

where $\gamma\equiv\sigma_1^2/(\sigma_0\sigma_2)$, the kurtosis coefficients are given in Eq. (B.6), and the G_{ijklm} specific form see in the codes.

We consider and test a two-dimensional field of the form

$$\kappa(\boldsymbol{\theta})=\kappa_g+\frac{\mathcal{A}}{\sigma_g}(\kappa_g^2-\langle\kappa_g^2\rangle)+\frac{\mathcal{B}}{\sigma_g}\left(\kappa_g^3-3\langle\kappa_g^2\rangle\kappa_g\right)$$

We set $\mathcal{A}$ to zero, so the field has negligible skewness at leading order in $\mathcal{A}$ and $\mathcal{B}$. The trispectrum is

$$
\begin{aligned}
T_\psi(\ell_1,\ell_2,\ell_3,\ell_4) &= 6\frac{\mathcal{B}}{\sigma_g}[P_\psi(\ell_1)P_\psi(\ell_2)P_\psi(\ell_3)+\text{cyc.}] \\
T_\kappa(\ell_1,\ell_2,\ell_3,\ell_4) &= \frac{\ell_1^2\ell_2^2\ell_3^2\ell_4^2}{16}T_\psi(\ell_1,\ell_2,\ell_3,\ell_4) = 24\frac{\mathcal{B}}{\sigma_g}\left[\frac{\ell_4^2}{\ell_1^2\ell_2^2\ell_3^2}P_\kappa(\ell_1)P_\kappa(\ell_2)P_\kappa(\ell_3)+\text{cyc.}\right] \\
\tilde{T}_{\kappa,\pm}(\ell_1,\ell_2,\ell_3,\ell_4) &= 24\frac{\mathcal{B}}{\sigma_g}\frac{\ell_{4,\pm}^2}{2\ell_1^2\ell_2^2\ell_3^2}P_\kappa(\ell_1)P_\kappa(\ell_2)P_\kappa(\ell_3)\times 4
\end{aligned}
$$

where $\ell_{4,\pm}^2 = \ell_1^2+\ell_2^2+\ell_3^2+2\ell_1\ell_2\mu+2\ell_1\ell_3\lambda+2\ell_2\ell_3\left[\mu\lambda\pm\sqrt{(1-\mu^2)(1-\lambda^2)}\right]$.

Results

Due to limited time, we did not use the trispectrum directly. Instead, we estimated the kurtosis parameters from the simulated fields. Figure B.1 shows the case with $\mathcal{A}=0$ and $\mathcal{B}=0.02$, where the skewness and kurtosis are measured directly from 300 simulated realizations. Since $\mathcal{A}=0$, the LO curve almost overlaps with the Gaussian prediction. From Δ, we see that $q-\text{LO}$ and $\text{NLO}-\text{LO}$ show similar trends as functions of ν, but their amplitudes differ substantially. The reason for this discrepancy is still unclear. At least the function weight for each CPs should be right. We leave the solution of the next-to-leading-order prediction to future work.

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